Occupation counts of rarely switching Markov chains: stratified local laws, crossings and the boundary layer

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Abstract

A Markov chain on finitely many states makes about $\boldsymbol{\tau \log N}$ switches in $\boldsymbol{N}$ steps. We ask which states its most likely vector of occupation counts uses. On each face of the simplex the law weighs a lattice entropy of one power of $\boldsymbol{N}$ per dimension against the Donsker and Varadhan cost of the chain killed on leaving the face, which is the restriction of the cost of the whole chain. A local limit theorem in each face gives an explicit constant. It locates every crossing between irreducible faces up to $\boldsymbol{o(1)}$, with a coefficient $\boldsymbol{1/2}$ of $\boldsymbol{\log\log N}$ independent of the chain, and describes the window of tied faces. Near a facet of a face the law is the facet law times an explicit kernel, up to a relative error $\boldsymbol{O(1/T)}$ for $\boldsymbol{T}$ switches. For a reversible chain the kernel is a Bessel kernel up to a relative error that is small when the time $\boldsymbol{x}$ in the rare state is $\boldsymbol{o(T)}$. Without reversibility we prove the Bessel form in logarithmic form, and in relative form only for $\boldsymbol{x}$ up to order $\boldsymbol{T^{1/3}}$, and numerics on a cycle of three states show that this restriction is real. On the complete graph all faces tie at first order. There each face crosses a vertex exactly once at every $\boldsymbol{N}$, the most likely compositions on each face are balanced whenever $\boldsymbol{T \leq N/d}$ on $\boldsymbol{d}$ states, and for large $\boldsymbol{N}$ the mode changes exactly once.

Keywords: occupation times of Markov chains, mode, local limit theorem, principal Dirichlet eigenvalue, Donsker and Varadhan functional, Bessel kernel

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1 Introduction

Which states does the most likely occupation-count vector of a slowly switching Markov chain use? Fix a generator Q on d states and a start law, and take N steps with transition matrix $I + (T/N)Q$, so that the chain makes about T switches. The occupation counts $n_i = \#\{t \le N : X_t = i\}$ have support $F = \{i : n_i > 0\}$, which we call a face. Write M_F for the largest probability on that exact support and λ_F for its principal exit rate (Definition 2.1). A face is admissible if a path of positive probability has exactly that support. When $T = \tau \log N$ the cost $e^{-T\lambda_F}$ of staying in F and the cost $N^{-(|F|-1)}$ of sharing the mass of F among its count vectors are both powers of N, and they compete. The paper describes the law of the count vector on every face at this scale, up to the boundary of the face, and reads off the face that carries the mode and the times at which it changes.

1.1 The law on a face

Put $\Phi_\tau(F) = \tau\lambda_F + |F| - 1$.

Theorem A (first-order face selection) *Fix the rates q_{ij} and the law of X_1, and let $T = \tau \log N$ with $\tau > 0$ fixed. Every admissible face F has $M_F = N^{-\Phi_\tau(F)+o(1)}$. For large N the support of every most likely composition is an admissible face that minimizes Φ_τ among the admissible faces. If the chain can pass between any two states of each such minimizing face without leaving it, then the visit fractions of every most likely composition are within $O(1/\log N)$ of the quasi-ergodic law of its support (Definition 2.1).*

Theorem A is Theorem 4.2(ii)–(iv). Theorem 4.4 reads the winners off the lower-left boundary of the convex hull of the points $(\lambda_F, |F| - 1)$, so as τ grows the winning size can only grow. The exponent is one functional. The rate J_F inside a face is the Donsker and Varadhan functional of the chain killed on leaving F, and Proposition 3.1 shows that J_F is the restriction to the face of the functional J_V of the whole chain, where $V = [d]$ is the face of all states. So Theorem 5.6 states the law of the count vector on every face in one display, as the lattice entropy $(|F| - 1)\log N$ of the face plus T times the Donsker and Varadhan cost of the point.

First order fixes a crossing only up to a factor $1 + o(1)$. To locate it up to $o(1)$ we need the constant in front. Say that a face F satisfies (H) if the chain can move between any two states of F without leaving F, and the start gives F positive mass. For such faces we prove a local limit theorem inside the face (Theorem 5.4), and it gives the following result.

Theorem B (face maxima and crossings) *Fix the rates q_{ij} and the law of X_1.*

(i) Let F satisfy (H)*, with $k = |F|$ states. Uniformly for $1 \le T \le N^{1/3}$,*

$$M_F = N^{-(k-1)}\, T^{(k-1)/2} K_F\, e^{-T\lambda_F} \big(1 + O(1/T + T^2/N)\big),$$

where $K_F > 0$ is the explicit constant of Definition 5.1 (for $k = 1$ it is the start mass of the state).

(ii) Let F and G satisfy (H)*, and suppose that F has $\Delta \geq 1$ more states than G and an exit rate smaller by $\delta > 0$. For large N the equation $M_F = M_G$ has a root T in $[1, N^{1/3}]$, and every such root satisfies*

$$\delta T + \frac{\Delta}{2} \log T = \Delta \log N - \log \frac{K_F}{K_G} + O\Big(\frac{1}{\log N}\Big).$$

Part (i) is Theorem 5.7 and part (ii) is Theorem 5.8. The factor $T^{(k-1)/2}$ appears because the visit fractions fluctuate on the scale $T^{-1/2}$ in each of the $k-1$ free directions. Solving the equation in (ii) gives $T = \frac{\Delta}{\delta}[\log N - \frac{1}{2}\log\log N + O(1)]$, so the coefficient $\frac{1}{2}$ of $\log\log N$ holds for every such pair, whatever the graph, the rates, the sizes or the start, and without reversibility. For reversible faces K_F involves the square root of a weighted count of spanning trees (Proposition 5.2), and on a general face two counts of trees (Proposition 5.11). When several faces tie at first order, Theorem 5.9 describes the window of T in which they compete, and a weak field shifts the line of each face in that window by the field averaged against its quasi-ergodic law (Proposition 5.15). The face maxima hold with a field v that weights a composition by $e^{Tv\cdot n/N}$, uniformly for v in a compact set, and over a compact class of reversible chains (Proposition 5.16).

The local law stays away from the boundary of the face. At the boundary a new scale appears. Let $x = Tn_i/N$ be the time spent in a state i whose count is small, and $S = T - x$. Section 6 pulls the visits to i out of the exact run expansion and writes the density on F as a contour integral against the density on the facet $F \setminus \{i\}$ with a complex weight. For a pair face the integral is a Bessel kernel, $e^{-\gamma x}\sqrt{abS/x}\, I_1(2\sqrt{abSx})$ times the density on the facet, which is the facet form of Pedler's formula for two-state chains, and we prove it with an explicit error for all x (Proposition 6.3). The constants a, b and γ are explicit for every irreducible chain through the left and right tilt vectors at the facet profile, and $\gamma - q_i$ is $ab/2$ times an effective resistance (Proposition 6.4). For every face, Theorem 6.9 describes the layer for x up to a small multiple of T. A generalized kernel holds there with relative error $O(1/T)$. The Bessel kernel holds up to a factor e^E with E of order $1/T + \sqrt{x/T}(1+x)$. For a reversible chain this improves to a relative error $O(1/T + \sqrt{x/T})$ on the whole range. Without reversibility we get the relative Bessel form only for x up to a constant times $T^{1/3}$, and a non-reversible cycle on three states shows that the restriction is real. The proof rests on a local law of the facet chain with a small complex weight (Theorem 6.6), which we prove on a torus through a complex critical point with a Laplace lemma for a complex Hessian. The near-vertex law of Section 8, where one count is large and a few counts are of order one and the crossing with the vertex moves to T of order $\sqrt{N}$, is the discrete pair-face case of this layer, in the regime where the Laguerre kernel cannot yet be replaced by the Bessel kernel (Theorem 8.2).

1.2 The complete graph

On the complete graph with the uniform start, $Q = J - dI$ with J the all-ones matrix, every face ties at first order and the constants decide. A face of k states has $K_F = \frac{1}{d}k^{k+1/2}(4\pi)^{-(k-1)/2}$, and the k^{k-2} spanning trees of K_k enter through it. Put

$a_k = \frac{1}{2}\log(4\pi) - \frac{k+1/2}{k-1}\log k$, write V_N for the probability of a vertex, and call a composition balanced if its positive coordinates differ by at most 1.

Theorem C (the complete graph) *Let $Q = J - dI$ and let the start be uniform.*

(i) Let F be a face of $2 \le k \le d$ states. For every $N \ge k$ the equation $M_F = V_N$ has exactly one root $T_{N,k}$ in $(0, N/(d-1))$, and the switch probability $T_{N,k}/N$ at the root decreases strictly as N grows. As $N \to \infty$,

$$T_{N,k} + \tfrac{1}{2}\log T_{N,k} = \log N + a_k + O\Big(\frac{1}{\log N}\Big).$$

(ii) For every N and every $0 < T \le N/d$, the most likely compositions with a given support are exactly the balanced ones.

(iii) Since $a_2 > a_3 > \cdots > a_d$, for large N and T up to any fixed multiple of $\log N$ the mode changes exactly once, from the d vertices straight to the balanced compositions with d positive coordinates.

Part (i) is Theorem 7.5, part (ii) is Theorem 7.6 and part (iii) is Corollary 7.7. Parts (i) and (ii) hold at every N, so on the complete graph we know the face maxima and their crossings with a vertex exactly, and part (iii) gives the mode for large N. Figure 1 shows the window. A start law changes the constant of a face through its start mass, and a weak field changes it through the field averaged over the face. Either can let a face of intermediate size win (Example 7.10 and Corollary 7.11), and one field makes every size win in turn.

1.3 Related work

The exact law of the count vector goes back to Whittle [38], Dawson and Good [15], Goodman [21] and Good [20]. Good's formula (3) is the run expansion of Lemma 2.3, and his formula (10) is the continuum density of Lemma 2.4. What our setting adds to (10) is the killing on leaving the face and the exponents $m_i - 1$ of the times spent in the states, where m_i is the number of runs in state i, and the proof of Lemma 2.4 checks them. He suggested the saddle-point analysis of the density and did not carry it out, and Theorem 5.4 is that analysis, uniform in T and with its constant. Polettini [35] and Carugno, Vivo and Coghi [9] derive the exact law from the BEST theorem and the matrix-tree theorem. Stadje [36] gives the generating functions of the joint numbers of visits and limit laws when one state becomes nearly absorbing. In continuous time, Brydges, van der Hofstad and König [8] give the joint density of the local times on the event that the range is a given set, and Huang, Kious, Sidoravicius and Tarrès [24] write it with Bessel functions and spanning trees for symmetric rates. The Bessel kernel of the boundary layer is Pedler's occupation-time density for two-state chains [31], continued by Kovchegov, Meredith and Nir [28], and Section 6 is its facet version. Hervé and Ledoux [23] prove a quantitative local limit theorem for densities of finite-state Markov additive processes, and Kontoyiannis and Meyn [27, Theorem 6.3] give exact large deviation asymptotics for a bounded functional of a fixed chain. Theorem 5.4 is a local form for the whole vector of counts, for a chain killed on leaving F whose switch probabilities shrink with N.

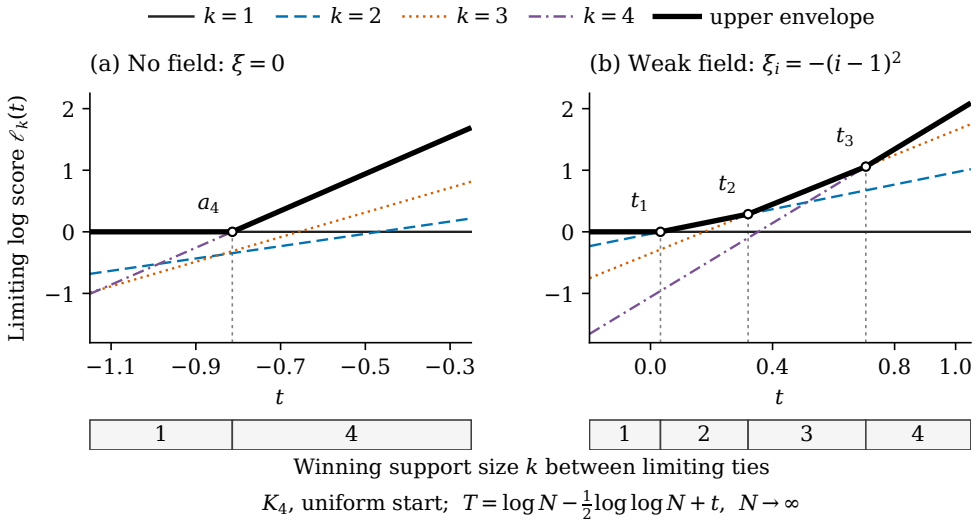


Fig. 1 Support selection on the complete graph with four states, unit rates and uniform start, in the window $T = \log N - \frac{1}{2}\log\log N + t$. Thin lines are limiting log face maxima relative to an untilted vertex, and the bold curve is their upper envelope. (a) Without a field the winner jumps from one state to all four. (b) Weighting a composition by $e^{\langle \xi, n\rangle/N}$, with $\xi_i = -(i-1)^2$, lets every size win in turn. The scores and marked crossings are derived in Section 7. Field weights are unnormalized, and marked ties concern the limit, not finite-N equalities

The rate inside a face is the Donsker and Varadhan functional of the killed chain [18]. Dupuis and Liu [19, Section 3.1] give it for reversible jump processes, Chen and Deng [11] use it for the principal Dirichlet eigenvalue, and Guillin, Nectoux and Wu [22] prove a large deviation principle for the occupation measure of a process conditioned to stay in a set. The quasi-ergodic law goes back to Darroch and Seneta [13, 14], with the book of Collet, Martínez and San Martín [12] as the standard reference. Dietz and Sethuraman [16, 17] and Bouguet and Cloez [7] study chains whose switching decays along the trajectory. Here the rate is fixed during each path and changes with N. Bertini, Gabrielli and Landim [5] and Landim [29] expand the level-two rate functions of finite chains in a parameter, and Kim and Landim [26] extend this to the joint rate of the empirical measure and the flow [3, 4]. These expansions concern the rate functional of a fixed chain. Here the functional enters through its restriction to each face, and the competition is between faces of different dimension.

The question grew out of Problem 131 of Kagey's Open Problem Collection [25], a two-direction crossing problem, whose two-state case is treated in [32]. This paper supersedes our preprint [33] on the complete graph. Its results on the crossings with a vertex and near a vertex appear here as cases of the general theory, in Sections 7 and 8. We cite the preprint for the results that we do not reprove.

1.4 Plan of the paper

Section 2 fixes the notation and gives the exact law of the composition as a sum over runs. Section 3 introduces the killed functional and the restriction identity. Section 4 proves first-order face selection, Section 5 the local law with a field, the face maxima, the crossing equation and the tie windows, and Section 6 the boundary layer. Section 7 treats the two-state walk and the complete graph, and Section 8 the compositions near a vertex. Section 9 describes the computer checks and the formal proofs. The appendices hold the torus and Laplace proofs of the local law, the proofs for the complete graph and the proofs near a vertex. From the persistent-walk paper [32] we use two results, the coefficientwise growth of the polynomials $S_{a,b}$ toward the middle in Proposition 7.1 and its bound on clipped products in Appendix D. We also recall its expansion of the crossing z_N and its fixed-edge expansion, for comparison only.

A companion paper [34] asks what the Donsker and Varadhan functional and the modal supports determine about the generator. It recovers a reversible generator on a known graph from modal supports, with a finite-horizon correction, and at low temperature it reads the Freidlin and Wentzell cycle tree off the modal supports. That paper rests on the killed functional and the first-order results of Sections 3 and 4, on Definition 5.1 with the face maxima of Theorem 5.7, and on Proposition 5.16. Nothing here depends on it.

2 Setting, notation and the exact law

2.1 Setting and notation

The chain. Fix $d \geq 2$ and a generator Q on $[d] = \{1, \dots, d\}$. Its rates are $q_{ij} \geq 0$ for $i \neq j$, the exit rates are $q_i = \sum_{j \neq i} q_{ij}$, $Q_{ii} = -q_i$, and $q_{\max} = \max_i q_i$. We also write q for the vector (q_i). (The scalar $q = 1 - p$ of the two-state walk appears only where we recall [32], and the scalar q of Proposition 7.1 is a common exit rate.) The start μ is a probability law on $[d]$, and $\mu(F) = \sum_{i \in F} \mu_i$. We write $V = [d]$ for the face of all states. For $N \geq 2$ and $T > 0$ put

$$h = \frac{T}{N}, \qquad L = \log N, \qquad \tau = \frac{T}{L}.$$

The chain $X_1, \dots, X_N$ has $X_1 \sim \mu$ and transition matrix $I + hQ$. So at each step it jumps from i to j with probability hq_{ij}, and N steps look like the continuous-time chain with generator Q run for time T. We assume $hq_{\max} \leq \frac{1}{2}$ unless we say otherwise. The *composition* is the vector of visit counts $n = (n_1, \dots, n_d)$, with $n_i = \#\{t \leq N : X_t = i\}$, so that $n_1 + \cdots + n_d = N$. We write $P_N(n)$ for the probability that the chain has composition n, $\operatorname{supp} n = \{i : n_i > 0\}$ for its support, and $\alpha = n/N$ for the visit fractions.

Faces. A face is a nonempty set $F \subseteq [d]$, of size $k = k_F = |F|$. Let Δ_F be the set of probability vectors on $[d]$ that vanish off F, and Δ_F° the set of those that are positive on F. We number the states of F as $1, \dots, k$ and use the coordinates $(\alpha_2, \dots, \alpha_k)$ on Δ_F and $(t_2, \dots, t_k)$ on $\{t \in \mathbb{R}^F : t_1 + \cdots + t_k = T\}$. We write $x \cdot y = \langle x, y \rangle = \sum_{i \in F} x_i y_i$ and $x \circ y = (x_i y_i)_{i \in F}$. Let $A_F = (q_{ij})_{i,j \in F}$ with zero diagonal, and $Q_F = A_F - \operatorname{diag}(q_i)_{i \in F}$.

Since q_i includes the rates of leaving F, Q_F generates the chain killed when it leaves F. The switching graph G_F has an edge $i \to j$ whenever $i, j \in F$, $i \neq j$ and $q_{ij} > 0$.

Perron roots. A square matrix B is Metzler if its off-diagonal entries are nonnegative, and then its Perron root $\varrho(B)$ is its largest real eigenvalue. If B is irreducible, $\varrho(B)$ is simple and has positive left and right eigenvectors [2].

Definition 2.1 (faces) Let F be a face. We call F *irreducible* if G_F is strongly connected. In particular every singleton is irreducible. We call F *admissible* if for some N some path $x_1 \cdots x_N$ of positive probability has support exactly F (for $0 < hq_{\max} < 1$ this does not depend on h). We say that F satisfies (H) if F is irreducible and $\mu(F) > 0$. The *exit rate* of F is $\lambda_F = -\varrho(Q_F)$, so $\lambda_{\{i\}} = q_i$. If F is irreducible, r_F and l_F are right and left eigenvectors of Q_F for $-\lambda_F$ with positive entries and $\langle l_F, r_F \rangle = 1$, and $\alpha_F^* = l_F \circ r_F$ is the *quasi-ergodic law* of F [13, 14]. The *face maximum* is $M_F = \max\{P_N(n) : \operatorname{supp} n = F\}$, with $\max \emptyset = 0$.

We call λ_F the exit rate for the following reason. If F is irreducible, then $e^{TQ_F} r_F = e^{-T\lambda_F} r_F$ and r_F is positive. So the chance that the continuous-time chain started at a state of F stays in F up to time T, which is a row sum of e^{TQ_F}, lies between $e^{-T\lambda_F} \min r_F / \max r_F$ and $e^{-T\lambda_F} \max r_F / \min r_F$. If F is reducible, we order the strongly connected components $C_1, \dots, C_r$ of G_F so that every edge between components goes forward. Then Q_F is block triangular with diagonal blocks Q_{C_j}, and $\lambda_F = \min_j \lambda_{C_j}$. In all cases $0 \le \lambda_F \le \min_{i \in F} q_i$, since the rows of Q_F have sums at most 0 and a Perron root is at least every diagonal entry. For a face F satisfying (H), the quasi-ergodic law α_F^* is the long-run law of the visit fractions of the chain conditioned to stay in F, and the best composition on F sits near $N\alpha_F^*$ (Theorem 5.7).

The two-state walk. The persistent random walk on $\mathbb{Z}$ makes its first step left or right with probability $\frac{1}{2}$, and each later step repeats the previous one with probability p and switches with probability $q = 1-p$. Its directions form the chain of Example 2.5(i). As on a Galton board, [32] calls a composition a *bin*. For two states we identify a composition with its number of right steps. For $a, b \ge 1$ let $S_{a,b}(u) = \sum_w u^{s(w)}$, the sum over the words w with a right steps and b left steps, where $s(w)$ is the number of switches of w. Each such word has probability $\frac{1}{2} p^{N-1} (q/p)^{s(w)}$ for the walk with $N = a + b$ steps. So $S_{a,b}(q/p)$ is the probability of composition a divided by the probability $\frac{1}{2} p^{N-1}$ of an endpoint. For $N \ge 2$ let $S_N = S_{\lfloor N/2 \rfloor, \lceil N/2 \rceil}$, which belongs to a middle composition. It is a nonzero polynomial with nonnegative coefficients and no constant term, so $S_N(u) = 1$ has exactly one positive root u_N, and $p_N = 1/(1+u_N)$ is the crossing of the middle and the endpoints. Put $z_N = N(1-p_N)$ and $c_0 = \frac{1}{2} \log(\pi/8)$. By [32], z_N satisfies the crossing equation $z_N + \frac{1}{2} \log z_N = L + c_0 + 1/(8z_N) + O(1/L^2)$. We recall this expansion only for comparison with Corollary 7.2.

Conventions. Unless we say otherwise, Q, μ and the faces under discussion are fixed, and every constant, every implied constant in $O(\cdot)$ and every threshold N_0 depends only on them and on the compact set $\mathcal{K}$ when one appears. A bound that holds "uniformly for $1 \le T \le N^{1/3}$" holds for all $N \ge N_0$ and all T in that range with one implied constant. Subscripts as in $O_{\mathcal{K}}$ stress a dependence.

2.2 Admissible faces

Lemma 2.2 (admissible faces) *Let F be a face and $C_1, \dots, C_r$ the strongly connected components of G_F.* (a) *F is admissible if and only if the components can be numbered so that $\mu(C_1) > 0$ and, for each $j < r$, $q_{xy} > 0$ for some $x \in C_j$ and $y \in C_{j+1}$.* (b) *An irreducible face is admissible if and only if $\mu(F) > 0$. So* (H) *holds exactly for the irreducible admissible faces.* (c) *If Q has symmetric support, that is $q_{ij} > 0$ exactly when $q_{ji} > 0$, then every admissible face is irreducible.*

Proof We translate paths into walks in G_F. Take a path of positive probability with support F and list the states of its maximal runs. This list is a walk in G_F that starts at a state of $\operatorname{supp}\mu$ and visits every state of F. Conversely, every such walk $v_1, \dots, v_J$ is the list of runs of a path of positive probability with support F, as soon as $N \geq J$. (The path follows the walk and then stays in its last state, and a stay has positive probability.) So F is admissible exactly when G_F has a walk that starts in $\operatorname{supp}\mu$ and visits every state of F.

(a) Suppose that there is such a walk. Once it has left a component C for a state $y \notin C$, it cannot come back to C. (If it did, y could be reached from C and C from y, so y would belong to C.) Hence the walk visits each component in one block of consecutive times, and it passes from one block to the next along a single edge. We number the components in the order of these blocks. The walk starts in C_1, so $\mu(C_1) > 0$, and the edge from block j to block $j+1$ is an edge $x \to y$ with $x \in C_j$ and $y \in C_{j+1}$. This is the condition in (a).

Conversely, suppose that the components are numbered as in (a), with edges $x_j \to y_{j+1}$ from C_j to C_{j+1} for $j < r$. Take $y_1 \in C_1$ with $\mu_{y_1} > 0$ and any $x_r \in C_r$. Since C_j is strongly connected, there is a walk inside C_j from y_j to x_j that visits every state of C_j. We join these r walks by the edges $x_j \to y_{j+1}$. The result is a walk that starts in $\operatorname{supp}\mu$ and visits every state of F.

(b) This is (a) with $r = 1$.

(c) If Q has symmetric support, an edge from one component to another would come with an edge back, and the two components would be the same. So no edge joins two components, and the condition in (a) forces $r = 1$. □

So $M_F > 0$ for large N if F is admissible, and $M_F = 0$ otherwise.

2.3 The exact law

For $m \in \mathbb{Z}^F_{\geq 1}$ put $M = \sum_{i \in F} m_i$ and $W(m) = \sum_c \mu_{c_1} \prod_{j<M} q_{c_j c_{j+1}}$, the sum over the words $c = c_1 \cdots c_M$ in F with $c_j \neq c_{j+1}$ in which each $i \in F$ occurs m_i times. Such a word lists the states of the runs of a path, and m_i is the number of runs in state i.

Lemma 2.3 (run expansion) *Let $hq_{\max} \leq 1$ and let n have support F. Then, with $\binom{n_i-1}{m_i-1} = 0$ for $m_i > n_i$ and $0^0 = 1$,*

$$P_N(n) = \sum_{m \in \mathbb{Z}^F_{\geq 1}} W(m)\, h^{M-1} \prod_{i \in F} \binom{n_i - 1}{m_i - 1} (1 - hq_i)^{n_i - m_i}.$$

This is Good's coefficient identity [20, (3)], with the exponents $n_i - 1$, for a chain of N letters with any initial law, here for the chain killed on leaving F. Grouping by runs is our way of writing it, and the proof is short.

Proof The states $c_1 \cdots c_M$ of the maximal runs of a path with composition n form a word as in $W(m)$. The path starts at c_1, of probability μ_{c_1}, it has $M-1$ switches, of probabilities $hq_{c_j c_{j+1}}$, and $n_i - m_i$ stays in state i, of probability $1 - hq_i$ each. The number of paths with the word c and composition n is $\prod_i \binom{n_i-1}{m_i-1}$, the number of ways to write each n_i as an ordered sum of m_i positive run lengths. Summing over the words c with given m gives $W(m)$, and summing over m gives the claim. □

Lemma 2.3 is checked in Lean (Section 9). For $F = \{i\}$ the only path is i^N, so uniformly for $1 \le T \le N^{1/2}$

$$M_{\{i\}} = \mu_i (1 - hq_i)^{N-1} = \mu_i e^{-q_i T} \big(1 + O(T^2/N)\big). \tag{1}$$

For $t \in (0,\infty)^F$ let $H(t) = \sum_m W(m) \prod_{i \in F} t_i^{m_i - 1}/(m_i - 1)!$, which is finite since $W(m) \le k^M q_{\max}^{M-1}$. In the continuum limit the binomial coefficients of Lemma 2.3 become the powers $t_i^{m_i-1}/(m_i-1)!$, and Lemma 5.5 below shows that the discrete law is close to the following density.

Lemma 2.4 (the continuum density) *Let $(Y_s)_{s \ge 0}$ be the continuous-time chain with generator Q and $Y_0 \sim \mu$, let $\mathcal{O}_i = \int_0^T \mathbf{1}\{Y_s = i\}\, ds$, and let E_F be the event that Y stays in F during $[0,T]$ and visits every state of F. Then $\mathbb{P}(E_F,\ \mathcal{O} \in \cdot\,)$ has the density $e^{-\langle q,t\rangle} H(t)$ in $(t_2, \dots, t_k)$ on $\{t \in (0,\infty)^F : \sum_i t_i = T\}$, and $\mathbb{P}(E_F,\ \mathcal{O}/T \in \cdot\,)$ has the density*

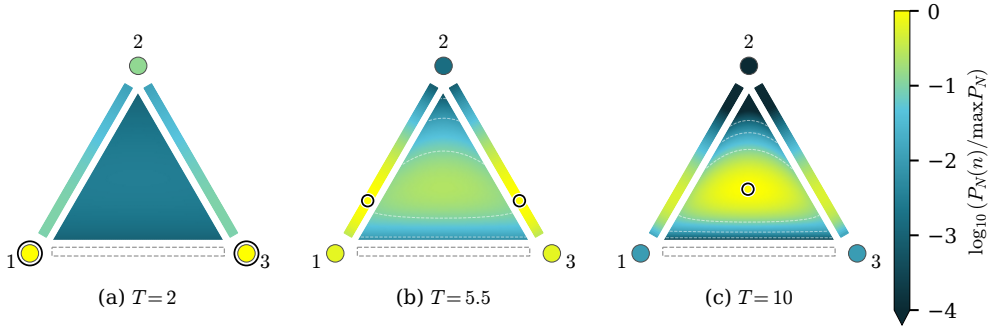
$$f_T(\alpha) = T^{k-1} e^{-T\langle q,\alpha\rangle} H(T\alpha)$$

in $(\alpha_2, \dots, \alpha_k)$. For n with support F and $t = hn$, $N^{-(k-1)} f_T(n/N) = h^{k-1} e^{-\langle q,t\rangle} H(t)$. For $F = \{i\}$, $f_T = \mathbb{P}(E_F) = \mu_i e^{-q_i T}$.

The density is Good's formula [20, (10)]. What we add is the killing, that is, the subprobability with the full exit rates q_i on the diagonal, and the exponents $m_i - 1$, which the proof checks.

Proof Up to a null set, E_F is the disjoint union of the events E_c that Y visits exactly $c_1, \dots, c_M$ in this order during $[0,T]$, over the words c in the sums $W(m)$. Fix c, and let s_j be the time that Y spends in its j-th run, so that $s_1 + \cdots + s_M = T$. For $j < M$ the run has the density $q_{c_j} e^{-q_{c_j} s_j}$ and ends with a jump to c_{j+1} of probability $q_{c_j c_{j+1}}/q_{c_j}$, and the last run survives with probability $e^{-q_{c_M} s_M}$. So on $\{s_j > 0\ \forall j\}$ the event E_c has the density $\mu_{c_1} \prod_{j<M} q_{c_j c_{j+1}}\, e^{-\langle q,\mathcal{O}\rangle}$ in $(s_1, \dots, s_{M-1})$, where $\mathcal{O}_i$ is the sum of the s_j with $c_j = i$. For each $i \ne c_M$ we replace the last holding time in state i by $\mathcal{O}_i$, a triangular change of variables with unit diagonal. With $\mathcal{O} = t$ fixed, the $m_i - 1$ other holding times in state i are positive with sum less than t_i, so they range over a simplex of volume $t_i^{m_i-1}/(m_i-1)!$. Integrating them out gives the density $\mu_{c_1} \prod_{j<M} q_{c_j c_{j+1}}\, e^{-\langle q,t\rangle} \prod_i t_i^{m_i-1}/(m_i-1)!$ in the coordinates t_i with $i \ne c_M$, and any $k-1$ of the coordinates give the same Lebesgue measure on $\{\sum_i t_i = T\}$. Summing over c and m gives $e^{-\langle q,t\rangle} H(t)$, and $t = T\alpha$ has Jacobian T^{k-1}. For $F = \{i\}$ the event E_F says that Y starts at i and does not jump before time T. □

Example 2.5 (two models) (i) For $d = 2$, $q_{12} = q_{21} = 1$ and uniform μ the chain is the persistent walk, with $p = 1 - h$. So $T = N(1-p)$, the faces $\{1\}$ and $\{2\}$ are the endpoints,



(a) $T = 2$ (b) $T = 5.5$ (c) $T = 10$

Fig. 2 The exact law of the composition for the path $1 - 2 - 3$ (rate 1 on each edge, uniform start, $N = 90$) on the simplex of visit fractions, with the faces drawn apart. The triangle holds the compositions with three states, each band those with two, and each disk a vertex. The color is the probability relative to the mode, the thin lines are spaced by a factor of ten, and the rings mark the modes. The mode is (a) at an end of the path, (b) on a pair of neighbors and (c) inside. The dashed band $\{1, 3\}$ has probability zero

and Lemma 2.3 is the run count for two states. (ii) For $Q = J - dI$, with J the all-ones matrix, the chain switches to each other state with probability h and keeps its state with probability $p = 1 - (d-1)h$. With uniform μ it is the model of [33], whose switch probability r is our h. A word $x_1 \cdots x_N$ then has probability $\frac{1}{d} h^{N-1} (p/h)^{\#\{s<N:\, x_s = x_{s+1}\}}$, so the path law is the one-dimensional d-state Potts chain with free boundaries. Here $q_i = d - 1$ for every i, so Lemma 2.3 reads

$$P_N(n) = p^{N-1} \sum_{a \in F} \mu_a S_n^{(a)}(u), \qquad u = \frac{h}{p}, \tag{2}$$

for $p > 0$ and $\operatorname{supp} n = F$, where $S_n^{(a)}(u)$ is the sum of u^j over the words with the counts n that start with a and have j switches. Indeed such a word has probability $\mu_a p^{N-1-j} h^j = \mu_a p^{N-1} u^j$. In both models Q has symmetric support and μ full support, so all admissible faces satisfy (H) by Lemma 2.2.

Figure 2 shows exact support competition for the three-state path. There λ_F is 1 for an end, $(3 - \sqrt{5})/2$ for a pair of neighbors and 0 for all three states, so to first order the mode moves at $\tau = (1 + \sqrt{5})/2$ and at $\tau = (3 + \sqrt{5})/2$, that is, at $T = 7.3$ and $T = 11.8$ for $N = 90$. At $N = 90$ it moves near $T = 4.9$ and $T = 8.5$. Corrections of this size are the subject of Section 5.

3 The killed functional

Every rate in this paper is the restriction of one functional, the Donsker and Varadhan occupation functional of the chain. This section defines it, proves the restriction identity and the variational formula with a field, and records the monotonicity of exit rates in the face.

3.1 The restriction identity

For a probability vector α on $[d]$ let

$$J_V(\alpha) = \sup_{u\in(0,\infty)^d} \sum_{i\in[d]} \alpha_i\Big(-\frac{(Qu)_i}{u_i}\Big) = \sup_{u\in(0,\infty)^d} \sum_{i\in[d]} \alpha_i\Big(q_i - \sum_{j\neq i} q_{ij}\frac{u_j}{u_i}\Big), \tag{3}$$

the Donsker and Varadhan rate of the empirical measure of the continuous-time chain with generator Q [18]. For a face F and $\alpha \in \Delta_F$ let

$$J_F(\alpha) = \sup_{u\in(0,\infty)^F} \sum_{i\in F} \alpha_i\Big(-\frac{(Q_F u)_i}{u_i}\Big) = \sup_{u\in(0,\infty)^F} \sum_{i\in F} \alpha_i\Big(q_i - \sum_{j\in F\setminus\{i\}} q_{ij}\frac{u_j}{u_i}\Big),$$

the same functional for the chain killed on leaving F. Let $q_i^{\mathrm{out}}(F) = \sum_{j\notin F} q_{ij}$. Taking $u = \mathbf{1}$ gives $J_F(\alpha) \geq \sum_i \alpha_i q_i^{\mathrm{out}}(F)$, and dropping the terms with q_{ij} gives $J_F(\alpha) \leq \sum_i \alpha_i q_i$.

Proposition 3.1 (the restriction identity) *For every face F and every $\alpha \in \Delta_F$, $J_F(\alpha) = J_V(\alpha)$. Hence $J_F = J_{F'}$ on $\Delta_{F'}$ for every nonempty $F' \subseteq F$.*

Proof Fix $\alpha \in \Delta_F$ and $u \in (0,\infty)^d$. The terms of (3) with $i \notin F$ vanish, since $\alpha_i = 0$ there. For $i \in F$ the terms with $j \notin F$ are nonnegative and are subtracted, so

$$\sum_{i\in F} \alpha_i\Big(q_i - \sum_{j\neq i} q_{ij}\frac{u_j}{u_i}\Big) \leq \sum_{i\in F} \alpha_i\Big(q_i - \sum_{j\in F\setminus\{i\}} q_{ij}\frac{u_j}{u_i}\Big) \leq J_F(\alpha).$$

So $J_V(\alpha) \leq J_F(\alpha)$. Conversely, let $w \in (0,\infty)^F$ and let u equal w on F and ε off F. The terms with $j \notin F$ are then $-q_{ij}\varepsilon/w_i$, which tend to zero as $\varepsilon \downarrow 0$, and the supremum over w gives $J_V(\alpha) \geq J_F(\alpha)$. The second claim is the first for F and for F'. □

So the cost of a point of Δ_F does not depend on the face in which we read it, and the faces compete through the lattice entropy $(|F|-1)L$ and the cost TJ_V of their best points (Theorem 5.6).

3.2 The variational formula

For a face F and a field $v \in \mathbb{R}^F$ let

$$\lambda_F(v) = -\varrho\big(Q_F + \operatorname{diag} v\big), \tag{4}$$

so that $\lambda_F(0) = \lambda_F$. If F is irreducible, let r and l be positive right and left Perron vectors of $Q_F + \operatorname{diag} v$ with $l \cdot r = 1$, and put $\alpha_F^*(v) = l \circ r$, so that $\alpha_F^*(0) = \alpha_F^*$. Since $\sum_i \alpha_i v_i = v \cdot \alpha$,

$$J_F(\alpha) - v\cdot\alpha = \sup_{u\in(0,\infty)^F} \sum_{i\in F} \alpha_i\Big(-\frac{((Q_F + \operatorname{diag} v)u)_i}{u_i}\Big). \tag{5}$$

So a field replaces the diagonal q_i by $q_i - v_i$ and changes nothing else.

Lemma 3.2 (monotonicity in the face) *If $F \subsetneq G$ are faces and Q_G is irreducible, then $\lambda_G(v_G) < \lambda_F(v_F)$ for every field $v \in \mathbb{R}^G$. In particular $\lambda_V(v) < \lambda_F(v_F)$ for every proper face F when Q is irreducible, and for every fixed $v \in \mathbb{R}^d$ there is $\gamma(v) > 0$ with $\lambda_F(v_F) \geq \lambda_V(v) + \gamma(v)$ for all proper F. So the score $\Phi_{\tau,v}(F) = |F| - 1 + \tau\lambda_F(v_F)$ is least at V for $\tau > (d-1)/\gamma(v)$.*

Proof Choose c with $B = Q_G + \operatorname{diag} v_G + cI \geq 0$. Its principal submatrix on F is $Q_F + \operatorname{diag} v_F + cI$, since the diagonal of a killed generator keeps the full exit rates. Let $\widetilde{B}$ be this submatrix padded with zeros to $G \times G$. Then $0 \leq \widetilde{B} \leq B$, and $\widetilde{B} \neq B$ because the irreducible matrix B has a positive entry in a row of F and a column off F. So $\varrho(\widetilde{B}) < \varrho(B)$ [2, Ch. 2], and $\varrho(\widetilde{B})$ is the Perron root of the submatrix on F, since the padding adds only the eigenvalue 0. This holds whether or not F is irreducible, and subtracting c gives the first claim. For fixed v there are finitely many proper faces, so $\gamma(v)$, the least of the numbers $\lambda_F(v_F) - \lambda_V(v)$, is positive. Finally $\Phi_{\tau,v}(F) - \Phi_{\tau,v}(V) = |F| - d + \tau(\lambda_F(v_F) - \lambda_V(v)) \geq -(d-1) + \tau\gamma(v)$. □

Proposition 3.3 (the rate on a face) *Let F be irreducible with $k = |F| \geq 2$, and let $v \in \mathbb{R}^F$.*

(a) On Δ_F, $J_F(\alpha) - v \cdot \alpha \geq \lambda_F(v)$, with equality exactly at $\alpha = \alpha_F^*(v)$. In particular $J_F \geq \lambda_F$, with equality exactly at $\alpha_F^* = l_F \circ r_F$.

(b) The rate J_F is convex on Δ_F and strictly convex on Δ_F°. For $\alpha \in \Delta_F^\circ$ the supremum defining $J_F(\alpha)$ is attained at a vector $u = r_\alpha$, which is unique up to a positive factor, and α is the stationary law of the generator on F with off-diagonal entries $q_{ij} r_{\alpha,j}/r_{\alpha,i}$.

(c) If there is $\nu \in (0,\infty)^F$ with $\nu_i q_{ij} = \nu_j q_{ji}$ for all $i, j \in F$, then for every $\alpha \in \Delta_F$

$$J_F(\alpha) = \sum_{\{i,j\} \subseteq F,\ i \neq j} \left(\sqrt{\alpha_i q_{ij}} - \sqrt{\alpha_j q_{ji}}\right)^2 + \sum_{i \in F} \alpha_i q_i^{\mathrm{out}}(F).$$

(d) For every nonempty face $F' \subsetneq F$ and every $\alpha \in \Delta_{F'}$, $J_F(\alpha) - v \cdot \alpha = J_{F'}(\alpha) - v \cdot \alpha \geq \lambda_{F'}(v_{F'}) > \lambda_F(v)$.

So the most likely way to stay on F is to spend the fractions α_F^* of the time in its states. Fractions that miss some states of F have the rate of the smaller face that they use, which is larger. For symmetric Q_F, $l_F = r_F$ and α_F^* is the entrywise square of the Perron vector normalized in ℓ^2. On a face that is not reversible the formula in (c) is only an upper bound, by its proof. The lower bound in (a) and the value at α_F^* for $v = 0$, and the interior form of (c) for symmetric Q_F, are checked in Lean.

Proof Put $B = Q_F + \operatorname{diag} v$, an irreducible Metzler matrix with the switching graph G_F and $\lambda_F(v) = -\varrho(B)$.

Setup. With $u = e^\psi$, (5) reads $J_F(\alpha) - v \cdot \alpha = \sup_\psi \Psi_\alpha(\psi)$ with

$$\Psi_\alpha(\psi) = \sum_i \alpha_i (q_i - v_i) - \sum_{i \neq j} \alpha_i q_{ij} e^{\psi_j - \psi_i},$$

where i, j run over F. It is concave in ψ, affine in α, and unchanged when a constant is added to ψ. Let G^ψ be the conservative generator on F with off-diagonal entries $q_{ij} e^{\psi_j - \psi_i}$.

It has the graph G_F, so it has exactly one stationary law, and it does not involve v. Since $\nabla_\psi \Psi_\alpha = -\alpha^\top G^\psi$ and a differentiable concave function attains its maximum exactly at its critical points, ψ attains the supremum if and only if α is the stationary law of G^ψ. We call this the criterion.

A lower bound on every face. Let F' be any face, irreducible or not, and $B' = Q_{F'} + \operatorname{diag} v_{F'}$. We show $J_{F'}(\alpha) - v \cdot \alpha \geq -\varrho(B')$ on $\Delta_{F'}$. Let $s > \varrho(B')$ and write $B' = B_0 - cI$ with $B_0 \geq 0$. Then $s + c > \varrho(B_0)$, so $u = (sI - B')^{-1}\mathbf{1} = \sum_{\ell \geq 0} (s+c)^{-\ell-1} B_0^\ell \mathbf{1}$ converges and is positive. Since $(sI - B')u = \mathbf{1}$, we get $-(B'u)_i/u_i = -s + 1/u_i > -s$ at every state, and letting $s \downarrow \varrho(B')$ gives the claim.

The value at $\alpha_F^*(v)$. Take $\psi = \log r$ and $R = \operatorname{diag} r$. Then $G^\psi = R^{-1}(B - \varrho(B)I)R$, since the off-diagonal entries agree and $Br = \varrho(B)r$. So $(l \circ r)^\top G^\psi = l^\top (B - \varrho(B)I)R = 0$, and by the criterion ψ attains the supremum at $\alpha_F^*(v)$. Every ratio $-(Br)_i/r_i$ equals $\lambda_F(v)$, so the value is $\lambda_F(v)$.

(b) Convexity holds because J_F is a supremum of affine functions. Let $\alpha \in \Delta_F^\circ$ and take $v = 0$.

The supremum is attained. Normalize $\min_i \psi_i = 0$ and put $\bar\psi = \max_i \psi_i$. Let a be the least positive rate q_{ij} in F. A shortest path in G_F from a state where $\psi = 0$ to a state where $\psi = \bar\psi$ has at most $k-1$ edges, so one of its edges $i \to j$ has $\psi_j - \psi_i \geq \bar\psi/(k-1)$. Keeping only this edge gives $\Psi_\alpha(\psi) \leq \sum_i \alpha_i q_i - a(\min_i \alpha_i) e^{\bar\psi/(k-1)}$, which tends to $-\infty$ as $\bar\psi \to \infty$. So the continuous function Ψ_α attains its supremum on a bounded set. Write $r_\alpha = e^\psi$ for a maximizer. The last claim of (b) is the criterion.

The maximizer is unique up to a factor. If ψ and $\psi + \delta$ both attain the supremum, Ψ_α is constant on the segment between them. There it is a constant minus a sum of the convex functions $\alpha_i q_{ij} e^{\psi_j - \psi_i + s(\delta_j - \delta_i)}$, so each of them is affine in s, and $\delta_j = \delta_i$ on every edge of G_F. As G_F is strongly connected, δ is constant.

Strict convexity. Let $\alpha^0 \neq \alpha^1$ lie in Δ_F°, put $\alpha^s = (1-s)\alpha^0 + s\alpha^1$ with $0 < s < 1$, and let ψ attain $J_F(\alpha^s)$. Since $\Psi_\alpha(\psi)$ is affine in α, $J_F(\alpha^s) = (1-s)\Psi_{\alpha^0}(\psi) + s\Psi_{\alpha^1}(\psi) \leq (1-s)J_F(\alpha^0) + sJ_F(\alpha^1)$. Equality would make α^0 and α^1 both stationary for G^ψ by the criterion, which is impossible.

(d) The equality is Proposition 3.1. The lower bound on F' gives $J_{F'}(\alpha) - v \cdot \alpha \geq \lambda_{F'}(v_{F'})$, and Lemma 3.2 with $G = F$ gives $\lambda_{F'}(v_{F'}) > \lambda_F(v)$.

(a) The lower bound and the value at $\alpha_F^*(v)$ are proved above. Let $\alpha \neq \alpha_F^*(v)$. If α lies on the boundary of Δ_F, then (d) with $F' = \operatorname{supp} \alpha$ gives $J_F(\alpha) - v \cdot \alpha > \lambda_F(v)$. If $\alpha \in \Delta_F^\circ$ and equality held, strict convexity would give a value below $\lambda_F(v)$ at $\frac{1}{2}(\alpha + \alpha_F^*(v))$.

(c) Since $q_i = \sum_{j \in F \setminus \{i\}} q_{ij} + q_i^{\mathrm{out}}(F)$, the function inside the supremum is $\sum_i \alpha_i q_i^{\mathrm{out}}(F)$ plus, for each pair $\{i, j\}$, the term $\alpha_i q_{ij}(1 - u_j/u_i) + \alpha_j q_{ji}(1 - u_i/u_j)$. By the AM–GM inequality this term is at most $(\sqrt{\alpha_i q_{ij}} - \sqrt{\alpha_j q_{ji}})^2$, which gives the upper bound. For $\alpha \in \Delta_F^\circ$ the vector $u_i = \sqrt{\alpha_i/\nu_i}$ gives $\alpha_i q_{ij} u_j/u_i = \alpha_j q_{ji} u_i/u_j$ by $\nu_i q_{ij} = \nu_j q_{ji}$, so there is equality for every pair at once. If some α_i vanish, take $u_i = \varepsilon$ for these i and let $\varepsilon \to 0$. A pair with $\alpha_i = 0 < \alpha_j$ then has the term $\alpha_j q_{ji}(1 - \varepsilon/u_j) \to \alpha_j q_{ji}$, which is its bound, and a pair with $\alpha_i = \alpha_j = 0$ has the term 0. □

Proposition 3.4 (the variational formula) *Let F be irreducible. For every $v \in \mathbb{R}^F$,*

$$\lambda_F(v) = \min_{\alpha \in \Delta_F} \big(J_F(\alpha) - v \cdot \alpha\big),$$

and the minimum is attained only at $\alpha_F^*(v)$. Equivalently, $-\lambda_F(v)$ is the Legendre transform of J_F at v, and J_F is the Legendre transform of $v \mapsto -\lambda_F(v)$ on Δ_F°. The map $v \mapsto \lambda_F(v)$

is concave and real-analytic, and $\lambda_F(v + c\mathbf{1}) = \lambda_F(v) - c$. *If* Q_F *is reversible with respect to* ν, *let* $S_F = D_{\sqrt{\nu}}(-Q_F)D_{\sqrt{\nu}}^{-1}$. *Then with* $\psi_i = \sqrt{\alpha_i}$,

$$J_F(\alpha) - v \cdot \alpha = \sum_{i \in F} (q_i - v_i)\alpha_i - \sum_{i \neq j \in F} \sqrt{q_{ij} q_{ji}}\, \psi_i \psi_j$$

is the Rayleigh quotient of the symmetric matrix $S_F - \operatorname{diag} v$ *at the unit vector* ψ, *and* $\lambda_F(v) = \lambda_{\min}(S_F - \operatorname{diag} v)$.

The formula is the finite form of the Donsker and Varadhan formula for the principal eigenvalue [18], close to the characterization of Collatz and Wielandt [2, Ch. 2].

Proof The formula and the unique minimizer are Proposition 3.3(a). As a minimum of affine functions of v, $\lambda_F(v)$ is concave. It is real-analytic because the Perron root of an irreducible Metzler matrix is a simple eigenvalue, and the shift is clear.

Legendre duality. The formula says $-\lambda_F(v) = \max_\alpha(v \cdot \alpha - J_F(\alpha))$. Conversely $J_F(\alpha) \geq v \cdot \alpha + \lambda_F(v)$ for all α and v. Let $\alpha \in \Delta_F^\circ$, take r_α from Proposition 3.3(b), and put $v = q - A_F r_\alpha / r_\alpha$ (entrywise). Then $Q_F + \operatorname{diag} v$ has the off-diagonal entries of Q_F and the diagonal $-A_F r_\alpha / r_\alpha$, so $r_\alpha > 0$ is a null vector of it. Only the Perron root has a positive eigenvector, so $\lambda_F(v) = 0$. Also $J_F(\alpha) = \langle q - A_F r_\alpha / r_\alpha, \alpha \rangle = v \cdot \alpha$, so equality holds at this v.

The reversible form. Reversibility gives $\sqrt{\nu_i}\, q_{ij} / \sqrt{\nu_j} = \sqrt{q_{ij} q_{ji}}$, so S_F is symmetric with diagonal q_i and off-diagonal entries $-\sqrt{q_{ij} q_{ji}}$. Expanding the squares in Proposition 3.3(c) gives $J_F(\alpha) = \langle \psi, S_F \psi \rangle$, and subtracting $v \cdot \alpha = \langle \psi, \operatorname{diag}(v) \psi \rangle$ gives the display. Every unit vector $\psi \geq 0$ arises from $\alpha = \psi \circ \psi$, and the off-diagonal entries of $S_F - \operatorname{diag} v$ are nonpositive, so passing to the entrywise modulus does not increase the quadratic form. Hence the minimum over Δ_F is $\lambda_{\min}(S_F - \operatorname{diag} v)$. □

4 First-order face selection

4.1 An example with four states on a cycle

Take $d = 4$ states on a cycle, with rate 1 between neighbors and the uniform start. Up to symmetry the admissible faces are a vertex, an adjacent pair, an arc of three states and the whole cycle, with exit rates 2, 1, $2 - \sqrt{2}$ and 0. (For the arc, $-Q_F = 2I - A$ with A the adjacency matrix of the path $1 - 2 - 3$, whose largest eigenvalue is $\sqrt{2}$.) The chain stays in a face F up to time T with probability about $e^{-T\lambda_F}$. Given that it stays, its visit fractions fluctuate on the scale $T^{-1/2}$ in each of the $|F| - 1$ free directions, so this mass is shared by about $(N/\sqrt{T})^{|F|-1}$ compositions. Hence $M_F \approx N^{-(|F|-1)} T^{(|F|-1)/2} e^{-T\lambda_F}$, and with $T = \tau \log N$ this is $N^{-\Phi_\tau(F)}$ up to powers of $\log N$. Figure 3(a) plots the four exponents. The vertices win for $\tau < 1$, the pairs for $1 < \tau < 2$ and the whole cycle for $\tau > 2$. The arcs never win, since they beat the pairs only for $\tau > 1 + \sqrt{2}$, where the cycle beats them. The whole cycle carries almost all the mass for every $\tau > 0$, and it carries the mode only for $\tau > 2$, because it spreads its mass over about N^3 compositions. Theorems 4.2 and 5.7 make this rule exact, and for a pair Corollary 5.3 gives the constant $1/\sqrt{2\pi}$ in front.

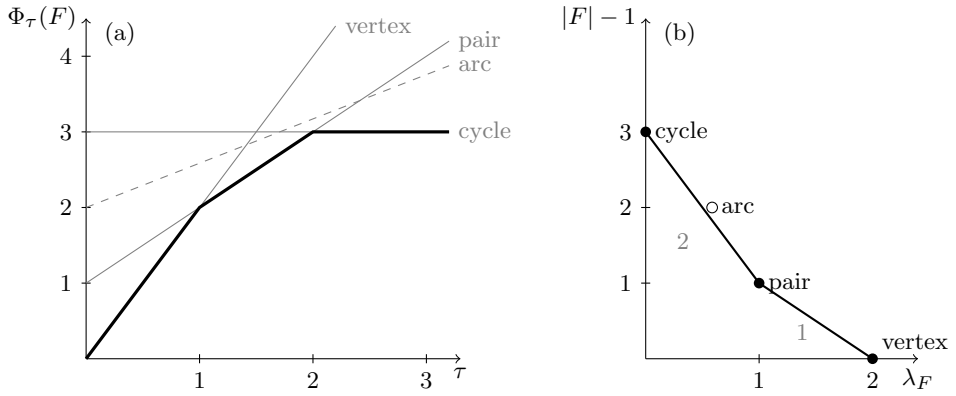


Fig. 3 The 4-cycle. (a) The exponents $\Phi_\tau(F) = \tau\lambda_F + |F| - 1$ of the four kinds of faces against τ, and their lower envelope (bold). The mode sits on the face whose line is lowest, namely a vertex for $\tau < 1$, an adjacent pair for $1 < \tau < 2$ and the whole cycle for $\tau > 2$. The dashed line of the arc of three states is never lowest. (b) The same faces as points $(\lambda_F, |F| - 1)$, with the lower-left boundary of their convex hull. The gray numbers are minus the slopes of its edges, the values of τ where the winner changes (Theorem 4.4). The arc (open) lies above the boundary

4.2 Face selection

We also need an upper bound on a whole face, including near its boundary, where the local law does not apply. For an irreducible face F put $V_F = \{\theta \in \mathbb{R}^F : \varrho(A_F - \operatorname{diag}\theta) = 0\}$. Each $\theta \in V_F$ gives an exponential bound with the linear rate $\alpha \mapsto \langle q - \theta, \alpha\rangle$, and J_F is the supremum of these linear rates. Indeed, for $u \in (0,\infty)^F$ the vector $\theta = A_F u/u$ (entrywise) lies in V_F, since u is a positive null vector of the irreducible Metzler matrix $A_F - \operatorname{diag}\theta$ and only its Perron root has a positive eigenvector. Then $\langle q - \theta, \alpha\rangle$ is the function of u inside the supremum that defines $J_F(\alpha)$. Conversely every $\theta \in V_F$ arises in this way from a positive Perron vector of $A_F - \operatorname{diag}\theta$. For a field $v \in \mathbb{R}^F$ the vector $\theta = q - v - \lambda_F(v)\mathbf{1}$ lies in V_F, since $A_F - \operatorname{diag}\theta = Q_F + \operatorname{diag} v + \lambda_F(v)I$.

Lemma 4.1 (a uniform upper bound) *Let F be irreducible with $k = |F| \geq 2$, and let $\theta \in V_F$. There is C_θ, depending on θ, Q and μ and bounded for θ in compact subsets of V_F, such that for every $n \in \mathbb{Z}^F_{\geq 0}$ with $\operatorname{supp} n = F$ and every h with $hq_{\max} \leq \frac{1}{2}$, the probability that $X_1, \dots, X_{|n|}$ have composition n is at most $C_\theta h^{k-1} \exp\bigl(-h\langle n, q - \theta\rangle + C_\theta h^2 |n|\bigr)$.*

The proof is in Appendix A. The lemma holds for a path of any length $|n|$, so it also applies to the blocks of a path. With $\theta = q - v - \lambda_F(v)\mathbf{1}$ and $|n| = N$ it gives

$$P_N(n) e^{Tv\cdot n/N} \leq C_\theta h^{k-1} e^{-T\lambda_F(v) + C_\theta T^2/N} \tag{6}$$

for every n with support F. For $v = 0$ this is the upper half of the rule $M_F \approx N^{-(k-1)} e^{-T\lambda_F}$ of Section 4.1, up to a power of T. The next theorem uses the local law and the face maxima of Section 5, which are proved there without it.

Theorem 4.2 (first-order face selection) *Fix Q and μ.*

(i) Let F satisfy (H), let $k = |F|$, and let $\mathcal{K} \subset \Delta_F^\circ$ be compact. There is N_0 such that, uniformly for $N \geq N_0$, $1 \leq T \leq N^{1/3}$ and $\operatorname{supp} n = F$ with $n/N \in \mathcal{K}$,

$$\log P_N(n) = -(k-1)L - T\, J_F(n/N) + O_{\mathcal{K}}(1 + \log T).$$

(ii) Let F be admissible with components $C_1, \dots, C_r$, and let k^* be the least size of a component C_j with $\lambda_{C_j} = \lambda_F$. There are κ, T_0 and N_0 such that for $N \geq N_0$ and $T_0 \leq T \leq N^{1/3}$

$$\log M_F \geq -(k_F - 1)L + (k_F - k^*)\log T - T\lambda_F - \kappa,$$
$$\log M_F \leq -(k_F - 1)L + (k_F - 1)\log T - T\lambda_F + \kappa.$$

If F is not admissible, $M_F = 0$.

(iii) Let $T = \tau L$ with $\tau > 0$ fixed, and let $\mathcal{F}_\tau$ be the set of admissible faces that minimize Φ_τ. Then $\log \max_n P_N(n) = -L \min_F \Phi_\tau(F) + O(\log L)$, with the minimum over admissible faces, and for large N every global mode $\hat{n}$ has $\operatorname{supp} \hat{n} \in \mathcal{F}_\tau$. This holds for every start μ.

(iv) If in (iii) every face in $\mathcal{F}_\tau$ is irreducible, then every global mode $\hat{n}$ satisfies $\hat{n}/N = \alpha_F^* + O(1/L)$ with $F = \operatorname{supp} \hat{n}$. This is the case when Q has symmetric support or μ has full support (Theorem 4.4(c)).

Proof (i) For $k = 1$ this is (1), since $J_{\{i\}} = q_i$. For $k \geq 2$, Lemma 5.5 gives $P_N(n) = N^{-(k-1)} f_T(n/N)(1 + O(T^2/N))$. For $T \geq T_1$ Theorem 5.4 gives $\log f_T(\alpha) = \frac{k-1}{2} \log T + \log C_F(\alpha) - T J_F(\alpha) + O_{\mathcal{K}}(1)$ on $\mathcal{K}$, and for $1 \leq T \leq T_1$ this holds because f_T and C_F are positive and continuous on $[1, T_1] \times \mathcal{K}$. As $\log C_F$ is bounded on $\mathcal{K}$, adding $\log N^{-(k-1)} = -(k-1)L$ gives (i).

(ii) Let F be admissible. A path visits each component in one block of consecutive times (proof of Lemma 2.2), and an irreducible face is one component, with $k^* = k_F$. For the upper bound, Lemma 4.1 (or a direct count if $|C| = 1$) bounds a block of length N_C in the component C by $O(h^{|C|-1} e^{-h\lambda_C N_C})$, and each of the $r - 1$ switches between blocks gives a factor h. As $\lambda_C \geq \lambda_F$, this gives $M_F = O(h^{k_F - 1} e^{-T\lambda_F})$. For the lower bound the path spends about N/T visits in each component except one of least exit rate, which loses a bounded factor per component, and we divide by the number of compositions. The details are in Appendix A.6.

(iii) For large N we have $T_0 \leq T = \tau L \leq N^{1/3}$, and both bounds in (ii) read $\log M_F = -L\,\Phi_\tau(F) + O(\log L)$ for every admissible face, while $M_F = 0$ for the others. Since $\max_n P_N(n) = \max_F M_F$ over finitely many faces, the first claim follows. Let $m_\tau = \min_F \Phi_\tau(F)$. An admissible $F \notin \mathcal{F}_\tau$ has $\Phi_\tau(F) \geq m_\tau + \varepsilon_0$ for a fixed $\varepsilon_0 > 0$, so $M_F \leq N^{-m_\tau - \varepsilon_0 + o(1)}$, which is smaller than $\max_n P_N(n) = N^{-m_\tau + o(1)}$ for large N. A global mode has positive probability, so its support is admissible, and hence it lies in $\mathcal{F}_\tau$.

(iv) By (iii), for large N a global mode $\hat{n}$ maximizes P_N on its face $F \in \mathcal{F}_\tau$, which satisfies (H) by Lemma 2.2(b). So Theorem 5.7 with $T = \tau L \leq N^{1/3}$ gives $\hat{n}/N = \alpha_F^* + O(1/L)$. □

For a reducible face we only have the bounds in (ii), and in Example 5.14 the power of T differs from $(k_F - 1)/2$.

Remark 4.3 (a field) Let $v \in \mathbb{R}^d$, put $M_F(T, v) = \max\{P_N(n) e^{Tv \cdot n/N} : \operatorname{supp} n = F\}$ and $\Phi_{\tau,v}(F) = |F| - 1 + \tau\lambda_F(v_F)$. On an admissible face, $Q_F + \operatorname{diag} v_F$ is block triangular in the order of the components, so $\lambda_F(v_F) = \min_j \lambda_{C_j}(v_{C_j})$. The upper bound in (ii) holds for

$M_F(T, v)$ with $\lambda_F(v_F)$ in place of λ_F. The proof in Appendix A.6 applies with (6) on each block and the exact weight $e^{hv_iN_j}$ on a block $\{i\}$. For F satisfying (H), Theorem 5.7 gives the matching lower bound. If μ has full support, a reducible admissible face F has a component C with $\lambda_C(v_C) = \lambda_F(v_F)$, which is admissible with fewer states, so $\Phi_{\tau,v}(C) \le \Phi_{\tau,v}(F) - 1$. So if μ has full support, then for fixed τ and v, (iii) and (iv) hold for the weighted law with $\Phi_{\tau,v}$ and $\alpha^*_F(v)$, every face that minimizes $\Phi_{\tau,v}$ is irreducible, and a strict minimizer carries every maximizer of the weighted law for large N.

4.3 Reading the winners off a convex hull

For an admissible face F put $P_F = (\lambda_F, |F| - 1)$, and let S be the set of these points. Then $\Phi_\tau(F) = \tau x + y$ at $(x, y) = P_F$. Picture a line of slope $-\tau$ moving up from far below. The first points of S that it touches are the winners at τ. For each $\tau > 0$ the points of $\operatorname{conv} S$ where $\tau x + y$ is least form a vertex or an edge of $\operatorname{conv} S$, and we call the union of these over all $\tau > 0$ the lower-left boundary of $\operatorname{conv} S$. Figure 3(b) shows it for the 4-cycle.

Theorem 4.4 (geometry of the winners) *Let $\mathcal{F}_\tau$ be the set of admissible faces that minimize Φ_τ.*

(a) The lower-left boundary of $\operatorname{conv} S$ *is a path whose vertices lie in S. Read from right to left, they are $v_i = (\lambda^{(i)}, y^{(i)})$, $0 \le i \le m$, with $\lambda^{(0)} > \cdots > \lambda^{(m)}$ and $y^{(0)} < \cdots < y^{(m)}$. Put $\tau_i = (y^{(i)} - y^{(i-1)})/(\lambda^{(i-1)} - \lambda^{(i)})$ for $1 \le i \le m$, $\tau_0 = 0$ and $\tau_{m+1} = \infty$. Then $\tau_1 < \cdots < \tau_m$. For $\tau_i < \tau < \tau_{i+1}$, $\mathcal{F}_\tau$ is the set of faces F with $P_F = v_i$. At $\tau = \tau_i$ it is the set of faces whose point lies on the closed edge $[v_{i-1}, v_i]$.*

(b) If $0 < \tau < \tau'$, $F \in \mathcal{F}_\tau$ and $G \in \mathcal{F}_{\tau'}$, then $\lambda_F \ge \lambda_G$ and $|F| \le |G|$. Both inequalities are strict when $\mathcal{F}_\tau \cap \mathcal{F}_{\tau'} = \emptyset$.

(c) If Q has symmetric support, or μ has full support, then every face in $\mathcal{F}_\tau$ is irreducible, for every $\tau > 0$.

Proof (a) We study the least value $g(\tau) = \min_{(x,y)\in S}(\tau x + y)$ for $\tau > 0$. Each point (x, y) of S gives the affine function $\tau \mapsto \tau x + y$, whose slope is x, and different points give different functions. So g is a minimum of finitely many affine functions. It is concave and piecewise affine. Let $\tau_1 < \cdots < \tau_m$ be the points of $(0, \infty)$ where its slope changes, and put $\tau_0 = 0$ and $\tau_{m+1} = \infty$.

Between two kinks. On (τ_i, τ_{i+1}) the function g is affine. The affine function of a point of S is nowhere below g. So if it equals g at one τ inside this interval, it equals g on the whole interval. Hence exactly one point of S attains g there. We call it $v_i = (\lambda^{(i)}, y^{(i)})$, and for $\tau_i < \tau < \tau_{i+1}$ the set $\mathcal{F}_\tau$ consists of the faces F with $P_F = v_i$. The slope of g on this interval is $\lambda^{(i)}$. The slope of a concave function decreases, and it changes at each τ_i, so $\lambda^{(0)} > \cdots > \lambda^{(m)}$. By continuity both v_{i-1} and v_i attain $g(\tau_i)$, namely

$$\tau_i \lambda^{(i-1)} + y^{(i-1)} = \tau_i \lambda^{(i)} + y^{(i)}.$$

This is the formula for τ_i in the statement, and since $\tau_i > 0$ it gives $y^{(i)} > y^{(i-1)}$.

At a kink. The points (x, y) of S that attain $g(\tau_i)$ are those on the line $\tau_i x + y = g(\tau_i)$, which passes through v_{i-1} and v_i. Let (x, y) be one of them. For τ slightly below

τ_i we have $\tau x + y \geq g(\tau) = \tau\lambda^{(i-1)} + y^{(i-1)}$, with equality at $\tau = \tau_i$. Subtracting gives $(\tau - \tau_i)(x - \lambda^{(i-1)}) \geq 0$, so $x \leq \lambda^{(i-1)}$. In the same way, τ slightly above τ_i gives $x \geq \lambda^{(i)}$. So (x, y) lies on the closed edge $[v_{i-1}, v_i]$. Conversely every point of S on this edge lies on the line and so attains $g(\tau_i)$.

The boundary. On $\operatorname{conv} S$ the function $\tau x + y$ is least exactly on the convex hull of the points of S where it is least. By the two steps above this hull is the vertex v_i for $\tau_i < \tau < \tau_{i+1}$ and the edge $[v_{i-1}, v_i]$ for $\tau = \tau_i$. So the lower-left boundary is the path $v_0 \cdots v_m$, and its vertices lie in S.

(b) We add two inequalities. Since F wins at τ and G wins at τ', we have $\Phi_\tau(F) \leq \Phi_\tau(G)$ and $\Phi_{\tau'}(G) \leq \Phi_{\tau'}(F)$. Adding them gives $(\tau' - \tau)(\lambda_F - \lambda_G) \geq 0$, so $\lambda_F \geq \lambda_G$. Then the first inequality gives $|F| - |G| \leq \tau(\lambda_G - \lambda_F) \leq 0$. If $\lambda_F = \lambda_G$, the two inequalities give $|F| = |G|$, so $\Phi_{\tau'}(F) = \Phi_{\tau'}(G)$ and $F \in \mathcal{F}_{\tau'}$. So if $\mathcal{F}_\tau$ and $\mathcal{F}_{\tau'}$ are disjoint, then $\lambda_F > \lambda_G$, and the first inequality gives $|F| < |G|$.

(c) With symmetric support every admissible face is irreducible by Lemma 2.2(c). Now let μ have full support, and let F be admissible and reducible. Take a component C of F with $\lambda_C = \lambda_F$. It is irreducible with $\mu(C) > 0$, so it is admissible by Lemma 2.2(b). It has the same exit rate as F and fewer states, so $\Phi_\tau(C) \leq \Phi_\tau(F) - 1$. Hence $F \notin \mathcal{F}_\tau$. □

Lean checks part (b), the fact that a face with a proper subface of the same exit rate never wins, and part (a) when the least exit rate of each size decreases strictly with the size. In that case the vertices of the hull are sizes, and Lean checks which size wins between consecutive kinks and that consecutive vertices tie at τ_i. The general path structure in (a) and the ties along a closed edge are proved only here. A face with small λ_F for its size is a set of states that the chain rarely leaves, a well-separated community. As τ grows the exit rate counts for more, and the mode moves to larger communities. It need not grow around its first vertex, and it can move to a disjoint community. For symmetric Q, λ_F is the least value of $\sum_{i<j} q_{ij}(f_i - f_j)^2$, the sum over all pairs in $[d]$, over unit vectors f supported in F (Rayleigh–Ritz), and it decreases strictly as a connected face grows, by Lemma 3.2. Since faces of one size share the term $|F| - 1$, only the faces with the least exit rate of each size can win. Figure 4(a) shows S for the 6-cycle.

5 The second-order theory

To locate a crossing up to $o(1)$ we need the power of T and the constant in front of $N^{-(k-1)}e^{-T\lambda_F}$ for every face that satisfies (H). Section 5.1 defines the constant K_F, Section 5.2 proves the local law and the face maxima with a field, and Section 5.3 compares two faces. Section 5.4 gives a tree formula for K_F, Section 5.5 shows what can change without (H), Section 5.6 treats a weak field, and Section 5.7 makes the face maxima uniform over a class of reversible chains. In this section gradients and Hessians in α are taken in the coordinates $\alpha' = (\alpha_2, \ldots, \alpha_k)$, $|\cdot|$ is the Euclidean norm, and $q = (q_i)_{i \in F}$.

5.1 The tilted chain and the constant

To find the probability of a composition with fractions α we change the rates so that α becomes typical. The optimal vector r_α in the definition of $J_F(\alpha)$ gives a new generator

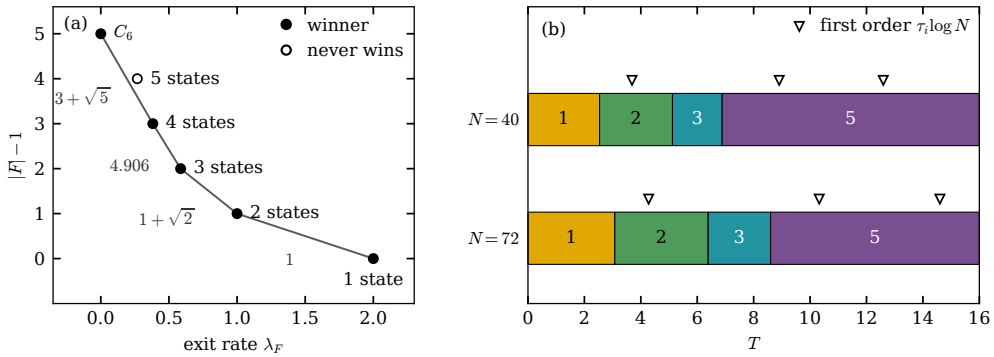


Fig. 4 (a) The points $(\lambda_F, |F|-1)$ of the arcs of the 6-cycle C_6 and of C_6 itself, with the lower-left boundary of their hull. The numbers beside the edges are the switch values τ_i of Theorem 4.4. The winners (filled) are the arcs of one to four states and C_6, and the arc of five states (open) never wins. (b) The support size of the exact mode on C_5 with the uniform start, against T, at $N = 40$ and $N = 72$. The mode skips the arc of four states. The triangles mark the first-order switch points $\tau_i \log N$ with $\tau_i = 1$, $1+\sqrt{2}$ and $2+\sqrt{2}$

$\widehat{Q}_\alpha$ on F, with the same switching graph, whose stationary law is exactly α. Under it the visit fractions satisfy a central limit theorem with covariance Γ_α/T, and the price of the change of rates is $e^{-TJ_F(\alpha)}$ times boundary factors for the start and the end of the path. The constant $C_F(\alpha)$ collects these factors and the Gaussian normalization.

Definition 5.1 (tilt and constants) Let F be irreducible with $k \geq 2$. For $\alpha \in \Delta_F^\circ$ let r_α attain the supremum defining $J_F(\alpha)$ (Proposition 3.3(b)), and put $\theta(\alpha) = A_F r_\alpha / r_\alpha$ and $l_\alpha = \alpha / r_\alpha$ (entrywise). The *tilted generator* is $\widehat{Q}_\alpha = R^{-1}(A_F - \operatorname{diag}\theta(\alpha))R$ with $R = \operatorname{diag} r_\alpha$. With $D = \operatorname{diag}\alpha$ let

$$Z_\alpha = (\mathbf{1}\alpha^\top - \widehat{Q}_\alpha)^{-1} - \mathbf{1}\alpha^\top, \qquad \Gamma_\alpha = DZ_\alpha + Z_\alpha^\top D,$$

and let Σ_α be Γ_α with the row and column of state 1 deleted. Finally let

$$C_F(\alpha) = \frac{(\mu \cdot r_\alpha)(l_\alpha \cdot \mathbf{1})}{(2\pi)^{(k-1)/2}\sqrt{\det \Sigma_\alpha}}, \qquad K_F = C_F(\alpha_F^*).$$

For a single state we put $K_{\{i\}} = \mu_i$. Nothing in Definition 5.1 changes when r_α is scaled, and nothing in it involves the diagonal of Q_F. The objects depend on A_F and μ only, so a field does not change them. For $v \in \mathbb{R}^F$ we put $K_F(v) = C_F(\alpha_F^*(v))$, so that $K_F(0) = K_F$, and $K_{\{i\}}(v) = \mu_i$. Note that $l_\alpha \cdot r_\alpha = 1$ and $J_F(\alpha) = \langle q - \theta(\alpha), \alpha\rangle$. Since r_α minimizes $u \mapsto \sum_{i\neq j} \alpha_i q_{ij} u_j / u_i$, its derivative in u_j, which is $\sum_i l_{\alpha,i} q_{ij} - l_{\alpha,j}\theta_j(\alpha)$, vanishes at r_α. Hence

$$l_\alpha^\top \big(A_F - \operatorname{diag}\theta(\alpha)\big) = 0. \tag{7}$$

Also $(A_F - \operatorname{diag}\theta(\alpha))r_\alpha = 0$ with $r_\alpha > 0$, so $\theta(\alpha) \in V_F$ [2]. So $\widehat{Q}_\alpha$ is an irreducible generator with the same off-diagonal zero pattern as A_F, and by (7) its stationary law is α. If $(\mathbf{1}\alpha^\top - \widehat{Q}_\alpha)v = 0$, multiplying by $\alpha^\top$ gives $\alpha^\top v = 0$, so $\widehat{Q}_\alpha v = 0$. Hence v is a

multiple of $\mathbf{1}$, and $\alpha^\top v = 0$ gives $v = 0$. Thus Z_α exists, and

$$Z_\alpha \mathbf{1} = 0, \qquad \alpha^\top Z_\alpha = 0, \qquad -\widehat{Q}_\alpha Z_\alpha = I - \mathbf{1}\alpha^\top. \tag{8}$$

So $\Gamma_\alpha \mathbf{1} = 0$. As Γ_α is symmetric, its adjugate is then a multiple of $\mathbf{1}\mathbf{1}^\top$, so deleting another state instead of state 1 gives the same $\det \Sigma_\alpha$. At α_F^* the vector $u = r_F$ gives the value $\lambda_F = J_F(\alpha_F^*)$, so $r_{\alpha_F^*}$ is a multiple of r_F, $\theta(\alpha_F^*) = q - \lambda_F \mathbf{1}$, and $l_{\alpha_F^*}$ is the matching multiple of l_F. In the local law below, $\mu \cdot r_\alpha$ accounts for the start of the path, $l_\alpha \cdot \mathbf{1}$ for its free end, and Σ_α for the Gaussian spread of the occupation fractions.

For conductances $c_{ij} = c_{ji}$ on F let $\mathrm{tree}(c) = \sum_{\mathcal{T}} \prod_{\{i,j\} \in \mathcal{T}} c_{ij}$, the sum over the spanning trees $\mathcal{T}$ of the complete graph on F.

Proposition 5.2 (the constant) *Let F be irreducible with $k \geq 2$.*

(a) The maps $\alpha \mapsto \theta(\alpha)$, $J_F(\alpha)$, Σ_α, r_α and l_α (with $\sum_i r_{\alpha,i} = 1$) are real-analytic on Δ_F°, Σ_α is positive definite, and $\nabla^2 J_F(\alpha) = \Sigma_\alpha^{-1}$. Hence $\det \Sigma_\alpha \cdot \det \nabla^2 J_F(\alpha) = 1$ and $C_F(\alpha) = (\mu \cdot r_\alpha)(l_\alpha \cdot \mathbf{1}) \sqrt{\det \nabla^2 J_F(\alpha)/(2\pi)^{k-1}}$.

(b) Suppose that $\nu_i q_{ij} = \nu_j q_{ji}$ for $i, j \in F$ and some $\nu \in (0,\infty)^F$. Then r_α is a multiple of $\sqrt{\alpha/\nu}$ and l_α of $\sqrt{\alpha\nu}$, and $\widehat{Q}_\alpha$ is reversible with conductances $c_{ij}(\alpha) = \sqrt{\alpha_i \alpha_j}\, q_{ij} \sqrt{\nu_i/\nu_j}$. Also $J_F(\alpha) = \sum_i q_i \alpha_i - \sum_{i \neq j} c_{ij}(\alpha)$, and

$$\det \Sigma_\alpha = \frac{2^{k-1} (\prod_i \alpha_i)^2}{\mathrm{tree}(c(\alpha))}, \qquad C_F(\alpha) = \frac{(\mu \cdot r_\alpha)(l_\alpha \cdot \mathbf{1})}{(4\pi)^{(k-1)/2}} \, \frac{\sqrt{\mathrm{tree}(c(\alpha))}}{\prod_i \alpha_i}.$$

Proof The proof of (a) is in Appendix A and uses Lemma A.1.

(b) We guess the vector r_α and check that the guess is a critical point. Put $u = \sqrt{\alpha/\nu}$ and $l = \alpha/u = \sqrt{\alpha\nu}$ (entrywise). As we saw before (7), the derivative of $\sum_{i \neq j} \alpha_i q_{ij} u_j / u_i$ in u_j is $\sum_i l_i q_{ij} - l_j (A_F u)_j / u_j$. Here

$$\sum_i l_i q_{ij} = \sum_i \sqrt{\alpha_i \nu_i}\, q_{ij} \qquad \text{and} \qquad \frac{l_j (A_F u)_j}{u_j} = \nu_j \sum_i q_{ji} \sqrt{\alpha_i/\nu_i} = \sum_i \sqrt{\alpha_i \nu_i}\, q_{ij},$$

where the last step uses $\nu_j q_{ji} = \nu_i q_{ij}$. So the derivative vanishes at u. The function is convex in $\log u$, so its critical point u is a minimizer, and the minimizer is unique up to a factor (Proposition 3.3(b)). Hence r_α is a multiple of u and l_α is the matching multiple of l. Then $\alpha_i (\widehat{Q}_\alpha)_{ij} = l_i q_{ij} u_j = c_{ij}(\alpha)$ for $i \neq j$, and $c_{ij}(\alpha) = c_{ji}(\alpha)$ because $q_{ij} \sqrt{\nu_i/\nu_j} = q_{ji} \sqrt{\nu_j/\nu_i}$. So $\widehat{Q}_\alpha$ is reversible with these conductances. Also $\alpha_i \theta_i(\alpha) = l_i (A_F u)_i = \sum_{j \neq i} c_{ij}(\alpha)$, and $J_F(\alpha) = \langle q - \theta(\alpha), \alpha \rangle$ gives the formula for J_F.

For the determinant we express the quadratic form of Σ_α through the Laplacian of the conductances, namely $\mathcal{L} = -D\widehat{Q}_\alpha$. It is symmetric, its off-diagonal entries are $-c_{ij}(\alpha)$, and $\mathcal{L}\mathbf{1} = 0$. Fix $\eta \in \mathbb{R}^F$ with $\eta_1 = 0$, and put $w = Z_\alpha \eta$ and $\zeta = D\eta - (\alpha \cdot \eta)\alpha$. By (8), $\mathcal{L} w = D(I - \mathbf{1}\alpha^\top)\eta = \zeta$ and $\alpha \cdot w = 0$. So $\eta^\top \Gamma_\alpha \eta = 2(D\eta)^\top w = 2\zeta^\top w$, where the second step uses $\alpha \cdot w = 0$. Next we remove state 1. Let $v = w - w_1 \mathbf{1}$, so that $v_1 = 0$. Since $\zeta \cdot \mathbf{1} = 0$ and $\mathcal{L}\mathbf{1} = 0$, we have $\zeta^\top w = \zeta^\top v$ and $\mathcal{L} v = \zeta$. We write primes for the coordinates $2, \ldots, k$, $D' = \mathrm{diag}\, \alpha'$, and $\mathcal{L}_{(1)}$ for $\mathcal{L}$ without row and column 1. As $v_1 = 0$, the rows $2, \ldots, k$ of $\mathcal{L} v = \zeta$ say that $\mathcal{L}_{(1)} v' = \zeta'$, and as $\eta_1 = 0$ we have $\zeta' = (D' - \alpha' \alpha'^\top)\eta'$. The matrix-tree

theorem [6] gives $\det \mathcal{L}_{(1)} = \operatorname{tree}(c(\alpha))$, which is positive because F is irreducible. So $\mathcal{L}_{(1)}$ is invertible, and

$$\eta'^{\top} \Sigma_\alpha \eta' = 2\zeta'^{\top} v' = 2\eta'^{\top} (D' - \alpha'\alpha'^{\top}) \mathcal{L}_{(1)}^{-1} (D' - \alpha'\alpha'^{\top}) \eta'$$

for every $\eta' \in \mathbb{R}^{k-1}$. Both matrices are symmetric, hence $\Sigma_\alpha = 2(D' - \alpha'\alpha'^{\top}) \mathcal{L}_{(1)}^{-1} (D' - \alpha'\alpha'^{\top})$. Now we take determinants. The matrix determinant lemma gives $\det(D' - \alpha'\alpha'^{\top}) = \prod_{i \ge 2} \alpha_i \, (1 - \sum_{i \ge 2} \alpha_i) = \prod_i \alpha_i$. So $\det \Sigma_\alpha = 2^{k-1} (\prod_i \alpha_i)^2 / \operatorname{tree}(c(\alpha))$, and putting this into Definition 5.1 gives the formula for $C_F(\alpha)$. □

Part (b) needs a reversible face. Proposition 5.11 gives a formula of the same kind for every irreducible face.

Corollary 5.3 (symmetric faces) *Let F satisfy* (H) *with $k \ge 2$, and let Q_F be symmetric with constant row sums. Then $\lambda_F = q_i - \sum_{j \in F \setminus \{i\}} q_{ij}$ for every $i \in F$, α_F^* is uniform on F, and*

$$K_F = \mu(F) \, k^{(k+1)/2} \sqrt{\operatorname{tree}(q_F)} \, (4\pi)^{-(k-1)/2},$$

where $\operatorname{tree}(q_F)$ *is* $\operatorname{tree}(c)$ *for the conductances $c_{ij} = q_{ij}$, $i, j \in F$.*

Proof The positive vector $\mathbf{1}$ is a right and left eigenvector of Q_F, so it belongs to the Perron root. So λ_F is minus the row sum, r_F and l_F are multiples of $\mathbf{1}$, and $\alpha_F^* = \mathbf{1}/k$. At $\alpha = \mathbf{1}/k$, Proposition 5.2(b) with $\nu = \mathbf{1}$ gives $r_\alpha = \mathbf{1}$, $l_\alpha = \alpha$ and $(\mu \cdot r_\alpha)(l_\alpha \cdot \mathbf{1}) = \mu(F)$. Also $c_{ij}(\alpha) = q_{ij}/k$, so $\operatorname{tree}(c(\alpha)) = k^{-(k-1)} \operatorname{tree}(q_F)$, and $\prod_i \alpha_i = k^{-k}$. □

5.2 The local law and the face maxima

A field $v \in \mathbb{R}^F$ enters the continuum density as an exact factor. By Lemma 2.4, $f_T(\alpha) e^{Tv \cdot \alpha}$ is the density f_T of the killed matrix $Q_F + \operatorname{diag} v$, which has the same A_F and μ and the diagonal $q - v$ in place of q. On the lattice the factor is $e^{Tv \cdot n/N} = e^{h\langle v, n \rangle}$.

Theorem 5.4 (local law) *Let F satisfy (H) with $k \ge 2$, and let $\mathcal{K} \subset \Delta_F^\circ$ be compact. Uniformly for $\alpha \in \mathcal{K}$, $T \ge 1$ and $v \in \mathbb{R}^F$,*

$$f_T(\alpha) e^{Tv \cdot \alpha} = T^{(k-1)/2} C_F(\alpha) e^{-T(J_F(\alpha) - v \cdot \alpha)} \big(1 + O_{\mathcal{K}}(1/T)\big).$$

For $F = \{i\}$, $f_T = \mu_i e^{-q_i T}$.

The field cancels from the two sides, so the case $v = 0$ is the whole content, and the error does not depend on v. Good [20, pp. 42–43] proposed a saddle-point analysis of his formula and did not carry it out. Theorem 5.4, with its explicit constant and its uniform relative error, completes that analysis.

Idea of the proof. The proof is in Appendix A. The generating function of the words in H is a matrix resolvent, so Cauchy's formula writes H as an integral over a torus with the radii $\theta_2(\alpha), \dots, \theta_k(\alpha)$ (Lemma A.2). All the coefficients are nonnegative, so the integrand is largest at the real point, and a Laplace lemma with a parameter (Lemma B.1) gives the expansion. The Hessian at the real point comes from the Perron

root of a perturbed generator (Lemma A.1), which brings in Σ_α. The hypothesis $\mu(F) > 0$ is needed, since $f_T = C_F = 0$ when $\mu(F) = 0$. Also $f_T(\alpha)$ is continuous and positive in (T, α), as H is a convergent power series with some $W(m) > 0$ by (H).

Lemma 5.5 (discrete and continuum) *Let F satisfy (H) with $k \geq 2$, and let $\mathcal{K} \subset \Delta_F^\circ$ be compact. Uniformly for $\operatorname{supp} n = F$ with $n/N \in \mathcal{K}$ and $1 \leq T \leq N^{1/2}$, $P_N(n) = N^{-(k-1)} f_T(n/N)(1 + O_{\mathcal{K}}(T^2/N))$.*

The proof is in Appendix A. A field multiplies both sides by the same factor $e^{Tv\cdot n/N}$. Together with the restriction identity the two results give one display for all faces.

Theorem 5.6 (the stratified local law) *Fix Q and a start μ. For every face F satisfying (H) with $|F| = k \geq 2$ and every compact $\mathcal{K} \subset \Delta_F^\circ$, uniformly for $\operatorname{supp} n = F$, $n/N \in \mathcal{K}$ and $1 \leq T \leq N^{1/3}$,*

$$-\log P_N(n) = (k-1)L + T\, J_V(n/N) - \frac{k-1}{2}\log T - \log C_F(n/N) + O_{\mathcal{K}}\Big(\frac{1}{T} + \frac{T^2}{N}\Big),$$

and for a singleton $-\log P_N(Ne_i) = Tq_i - \log \mu_i + O(T^2/N)$. With $T = \tau L$ the exponent of N is $(k-1) + \tau J_V(n/N)$, the lattice entropy of the face plus τ times the Donsker and Varadhan cost of the point.

Proof This is Theorem 5.4 with $v = 0$, Lemma 5.5 and Proposition 3.1, which replaces J_F by J_V on Δ_F. The singleton case is (1). □

The local law does not give the behaviour near the boundary of a face, where one coordinate of n/N tends to zero. Section 6 proves the layer at a facet, and Section 8 treats the complete graph when the small coordinates are of order one.

For a field $v \in \mathbb{R}^d$ we write $\lambda_F(v)$, $\alpha_F^*(v)$ and $K_F(v)$ for the quantities of v_F, and

$$M_F(T, v) = \max\{P_N(n) e^{Tv\cdot n/N} : \operatorname{supp} n = F\},$$

so that $M_F(T, 0) = M_F$.

Theorem 5.7 (face maxima) *Let F satisfy (H), and let $\mathcal{V} \subset \mathbb{R}^d$ be compact. If $k \geq 2$, then uniformly for $v \in \mathcal{V}$ and $1 \leq T \leq N^{1/3}$,*

$$M_F(T, v) = N^{-(k-1)} T^{(k-1)/2} K_F(v) e^{-T\lambda_F(v)}\big(1 + O_{\mathcal{V}}(1/T + T^2/N)\big),$$

and every maximizer $\hat{n}$ satisfies $\hat{n}/N = \alpha_F^*(v) + O_{\mathcal{V}}(1/T)$. For $k = 1$, $M_{\{i\}}(T, v) = e^{Tv_i} M_{\{i\}}$ with $M_{\{i\}}$ given by (1). In both cases, uniformly for $v \in \mathcal{V}$ and $1 \leq T \leq N^{1/3}$,

$$\log M_F(T, v) = -(k-1)L + \tfrac{k-1}{2}\log T + \log K_F(v) - T\lambda_F(v) + O_{\mathcal{V}}(1/T + T^2/N). \qquad (9)$$

The constants depend on $\mathcal{V}$ only through the compact set $\{\alpha_F^*(v) : v \in \mathcal{V}\}$, the bounds of λ_F and K_F on $\mathcal{V}$, and finitely many vectors of V_F.

Idea of the proof. Near $\alpha_F^*(v)$ we use the local law and Lemma 5.5, and maximize a Gaussian with a small linear perturbation. Away from $\alpha_F^*(v)$ the rate exceeds $\lambda_F(v)$ by a fixed amount, and Lemma 4.1, applied with finitely many vectors θ, shows that these compositions are exponentially less likely in T. For bounded T positivity and continuity suffice.

Proof For $k = 1$ the only composition is Ne_i. Let $k \geq 2$, and put $J^v(\alpha) = J_F(\alpha) - v \cdot \alpha - \lambda_F(v)$, which is nonnegative on Δ_F and vanishes only at $\alpha_F^*(v)$ (Proposition 3.3(a)). Put $X(v) = N^{-(k-1)}T^{(k-1)/2}K_F(v)e^{-T\lambda_F(v)}$, the value that the theorem claims, and write $x = \alpha' - \alpha_F^*(v)'$ for the displacement of α from $\alpha_F^*(v)$. Below, constants depend only on Q, μ and $\mathcal{V}$.

The balls. The Perron vectors of $Q_F + \operatorname{diag} v$ are real-analytic in v, so $\mathcal{A} = \{\alpha_F^*(v) : v \in \mathcal{V}\}$ is a compact subset of Δ_F°, and λ_F and $\log K_F$ are bounded on $\mathcal{V}$. By Propositions 3.3 and 5.2(a), J_F and $\log C_F$ are real-analytic on Δ_F° with $\nabla^2 J_F = \Sigma_\alpha^{-1}$ positive definite, and J^v has a critical point at $\alpha_F^*(v)$. Taylor's theorem with bounds on the derivatives of order up to three on a compact neighbourhood of $\mathcal{A}$ gives $\rho_0 > 0$ and positive constants c_U, C_U, C_1 and C_3 with the following properties. The set $\mathcal{K}_\mathcal{V}$ of points within ρ_0 of $\mathcal{A}$ lies in Δ_F°, and for $v \in \mathcal{V}$ the ball $U(v) = \{|x| \leq \rho_0\}$ satisfies

$$c_U|x|^2 \leq J^v \leq C_U|x|^2, \qquad \bigl|\log(C_F/K_F(v)) - g(v)\cdot x\bigr| \leq C_1|x|^2$$

on $U(v)$, where $g(v) = \nabla \log C_F(\alpha_F^*(v))$ has $|g(v)| \leq C_1$. Also $\log C_F$ is C_3-Lipschitz on $\mathcal{K}_\mathcal{V}$.

Outside $U(v)$. The set $\mathcal{E}$ of pairs $(v, \beta) \in \mathcal{V} \times \Delta_F$ with $\beta \notin U(v)^\circ$ is compact, and $J^v(\beta)$ is positive and lower semicontinuous on it, since J_F is a supremum of continuous functions. So $J^v \geq 3\eta$ on $\mathcal{E}$ for some $\eta > 0$. For each $(v, \beta) \in \mathcal{E}$ the definition of J_F gives $\theta \in V_F$ with $\langle q - \theta - v, \beta\rangle > \lambda_F(v) + 2\eta$, as in Section 4, and the same θ works near (v, β). By compactness finitely many vectors $\theta_j \in V_F$ suffice. Now let $\operatorname{supp} n = F$ with $(v, n/N) \in \mathcal{E}$, and take j that works at $(v, n/N)$. Lemma 4.1 with $\theta = \theta_j$, multiplied by $e^{h\langle v,n\rangle}$, gives

$$P_N(n)e^{Tv\cdot n/N} \leq C(T/N)^{k-1}e^{-T(\lambda_F(v)+2\eta)} = X(v)\cdot O\bigl(T^{(k-1)/2}e^{-2\eta T}\bigr).$$

Inside $U(v)$. By Theorem 5.4 and Lemma 5.5 with $\mathcal{K} = \mathcal{K}_\mathcal{V}$, for $\alpha = n/N \in U(v)$,

$$P_N(n)e^{Tv\cdot n/N} = X(v)\,\tfrac{C_F(\alpha)}{K_F(v)}\,e^{-TJ^v(\alpha)}(1+E), \qquad |E| \leq C_2(1/T + T^2/N). \tag{10}$$

For $T \geq 2C_1/c_U$ the logarithm of the two middle factors is at most

$$g\cdot x + C_1|x|^2 - c_U T|x|^2 \leq g\cdot x - \tfrac{c_U T}{2}|x|^2 \leq \frac{|g|^2}{2c_U T},$$

the maximum of a concave quadratic. So $P_N(n)e^{Tv\cdot n/N} \leq X(v)(1 + O(1/T + T^2/N))$ on $U(v)$. The bound outside $U(v)$ is smaller for large T, and hence $M_F(T, v) \leq X(v)(1 + O(1/T + T^2/N))$. For the lower bound let n^0 have support F and $|n_i^0 - N\alpha_F^*(v)_i| < 1$ for all i, which exists for $N \geq N_0$. Then $|x| \leq k/N$, $C_F(n^0/N) \geq K_F(v)e^{-kC_3/N}$ and $TJ^v(n^0/N) \leq C_U k^2 T/N^2$, so (10) gives $P_N(n^0)e^{Tv\cdot n^0/N} \geq X(v)(1 - O(1/T + T^2/N))$.

Bounded T. Fix T_2 such that the two bounds give $M_F(T, v)/X(v) \in [\frac{1}{2}, \frac{3}{2}]$ for $T \geq T_2$. For $1 \leq T \leq T_2$ it is enough to show $M_F(T, v)/X(v) \in [c, C]$ with $0 < c < C$. There $N^{k-1}X(v)$ lies between two positive constants, and so does $f_T(\alpha)e^{Tv\cdot\alpha}$ on $[1, T_2] \times \mathcal{V} \times \mathcal{K}_\mathcal{V}$, as it is positive and continuous. So Lemma 5.5 gives both bounds, and the bound outside $U(v)$ is $O(N^{-(k-1)})$. This proves the first display, and (9) is its logarithm.

The maximizer. Let $\hat{n}$ be a maximizer with displacement $\hat{x}$. For $T \geq T_3$ the bound outside $U(v)$ is below $X(v)/2 \leq P_N(n^0)e^{Tv\cdot n^0/N}$, so $\hat{n}/N \in U(v)$, and $|E| \leq \frac{1}{2}$ in (10). Taking logarithms in the comparison of $\hat{n}$ with n^0, and using $\log(C_F/K_F(v)) \leq C_3|\hat{x}|$ and $TJ^v \geq c_U T|\hat{x}|^2$, we get

$$c_U T|\hat{x}|^2 \leq C_3|\hat{x}| + C_4\Big(\frac{1}{T} + \frac{T^2}{N}\Big) \leq C_3|\hat{x}| + \frac{2C_4}{T}.$$

The last step uses $T \leq N^{1/3}$, which gives $T^2/N \leq 1/T$. This is the step that needs the exponent $\frac{1}{3}$. Solving the quadratic inequality gives $|\hat{x}| \leq \big(C_3 + \sqrt{C_3^2 + 8c_U C_4}\big)/(2c_U T)$. For $1 \leq T < T_3$ the claim holds since $|\hat{x}|$ is bounded. □

5.3 Crossings and tie windows

By Theorem 5.7, the difference $\log M_F - \log M_G$ is an explicit increasing function of T up to an error $O(1/T + T^2/N)$, and its zero is the crossing.

Theorem 5.8 (the crossing equation) *Let F and G satisfy (H), with $\Delta = k_F - k_G \geq 1$ and $\delta = \lambda_G - \lambda_F > 0$, and let $D_N(T) = \log M_F - \log M_G$.*

(i) For large N, D_N has a zero in $[\frac{\Delta}{2\delta}L, \frac{2\Delta}{\delta}L]$.

(ii) For large N, every zero $T_N \in [1, N^{1/3}]$ of D_N satisfies

$$\delta T_N + \frac{\Delta}{2}\log T_N = \Delta L - \log\frac{K_F}{K_G} + O\Big(\frac{1}{L}\Big). \tag{11}$$

(iii) All such zeros lie in one interval of length $O(1/L)$, and

$$T_N = \frac{\Delta}{\delta}\Big[L - \frac{1}{2}\log L + b_{FG}\Big] + O\Big(\frac{\log L}{L}\Big), \qquad b_{FG} = -\frac{1}{2}\log\frac{\Delta}{\delta} - \frac{1}{\Delta}\log\frac{K_F}{K_G}.$$

Proof We subtract (9) for G from (9) for F. Uniformly on $[1, N^{1/3}]$,

$$D_N(T) = -\Delta L + \Xi(T) + \log\frac{K_F}{K_G} + O\Big(\frac{1}{T} + \frac{T^2}{N}\Big), \qquad \Xi(T) = \delta T + \frac{\Delta}{2}\log T.$$

The function Ξ is increasing, with $\Xi' \geq \delta$.

(i) At $T = \frac{\Delta}{2\delta}L$ the display gives $D_N = -\frac{\Delta}{2}L + O(\log L) < 0$, and at $T = \frac{2\Delta}{\delta}L$ it gives $D_N = \Delta L + O(\log L) > 0$. For fixed N, M_F is a maximum of finitely many polynomials in h, and it is positive because F is admissible. The same holds for G. So D_N is continuous and has a zero in between.

(ii) First we locate a zero $T_N \in [1, N^{1/3}]$ roughly. The error term is $O(1)$ on $[1, N^{1/3}]$, so $\Xi(T_N) = \Delta L - \log(K_F/K_G) + O(1)$. This is larger than $\Xi(\frac{\Delta}{2\delta}L) = \frac{\Delta}{2}L + O(\log L)$, so $T_N > \frac{\Delta}{2\delta}L$ because Ξ is increasing. Also $\delta T_N \leq \Xi(T_N)$, so $T_N \leq \frac{2\Delta}{\delta}L$. With T_N of order L the error term is $O(1/T_N + T_N^2/N) = O(1/L)$, and $D_N(T_N) = 0$ is (11).

(iii) Let T° be the solution of $\Xi(T^\circ) = \Delta L - \log(K_F/K_G)$. By (ii) every zero has $\Xi(T_N) = \Xi(T^\circ) + O(1/L)$, and $\Xi' \geq \delta$, so every zero is within $O(1/L)$ of T°. To solve for T°, note that $T^\circ = \frac{\Delta}{\delta}L + O(\log L)$, so $\log T^\circ = \log L + \log\frac{\Delta}{\delta} + O(\log L/L)$. Putting this into $\delta T^\circ = \Delta L - \frac{\Delta}{2}\log T^\circ - \log(K_F/K_G)$ and dividing by δ gives the explicit form. □

In words, the larger face F pays $N^{-\Delta}$ for spreading and gains $e^{\delta T}$ for staying, and its Gaussian peak gives an extra factor $T^{\Delta/2}$. The crossing balances the three. On the 4-cycle the explicit form in (iii) is still off by 0.06 at $N = 10^{14}$, so computations should solve (11). We do not prove that the zero of D_N is unique in general.

At a switch value τ^* of Theorem 4.4 several faces minimize Φ_{τ^*}, sometimes more than two, as all faces do at $\tau^* = 1$ for the complete graph (Corollary 7.3). Then the competition plays out in a window $T = \tau^*(L - \frac{1}{2}\log L + t)$ with t of order 1. In this window each tied face has a line $\ell_F(t)$, and the mode follows the highest line.

Theorem 5.9 (tie windows) *Fix $\tau^* > 0$. Let m^* be the least value of Φ_{τ^*} over the admissible faces and $\mathcal{F}^*$ the set of minimizers. Assume that every face in $\mathcal{F}^*$ is irreducible. By Theorem 4.4(c) this holds if Q has symmetric support or μ has full support. Put $T = \tau^*(L - \frac{1}{2}\log L + t)$, and for $F \in \mathcal{F}^*$ let $\ell_F(t) = \log K_F + \frac{k_F-1}{2}\log\tau^* - \tau^*\lambda_F t$ and $E(t) = \max_{G\in\mathcal{F}^*} \ell_G(t)$. The function E is convex and piecewise linear. We call the finitely many points where its slope changes the* kinks *of E, and we write t_j for them.*

(i) Uniformly for t in compact sets, for $F \in \mathcal{F}^*$, $\log M_F = -m^*L + \frac{m^*}{2}\log L + \ell_F(t) + O(\log L/L)$.

(ii) There are $\varepsilon_1, \varepsilon_0 > 0$ such that for large N and $|T/L - \tau^*| \le \varepsilon_1$, every admissible $F \notin \mathcal{F}^*$ has $M_F \le N^{-\varepsilon_0} \max_{G\in\mathcal{F}^*} M_G$.

(iii) There are $\varepsilon, t_0 > 0$ such that the following holds for large N. If $t_0 \le t \le \varepsilon L$, then every global mode lies on a face $F \in \mathcal{F}^*$ with $\ell_F = E$ on $[t_0, \infty)$. If $-\varepsilon L \le t \le -t_0$, then every global mode lies on a face $F \in \mathcal{F}^*$ with $\ell_F = E$ on $(-\infty, -t_0]$. A face of the first kind has the least λ_F in $\mathcal{F}^*$, and a face of the second kind has the largest.

Consequently the following holds for every $\eta > 0$ and large N. If $|t| \le \varepsilon L$ and t is at distance at least η from every kink of E, then every global mode lies on a face whose line equals E near t. So for $|t| \le \varepsilon L$ the line of the face that carries the global mode changes only at $T = \tau^*(L - \frac{1}{2}\log L + t_j) + o(1)$, where t_j runs over the kinks of E, and it changes near each kink. Coincident lines do not distinguish their faces at this order.

Proof Every $F \in \mathcal{F}^*$ is admissible and irreducible, so it satisfies (H) by Lemma 2.2(b). We first put T into (9). Write $T = \tau^*L(1+y)$ with $y = (t - \frac{1}{2}\log L)/L$, so that $|y| \le \frac{1}{2}$ for $|t| \le L/4$ and large N. Then $\log T = \log\tau^* + \log L + \log(1+y)$ and $T\lambda_F = \tau^*\lambda_F(L - \frac{1}{2}\log L + t)$, and the error term of (9) is $O(1/L)$. Since $\tau^*\lambda_F + k_F - 1 = m^*$, we get, uniformly for $|t| \le L/4$,

$$\log M_F = -m^*L + \tfrac{m^*}{2}\log L + \ell_F(t) + \tfrac{k_F-1}{2}\log(1+y) + O(1/L) \qquad (F \in \mathcal{F}^*). \tag{12}$$

(i) For t in a compact set, $y = O(\log L/L)$, so $\log(1+y) = O(\log L/L)$ in (12).

(ii) Let $g > 0$ be the least value of $\Phi_{\tau^*}(F) - m^*$ over the admissible $F \notin \mathcal{F}^*$. (If there is no such face, (ii) says nothing, and we put $g = 1$.) Put $\varepsilon_1 = \min\bigl(\tau^*/2,\ g/(4q_{\max}+4)\bigr)$. (Any constant below $g/(2q_{\max})$ would do in the second entry. We use that $0 \le \lambda_F \le q_{\max}$ for every face, since the Perron root of Q_F lies between its largest diagonal entry and 0.) The entry $\tau^*/2$ keeps T of order L. Namely, if $|T/L - \tau^*| \le \varepsilon_1$, then $\frac{1}{2}\tau^*L \le T \le \frac{3}{2}\tau^*L$. So for large N we have $T_0 \le T \le N^{1/3}$, which is the range of Theorem 4.2(ii), and $\log T = O(\log L)$. Take an admissible $F \notin \mathcal{F}^*$ and any $G \in \mathcal{F}^*$. The difference $\Phi_\tau(F) - \Phi_\tau(G)$ is at least g at $\tau = \tau^*$, and its slope in τ is $\lambda_F - \lambda_G$, which is at most $q_{\max}$ in absolute value. So $\Phi_\tau(F) - \Phi_\tau(G) \ge g/2$ for $|\tau - \tau^*| \le \varepsilon_1$. With $\tau = T/L$ the upper bound of Theorem 4.2(ii) for F and the lower

bound for G give $\log M_F - \log M_G \le -(\Phi_\tau(F) - \Phi_\tau(G))L + O(\log L) \le -\frac{g}{4}L$ for large N. So (ii) holds with $\varepsilon_0 = g/4$.

(iii) Take $\varepsilon = \min(\frac{1}{4}, \varepsilon_1/(2\tau^*))$. Then for $|t| \le \varepsilon L$ and large N we have $|T/L - \tau^*| = \tau^*|y| \le \varepsilon_1$, so (ii) applies and no global mode lies on a face outside $\mathcal{F}^*$. It remains to compare the tied faces. Let $t_0 \ge 1$ be larger than every $|t_j|$, so that E is a single line on $[t_0, \infty)$. Let $F \in \mathcal{F}^*$ have this line, and let $G \in \mathcal{F}^*$ have another line. We show that $M_F > M_G$ for $t_0 \le t \le \varepsilon L$ if t_0 and N are large. Since $\ell_F \ge \ell_G$ on $[t_0, \infty)$, the slopes satisfy $\lambda_F \le \lambda_G$. As both faces are tied,

$$\Delta = k_F - k_G = \tau^*(\lambda_G - \lambda_F) \ge 0.$$

There are two cases.

First let $\Delta = 0$. Then the two lines are parallel, and $\ell_F - \ell_G = \log(K_F/K_G) > 0$. The terms with $\log(1+y)$ in (12) are the same for F and G, so $\log(M_F/M_G) = \log(K_F/K_G) + O(1/L) > 0$ for large N.

Now let $\Delta > 0$. Then $\ell_F(t) - \ell_G(t) = \Delta t + \ell_F(0) - \ell_G(0)$, and (12) gives

$$\log \frac{M_F}{M_G} = \Delta t + \ell_F(0) - \ell_G(0) + \frac{\Delta}{2}\log(1+y) + O\Big(\frac{1}{L}\Big).$$

For large N and $t \ge 1$ we have $|\log(1+y)| \le 2|y| \le (2t + \log L)/L \le t/2$. So the right side is at least $\frac{3}{4}\Delta t - C$, which is positive for $t \ge t_0$ once t_0 is large.

So for $t_0 \le t \le \varepsilon L$ every global mode lies on a face whose line is E on $[t_0, \infty)$. The slope of ℓ_F is $-\tau^*\lambda_F$, and E follows the line of largest slope on the far right, so such a face has the least λ_F in $\mathcal{F}^*$. The case $t \le -t_0$ is the same with the signs reversed.

The consequence. Fix $\eta > 0$ and let $S_\eta = [-t_0, t_0] \setminus \bigcup_j (t_j - \eta, t_j + \eta)$, a finite union of closed intervals without kinks. Take $G \in \mathcal{F}^*$. The function $E - \ell_G$ is convex, piecewise linear and nonnegative, and its slope changes only at kinks of E. So its zero set is empty, or a single kink of E, or an interval whose finite endpoints are kinks of E. Hence each interval of S_η lies inside the zero set or is disjoint from it. In the first case $\ell_G = E$ on the interval, and in the second $E - \ell_G \ge g_\eta$ there for some $g_\eta > 0$. Now let $t \in S_\eta$ and let $F \in \mathcal{F}^*$ have $\ell_F(t) = E(t)$. By (i), every $G \in \mathcal{F}^*$ whose line is not E near t has $\log(M_F/M_G) \ge g_\eta - O(\log L/L) > 0$, and by (ii) the faces outside $\mathcal{F}^*$ are smaller still. This is the claim for $t \in S_\eta$. For $t_0 \le |t| \le \varepsilon L$ it is (iii). Since η is arbitrary, the line of the face that carries the mode can change only at $t = t_j + o(1)$. It does change near each kink, because E is a different line on the two sides of a kink. □

Remark 5.10 (kinks are crossings) At a kink of E where the line of G gives way to that of F as t increases, $\Delta = \tau^*\delta$ and $t_j = b_{FG}$. So every change in a tie window is a crossing of Theorem 5.8. The coefficient $\frac{1}{2}$ of $\log\log N$ in both theorems comes from the power $T^{(k-1)/2}$ in Theorem 5.7, so it needs only (H). Example 5.14 shows that it can fail on a reducible face.

5.4 A tree formula for every face

The next proposition extends Proposition 5.2(b) from reversible faces to every irreducible face. It uses two counts of spanning trees in place of one, namely tree as before and the count arb of spanning trees oriented toward a root, which is defined just before Lemma A.5. Its proof rests on Lemma A.5 in Appendix A.

Proposition 5.11 (the constant on a general face) *Let F be irreducible with $k \geq 2$. For $\alpha \in \Delta_F^\circ$ put*

$$f_{ij}(\alpha) = l_{\alpha,i}\, q_{ij}\, r_{\alpha,j} \qquad (i \neq j \text{ in } F),$$

which does not change when r_α is scaled.

(a) The weights $f(\alpha)$ are the stationary flow of the tilted chain, that is, $f_{ij}(\alpha) = \alpha_i(\widehat{Q}_\alpha)_{ij}$, and $\sum_{j\neq i} f_{ij}(\alpha) = \sum_{j\neq i} f_{ji}(\alpha) = \alpha_i\theta_i(\alpha)$ for every i. The number $\operatorname{arb}_i(f(\alpha))$ does not depend on the root i, and

$$\det \Sigma_\alpha = \frac{(\prod_i \alpha_i)^2\ \operatorname{tree}\big(f(\alpha) + f(\alpha)^\top\big)}{\operatorname{arb}(f(\alpha))^2}$$

and

$$C_F(\alpha) = \frac{(\mu \cdot r_\alpha)(l_\alpha \cdot \mathbf{1})}{(2\pi)^{(k-1)/2}}\, \frac{\operatorname{arb}(f(\alpha))}{\prod_i \alpha_i\, \sqrt{\operatorname{tree}(f(\alpha) + f(\alpha)^\top)}}.$$

(b) Also $\operatorname{arb}(f(\alpha))/\prod_i \alpha_i = \sum_{i\in F} \det\big(\operatorname{diag}\theta(\alpha) - A_F\big)_{(i)}$. At $\alpha = \alpha_F^*$ this is $\prod_\rho(\rho - \lambda_F)$, the product over the other $k-1$ eigenvalues ρ of $-Q_F$, counted with multiplicity. So with $l_F \cdot r_F = 1$ and $f_{ij} = l_{F,i}\, q_{ij}\, r_{F,j}$,

$$K_F = \frac{(\mu \cdot r_F)(l_F \cdot \mathbf{1})}{(2\pi)^{(k-1)/2}}\, \frac{\operatorname{arb}(f)}{\prod_i \alpha^*_{F,i}\, \sqrt{\operatorname{tree}(f + f^\top)}} = \frac{(\mu \cdot r_F)(l_F \cdot \mathbf{1})}{(2\pi)^{(k-1)/2}}\, \frac{\prod_\rho(\rho - \lambda_F)}{\sqrt{\operatorname{tree}(f + f^\top)}}.$$

(c) Let $F = \{1, \dots, k\}$ be a directed cycle, that is, $q_{i,i+1} = a_i > 0$ with indices modulo k, and $q_{ij} = 0$ for all other $i \neq j$ in F. Then for every $\alpha \in \Delta_F^\circ$

$$\det \Sigma_\alpha = \frac{k\,(\prod_i \alpha_i)^2}{\big(\prod_i \alpha_i a_i\big)^{(k-1)/k}}.$$

If moreover $q_i = a_i$ for all i, then $\alpha^*_{F,i} = \varphi^*/a_i$ with $\varphi^* = (\sum_i 1/a_i)^{-1}$, and

$$\det \Sigma_{\alpha_F^*} = \frac{k\,(\varphi^*)^{k+1}}{\prod_i a_i^2}, \qquad K_F = \frac{\mu(F)\, \prod_i a_i}{(2\pi)^{(k-1)/2}\, \sqrt{k}\, (\varphi^*)^{(k+1)/2}}.$$

In the reversible case of Proposition 5.2(b), $f(\alpha) = c(\alpha)$ is symmetric, $\operatorname{arb}(f) = \operatorname{tree}(c)$ and $\operatorname{tree}(2c) = 2^{k-1}\operatorname{tree}(c)$, so (a) reduces to that proposition.

Proof (a) By Definition 5.1, $\alpha_i(\widehat{Q}_\alpha)_{ij} = l_{\alpha,i} q_{ij} r_{\alpha,j} = f_{ij}(\alpha)$ for $i \neq j$, and the diagonal of $\widehat{Q}_\alpha$ is $-\theta(\alpha)$. So $\widehat{Q}_\alpha \mathbf{1} = 0$ and $\alpha^\top \widehat{Q}_\alpha = 0$ give the two flow identities. Lemma A.5 for $G = \widehat{Q}_\alpha$ and $\pi = \alpha$, for which $\Sigma = \Sigma_\alpha$, gives $\det \Sigma_\alpha$, and Definition 5.1 gives $C_F(\alpha)$.

(b) Since $-\widehat{Q}_\alpha = R^{-1}(\operatorname{diag}\theta(\alpha) - A_F)R$ and a diagonal conjugation keeps principal minors, Lemma A.5(i) gives the first claim. At α_F^*, $\operatorname{diag}\theta - A_F = -Q_F - \lambda_F I$, whose eigenvalues are the numbers $\rho - \lambda_F$, exactly one of them 0. The sum of its principal minors of order $k-1$ is the elementary symmetric polynomial of degree $k-1$ in these eigenvalues, which is the product of the other $k-1$. With $l_{\alpha_F^*} \cdot r_{\alpha_F^*} = 1$, (a) gives both forms of K_F.

(c) The flow out of i equals the flow into i, so all $f_{i,i+1}(\alpha)$ equal one number φ, and their product $\prod_i l_{\alpha,i} a_i r_{\alpha,i+1} = \prod_i \alpha_i a_i$ gives $\varphi = (\prod_i \alpha_i a_i)^{1/k}$. The only spanning tree oriented toward i with nonzero weight is the cycle without the edge out of i, so $\operatorname{arb}(f(\alpha)) = \varphi^{k-1}$, and the symmetrized flow has conductance φ on each edge of the undirected cycle (for $k = 2$ one edge of conductance 2φ), so $\operatorname{tree}(f + f^\top) = k\varphi^{k-1}$. Now (a) gives $\det \Sigma_\alpha$. If $q_i = a_i$, then Q_F is a generator, $\lambda_F = 0$, $r_F = \mathbf{1}$ and $l_F = \alpha_F^*$ has constant flow $l_{F,i}a_i$, so $\alpha^*_{F,i} = \varphi^*/a_i$, $\prod_i \alpha^*_{F,i} = (\varphi^*)^k/\prod_i a_i$ and $(\mu \cdot r_F)(l_F \cdot \mathbf{1}) = \mu(F)$. □

On the complete graph the tree sum of Proposition 5.2(b) has a closed form. This is classical, and we include a short proof.

Lemma 5.12 (weighted Cayley formula) *For $x_1, \dots, x_k > 0$ and the conductances $c_{ij} = x_i x_j$,*

$$\operatorname{tree}(c) = x_1 x_2 \cdots x_k \, (x_1 + \cdots + x_k)^{k-2}.$$

With every $x_i = 1$ this is Cayley's count k^{k-2} of the spanning trees of the complete graph K_k.

Proof Put $X = \sum_i x_i$. The Laplacian of the conductances is $\operatorname{diag}(Xx_i) - xx^{\top}$, so without its first row and column it is $\operatorname{diag}(Xx_2, \dots, Xx_k) - x' x'^{\top}$ with $x' = (x_2, \dots, x_k)$. The matrix determinant lemma gives the determinant $X^{k-1} \prod_{i \geq 2} x_i \, (1 - \sum_{i \geq 2} x_i / X) = X^{k-2} \prod_i x_i$, and this is $\operatorname{tree}(c)$ by the matrix-tree theorem [6]. (For integer conductances this is the multigraph case, and both sides are polynomials in the c_{ij}.) □

Example 5.13 (the complete face) Let F be a face of $k \geq 2$ states with $q_{ij} = 1$ for all $i \neq j$ in F, and any exit rates, for example a face of $Q = J - dI$, where $q_i = d - 1$. Then F is reversible with $\nu = \mathbf{1}$, so by Proposition 5.2(b) we may take $r_\alpha = l_\alpha = \sqrt{\alpha}$ (entrywise), and the conductances $c_{ij}(\alpha) = \sqrt{\alpha_i \alpha_j}$ are products. Put $A = \sum_{i \in F} \sqrt{\alpha_i}$. Lemma 5.12 gives $\operatorname{tree}(c(\alpha)) = (\prod_i \alpha_i)^{1/2} A^{k-2}$, so for every $\alpha \in \Delta_F^\circ$

$$C_F(\alpha) = \frac{(\mu \cdot \sqrt{\alpha})\, A^{k/2}}{(4\pi)^{(k-1)/2} \, (\prod_i \alpha_i)^{3/4}} = \frac{(\mu \cdot \sqrt{\alpha})\, A}{\sqrt{\det(2\pi \Sigma_\alpha)}}, \qquad J_F(\alpha) = \sum_{i \in F} q_i \alpha_i - (A^2 - 1).$$

The second form of $C_F(\alpha)$ uses $\det \Sigma_\alpha = 2^{k-1} (\prod_i \alpha_i)^2 / \operatorname{tree}(c(\alpha))$. For $Q = J - dI$ the rate is $J_F(\alpha) = d - A^2$, the rate of Dupuis and Liu [19, Eq. (3.2)]. At the center $\alpha = \mathbf{1}/k$ we have $A^2 = k$ and $C_F = \mu(F) k^{k-1/2} (4\pi)^{-(k-1)/2}$, which is Corollary 5.3 with $\operatorname{tree}(q_F) = k^{k-2}$. With the uniform start, $dC_F(\alpha)$ is the constant $D(\alpha)$ of [33, Theorem 3.3 and Proposition 3.6], written there as a mass over a Gaussian volume. Section 7 uses this example.

5.5 A reducible face

The best composition of a reducible face can visit a state only once, with no Gaussian window in that direction. Then the coefficient $\frac{1}{2}$ of Remark 5.10 can fail, and in the next example it is 1 and then 0.

Example 5.14 (a directed 3-cycle) Let $d = 3$, $q_{12} = b > 2$, $q_{23} = q_{31} = 1$, all other rates 0, and $\mu_1 = 1$. From state 1 the chain moves only along $1 \to 2 \to 3 \to 1$, so the admissible faces are $\{1\}$, $\{1, 2\}$ (reducible, $\lambda = 1$) and $\{1, 2, 3\}$ (irreducible, $\lambda = 0$).

(i) Exactly $M_{\{1\}} = (1 - bh)^{N-1}$ and $M_{\{1,2\}} = bh(1-h)^{N-2}$, and the latter is attained only at $n = (1, N-1, 0)$. For the full face, $\alpha^* = (1, b, b)/(1 + 2b)$ and $K_{\{1,2,3\}} = (1 + 2b)^2 / (2\sqrt{3}\, \pi b)$.

(ii) The equation $M_{\{1\}} = M_{\{1,2\}}$ has exactly one root $T_N^{(1)}$ with $bh < 1$, and $T_N^{(1)} = \frac{1}{b-1}[L - \log L + \log \frac{b-1}{b}] + O(\log L / L)$. The coefficient of $\log \log N$ in the bracket is 1.

(iii) Every root $T \in [L/2, 2L]$ of $M_{\{1,2\}} = M_{\{1,2,3\}}$ satisfies $T = L + \log \frac{2\sqrt{3}\,\pi b^2}{(1+2b)^2} + O(1/L)$, so the coefficient of $\log\log N$ is 0.

Proof (i) A path with support $\{1,2\}$ is $1^{n_1}2^{n_2}$, of probability $(1-bh)^{n_1-1}\,bh\,(1-h)^{n_2-1}$. Moving one visit from state 2 to state 1 multiplies it by $(1-bh)/(1-h) < 1$, so the maximum is at $n_1 = 1$. On the full face Q is a generator, so $\lambda = 0$, $r = \mathbf{1}$ and α^* is the stationary law $(1, b, b)/(1+2b)$. Proposition 5.11(c) with $(a_1, a_2, a_3) = (b, 1, 1)$ and $q_i = a_i$ gives $\varphi^* = b/(1+2b)$ and the value of $K_{\{1,2,3\}}$.

(ii) The function $\log(M_{\{1,2\}}/M_{\{1\}}) = \log(bh) + (N-2)\log(1-h) - (N-1)\log(1-bh)$ has the derivative $1/h - (N-2)/(1-h) + (N-1)b/(1-bh) > 0$ in h, since $b > 1$ and $1-bh < 1-h$. It tends to $-\infty$ as $h \to 0$ and to $+\infty$ as $bh \to 1$, so it has exactly one zero. Its expansion $\log b + \log T - L + (b-1)T + O(T^2/N)$ gives $(b-1)T + \log T = L - \log b + O(T^2/N)$ at the zero, and with $\log T = \log L - \log(b-1) + O(\log L/L)$ this gives the formula.

(iii) For the full face $k = 3$, so Theorem 5.7 gives $\log M_{\{1,2,3\}} = -2L + \log T + \log K_{\{1,2,3\}} + O(1/L)$ on $[L/2, 2L]$, with the same power T^1 as $\log M_{\{1,2\}} = \log b + \log T - L - T + O(T^2/N)$. So $\log T$ cancels, and equating the two gives $T = L + \log b - \log K_{\{1,2,3\}} + O(1/L)$. □

The face $\{1,2\}$ has the power T^1 in its maximum where an irreducible pair would have $T^{1/2}$, and this changes the coefficient of $\log\log N$ at both crossings. The exact parts of (i) and the uniqueness of the root in (ii) are checked in Lean.

5.6 A weak field

Give each visit to state i the extra weight $e^{\xi_i/N}$, for a fixed vector $\xi \in \mathbb{R}^d$. Over N steps these weights multiply to $e^{\langle \xi, n\rangle/N}$, a factor of order 1, which is the order of the constants K_F. So such a field cannot change the first-order rule, but it can change the winner inside a tie window. For a composition n put

$$w_\xi(n) = e^{\langle \xi, n\rangle/N} P_N(n), \qquad Z_{N,\xi} = \sum_n w_\xi(n), \qquad \widetilde{P}_{N,\xi}(n) = \frac{w_\xi(n)}{Z_{N,\xi}},$$

and for a face F let $M_F^\xi = \max\{w_\xi(n) : \operatorname{supp} n = F\}$, which is $M_F(T, \xi/T)$ in the notation of Theorem 5.7. Write $\|\xi\| = \max_i |\xi_i|$. On the complete graph with the uniform start, $\widetilde{P}_{N,\xi}$ is the law of the counts of the Potts chain in a field of size ξ_i/N at each site in state i [33].

Proposition 5.15 (a weak field) *Fix Q, μ and a compact set $\mathcal{X} \subset \mathbb{R}^d$.*

(a) $\widetilde{P}_{N,\xi}$ is the law of the composition when the law of the path $X_1, \dots, X_N$ is multiplied by $\exp(N^{-1}\sum_{s=1}^N \xi_{X_s})$ and normalized. For every face F, $e^{-\|\xi\|} M_F \le M_F^\xi \le e^{\|\xi\|} M_F$.

(b) Let F satisfy (H). *Uniformly for $\xi \in \mathcal{X}$ and $1 \le T \le N^{1/3}$,*

$$M_F^\xi = e^{\langle \xi, \alpha_F^*\rangle} M_F\big(1 + O(1/T + T^2/N)\big),$$

and every maximizer $\hat{n}$ of w_ξ on F satisfies $\hat{n}/N = \alpha_F^* + O(1/T)$.

(c) *Fix ξ and $\tau^* > 0$, and let m^*, $\mathcal{F}^*$ and ℓ_F be as in Theorem 5.9, with every face in $\mathcal{F}^*$ irreducible. Then Theorem 5.9 holds for the modes of $\widetilde{P}_{N,\xi}$, with M_F replaced by M_F^ξ and ℓ_F by $\ell_F^\xi = \ell_F + \langle \xi, \alpha_F^* \rangle$. Thus it is the constants $K_F e^{\langle \xi, \alpha_F^* \rangle}$ that determine whether two field-shifted lines coincide.*

On a face satisfying (H) a weak field shifts the line by the field averaged over the quasi-ergodic law. On the complete graph α_F^* is uniform on F, and the tilt is the average of ξ over F (Corollary 7.11). By (a) the first-order rule of Theorem 4.2(iii) holds for $\widetilde{P}_{N,\xi}$ without change, and by (b) so does the location in Theorem 4.2(iv).

Proof (a) A path $x_1 \cdots x_N$ has probability $\mu_{x_1} \prod_{s<N} (I + hQ)_{x_s x_{s+1}}$, and $\sum_s \xi_{x_s} = \langle \xi, n \rangle$ depends on the path only through its composition n. So the multiplied law, summed over the paths with composition n, is $w_\xi(n)$. Since n/N is a probability vector, $|\langle \xi, n \rangle|/N \le \|\xi\|$.

(b) For a singleton the claim is exact. Let $k \ge 2$. For $T \ge 1$ the field ξ/T lies in the compact set $\mathcal{V} = \{s\xi : 0 \le s \le 1,\ \xi \in \mathcal{X}\}$, so Theorem 5.7 applies to $M_F^\xi = M_F(T, \xi/T)$ uniformly. The map $v \mapsto \lambda_F(v)$ is real-analytic with $\nabla \lambda_F(0) = -\alpha_F^*$, the derivative of a simple eigenvalue, so $T\lambda_F(\xi/T) = T\lambda_F - \langle \xi, \alpha_F^* \rangle + O(1/T)$. Also $\alpha_F^*(\xi/T) = \alpha_F^* + O(1/T)$ and $K_F(\xi/T) = K_F + O(1/T)$, since Perron vectors and C_F are real-analytic. Dividing by the face maximum of Theorem 5.7 at $v = 0$ gives (b).

(c) The proof of Theorem 5.9 uses (9) for the faces in $\mathcal{F}^*$ and the bounds of Theorem 4.2(ii) for the other faces. By (b), (9) holds for M_F^ξ with $\log K_F$ replaced by $\log K_F + \langle \xi, \alpha_F^* \rangle$, so (12) and part (i) of that theorem hold with ℓ_F^ξ. By (a), the bounds of Theorem 4.2(ii) hold for M_F^ξ with κ replaced by $\kappa + \|\xi\|$, so part (ii) holds. Part (iii) and the consequence use only (12) and part (ii). Finally the modes of $\widetilde{P}_{N,\xi}$ are the maximizers of w_ξ, and $\max_n w_\xi(n) = \max_F M_F^\xi$. □

5.7 Face maxima uniformly over a class

The constants of Theorem 5.7 depend on Q and μ. A statement about many chains at once needs them uniform over a compact class. We give this for reversible chains on a fixed graph. The proof uses only the compactness of the set $\mathcal{V}$ of fields below, and we fix this set because the companion paper [34] uses it.

Fix a connected graph $G = ([d], E)$ and numbers $0 < b \le R$ and $0 < \mu_* \le 1/d$. A generator Q is *on* G if $q_{ij} > 0$ exactly when $\{i, j\} \in E$. For a reversible irreducible Q with stationary law π put $w_{ij} = q_{ij} q_{ji}$ and $S = D_{\sqrt{\pi}}(-Q) D_{\sqrt{\pi}}^{-1}$, a symmetric matrix with $S_{ii} = q_i$ and $S_{ij} = -\sqrt{w_{ij}}$ on edges. Let $\mathcal{C}$ be the set of reversible irreducible generators on G with $\|S\|_2 \le R$ and $\sqrt{w_{ij}} \ge b$ on every edge, and let $\mathcal{M}$ be the set of starts with $\min_i \mu_i \ge \mu_*$. Put $U = 8R/b^2$ and $\bar{v} = 4R + 4UR^2$. For a pair (i, j) and $\delta \in \mathbb{R}$ let $v^{ij}(\delta)$ be the field with $v_i = \delta$, $v_j = 0$ and $v_l = -\bar{v}$ for $l \ne i, j$, and let $\mathcal{V}$ be the compact set of these fields with $|\delta| \le 3R$. We use $M_F(T, v)$, $\alpha_F^*(v)$ and $K_F(v)$ as defined before Theorem 5.7.

Proposition 5.16 (uniform face maxima on the class) *Let G induce a connected graph on the face F, with $k = |F| \ge 2$. There are N_0 and C, depending only on G, R, b and μ_*, such that for all $Q \in \mathcal{C}$, $\mu \in \mathcal{M}$ and $v \in \mathcal{V}$ the following hold.*

(a) For all $N \geq 2$ and $T > 0$ with $TR \leq N/2$, $M_F(T,v) \leq C(T/N)^{k-1} e^{-T\lambda_F(v)+CT^2/N}$.

(b) For $N \geq N_0$ and $1 \leq T \leq N^{1/3}$, we have $C^{-1} \leq K_F(v) \leq C$ and

$$M_F(T,v) = K_F(v)\, N^{-(k-1)} T^{(k-1)/2} e^{-T\lambda_F(v)}(1+E), \qquad |E| \leq C\Big(\frac{1}{T} + \frac{T^2}{N}\Big).$$

For a singleton, $M_{\{i\}}(T,v) = \mu_i(1 - Tq_i/N)^{N-1} e^{Tv_i}$ exactly.

Proof For fixed (Q,μ), part (b) is Theorem 5.7 with the compact set $\mathcal{V}$ of fields, whose constants already do not depend on $v \in \mathcal{V}$. Part (a) is (6), that is, Lemma 4.1 with the Perron tilt $\theta = q - v - \lambda_F(v)\mathbf{1} \in V_F$, and the singleton is the path i^N. It remains to show that no constant depends on (Q,μ) either.

The parameter set is compact. Let $\mathcal{P} = \mathcal{C} \times \mathcal{M} \times \mathcal{V}$. Since S is symmetric and similar to $-Q$, it is positive semidefinite, so $0 \leq q_l \leq R$ for every l and $\sqrt{w_{ij}} \leq R$ on every edge. On an edge $q_{ij} \geq b^2/R$, since $q_{ij}q_{ji} \geq b^2$ and $q_{ij} \leq R$, and off the edges $q_{ij} = 0$. Reversibility is the closed condition of Kolmogorov on cycle products, and $\|S\|_2 \leq R$ is closed since π is continuous. So $\mathcal{P}$ is compact, and every connected face is irreducible for every $Q \in \mathcal{C}$.

Joint regularity. The Perron root is simple, so $\lambda_F(v)$, the normalized Perron vectors and $\alpha_F^*(v)$ are real-analytic in (Q,v). The proof of Proposition 5.2(a), run with Q as an extra variable, makes $\theta(\alpha)$, r_α, l_α, Σ_α, J_F and C_F real-analytic in (α,Q), and C_F is linear in μ. So $K_F(v)$ is continuous and positive on $\mathcal{P}$, which gives its bounds, and there is $\varepsilon_0 > 0$ with $\alpha_{F,i}^*(v) \geq 2k\varepsilon_0$ on $\mathcal{P}$. Put $\mathcal{K}_0 = \{\alpha \in \Delta_F : \alpha_i \geq \varepsilon_0\ \forall i\}$.

The constants. They are the following, and each is continuous on $\mathcal{P}$ or on $\mathcal{P} \times \mathcal{K}_0$, or comes from a compactness argument run there. (i) The constant C_θ of Lemma 4.1 is built from the factor $b(z')$ of Lemma A.2 (not the edge bound b), which is continuous where the radius condition (A3) holds strictly, as it does on V_F. So it is continuous along the Perron tilt. (ii) In Theorem 5.4 the Perron root in (A3) at $\theta'(\alpha)$ is continuous and negative on $\mathcal{P} \times \mathcal{K}_0$, so one radius margin serves, and Lemma B.1 applies on the compact space $\mathcal{P} \times \mathcal{K}_0$, since the matrix $\mathcal{J}$ of Appendix A is continuous and positive definite and $\chi(\alpha,0) = (\mu \cdot r_\alpha)(l_\alpha \cdot \mathbf{1})/\alpha_1$ is continuous and positive. (iii) In Step 3 of the proof of Lemma 5.5 the lower bound for H comes from one word c^0 with at most $2k-1$ letters that walks along edges of G through every state of F. Its weight is at least $\mu_*(b^2/R)^{|c^0|-1}$, which bounds $H(T\alpha)$ below uniformly for $T \geq 1$ and $\alpha \in \mathcal{K}_0$. Step 4 uses only $q_{\max} \leq R$ and $n_{\min} \geq \varepsilon_0 N$. (iv) In Theorem 5.7 the bounds for Σ_α^{-1}, for the third derivatives of J_F and for two derivatives of $\log C_F$ are continuous on $\mathcal{P} \times \mathcal{K}_0$, which gives one radius $\rho_0 \leq \varepsilon_0$ and uniform c_U, C_U, C_1 and C_3. Away from the ball, $J_F(\beta) - v \cdot \beta - \lambda_F(v)$ is lower semicontinuous and positive on a compact subset of $\mathcal{P} \times \Delta_F$, hence at least 3η, and finitely many vectors $\theta_u = A_F u/u \in V_F$ serve, with constants bounded by (i). For bounded T, f_T is continuous and positive on $[1,T_2] \times \mathcal{P} \times \mathcal{K}_0$. □

6 The boundary layer at a facet

The local law of Theorem 5.4 holds on compact subsets of the open face. This section follows the law of the occupation times from the inside of a face F to one of its facets $G = F \setminus \{i\}$, where the time spent in i is small. We prove an exact formula for the law near the facet, which is the run expansion with the runs in i pulled out (Lemmas 6.1 and 6.2). It reduces the layer to one statement about a weighted chain on G, a local law with a small complex parameter. We prove that statement for every facet (Theorem 6.6), in closed form when G has one or two states (Lemma 6.7), and

the layer theorem (Theorem 6.9) follows for every face with at least two states. We compute the constants of the layer for every irreducible Q (Proposition 6.4). Their exponent matches the rate of the local law for every irreducible Q. Their prefactor matches the constant of the local law when Q is reversible (Proposition 6.5), and we do not prove the prefactor match without reversibility. The near-vertex theorem of Section 8 is the pair-face case of this picture on the lattice (Section 6.7).

6.1 Setting

Let F be an irreducible face with $k = |F| \geq 2$, let $i \in F$, and put $G = F \setminus \{i\}$. We assume that G satisfies (H), so G is irreducible and $\mu(G) > 0$. The start mass μ_i may be zero. For $j \in G$ put

$$c_j^- = q_{ji}, \qquad c_j^+ = q_{ij},$$

the rates into i and out of i. Both vectors are nonnegative and nonzero, since F is irreducible. For $z \in \mathbb{C}$ let

$$Q_G(z) = Q_G + z\, c^-(c^+)^\top, \qquad \mu^{(z)} = \mu_G + z\mu_i c^+, \qquad e^{(z)} = \mathbf{1} + zc^-,$$

and write $Q_G(z) = A(z) - \operatorname{diag} d(z)$ with $A(z)$ the off-diagonal part and $d(z) = q_G - z\, c^- \circ c^+$. A jump $j \to j'$ of $Q_G(z)$ of rate $zc_j^- c_{j'}^+$ stands for a visit to i that is entered from j and left to j'.

Occupation coordinates. Let E_F be the event of Lemma 2.4. On E_F let $s \in (0,\infty)^G$ be the times spent in the states of G, $S = \sum_{j\in G} s_j$, and $x = T - S$ the time spent in i. We write

$$f_F(s,x) = e^{-\langle q_G, s\rangle - q_i x} H_F(s,x), \qquad f_G(s) = e^{-\langle q_G, s\rangle} H_G(s),$$

where H_F and H_G are the sums H of Lemma 2.4 for the faces F and G. By that lemma f_F is the density of the occupation times on E_F in any $k-1$ of the coordinates, and $f_G(s)$ is the same density for G at total time S. In this notation Theorem 5.4 for the face G reads

$$f_G(S\beta) = S^{-(k-2)/2} C_G(\beta) e^{-SJ_G(\beta)} \big(1 + O_{\mathcal{K}}(1/S)\big) \tag{13}$$

uniformly for β in a compact $\mathcal{K} \subset \Delta_G^\circ$ and $S \geq 1$, with $C_G(\beta) = \mu_j$ and $J_G = q_j$ when $G = \{j\}$. The profile on the facet is $\beta = s/S \in \Delta_G^\circ$.

The weighted chain. For real $z \geq 0$ the matrix $Q_G(z)$ is Metzler and irreducible. We let $J_G(\beta, z)$ be its rate, the supremum in Proposition 3.3 with $Q_G(z)$ in place of Q_F, and $C_G(\beta, z)$ its local-law constant, which is Definition 5.1 for the off-diagonal part $A(z)$ with the start $\mu^{(z)}$ in place of μ and the end vector $e^{(z)}$ in place of $\mathbf{1}$, namely

$$C_G(\beta, z) = \frac{(\mu^{(z)} \cdot r_\beta(z))(l_\beta(z) \cdot e^{(z)})}{(2\pi)^{(k-2)/2} \sqrt{\det \Sigma_\beta(z)}}.$$

So $J_G(\beta, 0) = J_G(\beta)$ and $C_G(\beta, 0) = C_G(\beta)$.

The facet constants. Let $r_\beta = r_\beta(0)$ and $l_\beta = \beta / r_\beta$ be the vectors of Definition 5.1 for the face G, so $l_\beta \cdot r_\beta = 1$. Put

$$a_\beta = c^- \cdot l_\beta, \qquad b_\beta = c^+ \cdot r_\beta, \qquad \omega_\beta = a_\beta b_\beta, \qquad \pi^{\mathrm{in}}_{\beta,j} = \frac{c^-_j l_{\beta,j}}{a_\beta}, \qquad \pi^{\mathrm{out}}_{\beta,j} = \frac{c^+_j r_{\beta,j}}{b_\beta}.$$

Only ω_β is intrinsic, since a_β and b_β move in opposite ways when r_β is scaled. The law π^{in}_β is the law of the state from which the tilted chain enters i, and π^{out}_β the law of the state into which it leaves i. Let $\mathcal{L}_\beta$ be the Laplacian on G with the conductances $\kappa_{jj'} = l_{\beta,j} q_{jj'} r_{\beta,j'} + l_{\beta,j'} q_{j'j} r_{\beta,j}$ for $j \neq j'$, which are those of the additive reversibilization of the tilted generator $\widehat{Q}_\beta$. Its kernel is $\mathbb{R}\mathbf{1}$, since G is irreducible. Let $\mathcal{L}^+_\beta$ be its pseudo-inverse and

$$\gamma_\beta = q_i + \frac{\omega_\beta}{2} (\pi^{\mathrm{out}}_\beta - \pi^{\mathrm{in}}_\beta)^\top \mathcal{L}^+_\beta (\pi^{\mathrm{out}}_\beta - \pi^{\mathrm{in}}_\beta). \tag{14}$$

The quadratic form is the effective resistance of the network κ between the laws π^{in}_β and π^{out}_β. So $\gamma_\beta \geq q_i$, with equality exactly when the two laws agree. Finally put

$$\zeta = 2\sqrt{\omega_\beta S x}, \qquad z_* = \sqrt{\frac{x}{\omega_\beta S}}, \qquad \mathbf{B}_\beta(x, S) = e^{-\gamma_\beta x} \sqrt{\frac{\omega_\beta S}{x}}\, I_1(\zeta),$$

with the modified Bessel function I_1. We call $\mathbf{B}_\beta$ the Bessel kernel.

6.2 The exact representation

Lemma 6.1 (the runs in i pulled out) *For $z \in \mathbb{C}$ and $s \in (0,\infty)^G$ let*

$$f^{(z)}_G(s) = e^{-\langle d(z), s\rangle} \sum_{m \in \mathbb{Z}^G_{\geq 1}} W_z(m) \prod_{j \in G} \frac{s_j^{m_j - 1}}{(m_j - 1)!},$$

$$W_z(m) = \sum_w \mu^{(z)}_{w_1} \prod_{l < |w|} A(z)_{w_l w_{l+1}}\, e^{(z)}_{w_{|w|}},$$

the inner sum over the words w in G with no two equal neighbours and m_j letters j. Then $z \mapsto f^{(z)}_G(s)$ is entire, $f^{(0)}_G = f_G$, and for $x > 0$ and every $\rho > 0$

$$f_F(s,x) = e^{-q_i x} \sum_{K \geq 1} \frac{x^{K-1}}{(K-1)!} [z^K] f^{(z)}_G(s) \tag{15}$$

$$= e^{-q_i x} \frac{1}{2\pi \mathrm{i}} \oint_{|z| = \rho} e^{x/z} f^{(z)}_G(s)\, \frac{dz}{z^2}. \tag{16}$$

So $f^{(z)}_G$ is the killed density on the facet of the chain in which each visit to i is replaced by an instantaneous jump $j \to j'$ of rate $z c^-_j c^+_{j'}$, and z counts the visits. The K visits have total length x, with density $q_i^K x^{K-1} e^{-q_i x}/(K-1)!$, and each return carries a factor $c^+_{j'}/q_i$. So the powers of q_i cancel.

Proof Fix a word c in F with no equal neighbours, every letter of F, and $K = m_i(c) \ge 1$ letters i. Its term in $H_F(s,x)$ is $\mu_{c_1} \prod q_{c_l c_{l+1}}$ times $x^{K-1}/(K-1)!$ times $\prod_{j\in G} s_j^{m_j-1}/(m_j-1)!$. The letters i are isolated in c. We delete them and mark the places where they were. The result is a marked word c' in G, in which two equal neighbours occur exactly where c had $j\,i\,j$. A mark between letters j and j' carries $q_{ji}q_{ij'} = c_j^- c_{j'}^+$. A mark before the first letter j' carries $\mu_i c_{j'}^+$, and a mark after the last letter j carries c_j^-. Two neighbours without a mark are different and carry $q_{jj'}$. So the weight of c is the product of these weights. Between two different letters of c' there is at most one mark, and between two equal letters exactly one.

We group the marked words by their reduced word w, obtained by merging each block of equal letters of c' into one letter, and by the numbers $n_1, n_2, \dots$ of marks inside the blocks. If j forms ϱ blocks with n inner marks in all, it has $\varrho+n$ letters in c', and there are $\binom{n+\varrho-1}{\varrho-1}$ ways to spread the marks. Since $\binom{n+\varrho-1}{\varrho-1}/(n+\varrho-1)! = 1/(n!(\varrho-1)!)$,

$$\sum_{n\ge 0} \binom{n+\varrho-1}{\varrho-1} \frac{y^n s^{n+\varrho-1}}{(n+\varrho-1)!} = e^{ys}\,\frac{s^{\varrho-1}}{(\varrho-1)!}.$$

With $y = z c_j^- c_j^+$, one power of z for each mark, the inner marks give the factor $e^{z c_j^- c_j^+ s_j}$, which is the diagonal part of $d(z)$. The marks between different letters give $A(z)$, and the first and last marks give $\mu^{(z)}$ and $e^{(z)}$. Collecting the powers of z gives (15). The entries of $A(z)$, $\mu^{(z)}$ and $e^{(z)}$ are at most $q_{\max}(1+|z|q_{\max})$, $1+|z|q_{\max}$ and $1+|z|q_{\max}$ in modulus. So the bound $W(m) \le k^M q_{\max}^{M-1}$ of Section 2 holds for W_z with $q_{\max}$ replaced by $q_{\max}(1+|z|q_{\max})$, up to a factor $(1+|z|q_{\max})^2$, and the series for $f_G^{(z)}(s)$ converges locally uniformly in z. Hence it is entire. For $z=0$ no word carries a mark, and $f_G^{(0)}$ is the sum of Lemma 2.4 for G. Finally we expand $e^{x/z} = \sum_{n\ge 0} x^n z^{-n}/n!$ and integrate term by term on the circle. This gives $\sum_{n\ge 0}(x^n/n!)[z^{n+1}] f_G^{(z)}(s)$, which is (15). □

The discrete law has the same structure. The only change is that the visits to i come in blocks of steps, whose lengths form a composition of n_i.

Lemma 6.2 (the discrete form) *Let $hq_{\max} \le 1$, $\operatorname{supp} n = F$, $N' = N - n_i$, and let $\hat n$ be n restricted to G. For $z \in \mathbb{C}$ let*

$$P_{N'}^{(z)}(\hat n) = \sum_x \mu_{x_1}^{h,(z)} \prod_{t<N'} \bigl(I + hQ_G + zh^2 c^-(c^+)^\top\bigr)_{x_t x_{t+1}} e_{x_{N'}}^{h,(z)},$$

the sum over the words $x_1 \cdots x_{N'}$ in G with composition $\hat n$, where $\mu^{h,(z)} = \mu_G + zh\mu_i c^+$ and $e^{h,(z)} = \mathbf{1} + zhc^-$. Then $P_{N'}^{(0)}(\hat n)$ is the probability of the composition $\hat n$ for the chain run for N' steps, and

$$P_N(n) = \sum_{K\ge 1} \binom{n_i-1}{K-1} (1-hq_i)^{n_i-K}\,[z^K] P_{N'}^{(z)}(\hat n).$$

Proof We delete the maximal blocks of i from a path with composition n. What is left is a path $x_1 \cdots x_{N'}$ in G with composition $\hat n$. A block of length ℓ between $x_t = j$ and $x_{t+1} = j'$ replaces the factor $(I+hQ)_{jj'}$ by $hq_{ji}(1-hq_i)^{\ell-1} hq_{ij'}$. A block at the start replaces μ_{x_1} by $\mu_i(1-hq_i)^{\ell-1} hq_{ix_1}$, and a block at the end adds the factor $hq_{x_{N'}i}(1-hq_i)^{\ell-1}$. We count the blocks by z. If there are K blocks, the factors $(1-hq_i)^{\ell-1}$ give $(1-hq_i)^{n_i-K}$, and the block lengths form one of the $\binom{n_i-1}{K-1}$ compositions of n_i into K positive parts. □

6.3 Pair faces

For $k = 2$ the facet is one state and the weighted chain has no words to sum. The contour formula then gives the law in closed form.

Proposition 6.3 (pair faces) *Let $F = \{j, i\}$ with $q_{ji}, q_{ij} > 0$ and $\mu_j > 0$, and put $\omega = q_{ji}q_{ij}$ and $\zeta = 2\sqrt{\omega s x}$. For all $s, x > 0$*

$$f_F(s,x) = \mu_j e^{-q_j s - q_i x}\Big[\sqrt{\frac{\omega s}{x}}\, I_1(\zeta) + \Big(q_{ji} + \frac{\mu_i}{\mu_j} q_{ij}\Big) I_0(\zeta) + \frac{\mu_i}{\mu_j}\sqrt{\frac{\omega x}{s}}\, I_1(\zeta)\Big].$$

Here $\omega_\beta = \omega$, $\gamma_\beta = q_i$ and $f_G(s) = \mu_j e^{-q_j s}$, and for all $s, x > 0$

$$0 \le \frac{f_F(s,x)}{f_G(s)\mathbf{B}_\beta(x,s)} - 1 \le \Big(q_{ji} + \frac{\mu_i}{\mu_j} q_{ij}\Big)\Big(\frac{1}{\omega s} + \sqrt{\frac{x}{\omega s}}\Big) + \frac{\mu_i}{\mu_j}\,\frac{x}{s}.$$

This is the facet version of the formula of Pedler for the occupation times of a two-state chain [31], and Kovchegov, Meredith and Nir give Bessel densities of the same kind [28]. What is new is the reading as a layer with an explicit two-sided bound. For $x \le T/2$ the bound is $O(1/T + \sqrt{x/T})$, and the term $\sqrt{x/T}$ is attained, since the middle term of the formula has exactly this size.

Proof For $G = \{j\}$ the only word is j, so $f_G^{(z)}(s) = (\mu_j + z\mu_i q_{ij})(1 + z q_{ji}) e^{-(q_j - z\omega)s}$. With $z = z_* t$ and $z_* = \sqrt{x/(\omega s)}$ we have $x/z + \omega s z = \frac{\zeta}{2}(t + 1/t)$, and the generating function $e^{(\zeta/2)(t+1/t)} = \sum_{n\in\mathbb{Z}} I_n(\zeta) t^n$ [30, Eq. 10.35.1] gives

$$\frac{1}{2\pi \mathrm{i}} \oint e^{x/z + \omega s z} z^{n-2}\, dz = z_*^{n-1} I_{n-1}(\zeta) \qquad (n \in \mathbb{Z}). \tag{17}$$

We put this into (16) for $n = 0, 1, 2$ and use $I_{-1} = I_1$. This gives the formula. Each term is nonnegative, which gives the lower bound. For the upper bound we divide the last two terms by the first. This gives $(q_{ji} + \frac{\mu_i}{\mu_j} q_{ij}) z_* I_0(\zeta)/I_1(\zeta) + \frac{\mu_i}{\mu_j} x/s$. The lower bound $I_1(\zeta)/I_0(\zeta) \ge \zeta/(1 + \sqrt{1 + \zeta^2})$ of Amos [1] gives $I_0/I_1 \le (1 + \sqrt{1+\zeta^2})/\zeta \le 1 + 2/\zeta$, and $z_*(1 + 2/\zeta) = \sqrt{x/(\omega s)} + 1/(\omega s)$. For a pair $\beta = (1)$, the vectors of the facet are $r_\beta = l_\beta = 1$, so $\omega_\beta = \omega$ and $\pi_\beta^{\mathrm{in}} = \pi_\beta^{\mathrm{out}}$, which gives $\gamma_\beta = q_i$. □

6.4 The constants for every irreducible chain

The rate on F is homogeneous of degree one in the occupation times, and we write $J_F(s,x) = T J_F((s,x)/T)$. For $u \in (0,\infty)^G$ put

$$\Psi_\beta(u) = \sum_{j\in G} \beta_j \Big(q_j - \sum_{j' \in G\setminus\{j\}} q_{jj'} \frac{u_{j'}}{u_j}\Big), \qquad P_\beta(u) = (c^+ \cdot u) \sum_{j\in G} \frac{c_j^- \beta_j}{u_j}.$$

Since $\sum_j \beta_j(-(Q_G(z)u)_j/u_j) = \Psi_\beta(u) - z P_\beta(u)$, the rate of the weighted chain is $J_G(\beta, z) = \sup_u[\Psi_\beta(u) - z P_\beta(u)]$ for real $z \ge 0$.

Proposition 6.4 (the facet expansion of the rate) *Let F be irreducible, $i \in F$, and $G = F \setminus \{i\}$ irreducible.*

(i) For $s = S\beta$ with $\beta \in \Delta_G^\circ$ and $x > 0$,

$$J_F(s,x) = q_i x + \sup_{z>0}\big[S\,J_G(\beta,z) - x/z\big].$$

(ii) The map $z \mapsto J_G(\beta,z)$ is real-analytic near $z = 0$, jointly with β, and

$$J_G(\beta,0) = J_G(\beta), \qquad \partial_z J_G(\beta,0) = -\omega_\beta, \qquad \partial_z^2 J_G(\beta,0) = 2\omega_\beta(\gamma_\beta - q_i).$$

(iii) There are $\delta > 0$ and C such that, uniformly for β in a compact subset of Δ_G° and $0 < x \le \delta S$,

$$\Big|J_F(s,x) - \big(S\,J_G(\beta) + q_i x - 2\sqrt{\omega_\beta S x} + (\gamma_\beta - q_i)x\big)\Big| \le C\,x^{3/2}S^{-1/2}.$$

(iv) If $\nu_j q_{jj'} = \nu_{j'} q_{j'j}$ on F for some $\nu \in (0,\infty)^F$, then $\pi_\beta^{\mathrm{in}} = \pi_\beta^{\mathrm{out}}$, $\gamma_\beta = q_i$, $J_G(\beta,z) = J_G(\beta) - z\omega_\beta$ for all real $z \ge 0$, $\omega_\beta = \big(\sum_{j\in G}\sqrt{q_{ij}q_{ji}\beta_j}\big)^2$, and for all $s = S\beta$ and $x > 0$

$$J_F(s,x) = S\,J_G(\beta) + q_i x - 2\sqrt{\omega_\beta S x}.$$

So in the visit fraction α_i the rate has a square-root singularity $-2\sqrt{\omega_\beta \alpha_i}$ at the facet, and the time in i is charged at the rate γ_β, not q_i.

Proof (i) In the supremum that defines $J_F(s,x)$ we write $u = (u_G, \epsilon)$ with $\epsilon = u_i$. The terms with ϵ are $-\epsilon\sum_j s_j c_j^-/u_j - (x/\epsilon)(c^+ \cdot u_G)$, and both coefficients are positive. Their supremum over $\epsilon > 0$ is $-2\sqrt{xS\,P_\beta(u_G)}$. So $J_F(s,x) = q_i x + \sup_u[S\Psi_\beta(u) - 2\sqrt{xSP_\beta(u)}]$. Now $2\sqrt{xSP} = \inf_{z>0}[SzP + x/z]$, and we exchange the two suprema.

(ii) Write $u = e^\psi$. For real $z \ge 0$ the function $\psi \mapsto \Psi_\beta(e^\psi) - zP_\beta(e^\psi)$ is concave, since $P_\beta(e^\psi)$ is a positive combination of the convex functions $e^{\psi_m - \psi_j}$, and it does not change when a constant is added to ψ. At $z = 0$ its maximizer is $\psi_0 = \log r_\beta$, unique up to a constant (Proposition 3.3(b)). The Hessian of $\sum_{j\ne j'}\beta_j q_{jj'}e^{\psi_{j'}-\psi_j}$ at ψ_0 is the Laplacian with the conductances $\kappa_{jj'}$, because $\beta_j q_{jj'} r_{\beta,j'}/r_{\beta,j} = l_{\beta,j}q_{jj'}r_{\beta,j'}$. So the Hessian of Ψ_β at ψ_0 is $-\mathcal{L}_\beta$, which is negative definite on the complement of the constants. By the implicit function theorem on that complement, the critical point $\psi(z)$ is real-analytic in (β,z) near $(\beta,0)$, and for real $z \ge 0$ it is the maximizer by concavity. So $J_G(\beta,z) = \Psi_\beta(\psi(z)) - zP_\beta(\psi(z))$ is real-analytic. The envelope theorem gives $\partial_z J_G = -P_\beta(\psi(z))$, so $\partial_z J_G(\beta,0) = -(c^+\cdot r_\beta)(c^-\cdot l_\beta) = -\omega_\beta$. Differentiating the critical equation gives $-\mathcal{L}_\beta\psi'(0) = \nabla P$, with ∇P the gradient of $P_\beta(e^\psi)$ at ψ_0. Its entries are

$$\partial_{\psi_m} P = c_m^+ r_{\beta,m}\,(c^-\cdot l_\beta) - (c^+\cdot r_\beta)\,c_m^- l_{\beta,m} = \omega_\beta(\pi_{\beta,m}^{\mathrm{out}} - \pi_{\beta,m}^{\mathrm{in}}),$$

which sum to zero. So $\partial_z^2 J_G(\beta,0) = -\nabla P\cdot\psi'(0) = \nabla P^\top \mathcal{L}_\beta^+ \nabla P$, and this is $2\omega_\beta(\gamma_\beta - q_i)$ by (14).

(iii) By the proof of (i), $J_F(s,x) - q_i x = S\sup_\psi[\Psi_\beta(\psi) - 2\varepsilon P_\beta(\psi)^{1/2}]$ with $\varepsilon = \sqrt{x/S}$. The maximizer lies in a bounded set of ψ modulo constants, uniformly for β in a compact set, as in the proof of Proposition 3.3(b). Comparing its value with the value at ψ_0 gives $\Psi_\beta(\psi_0) - \Psi_\beta(\psi^*) \le 2\varepsilon\sup P_\beta^{1/2}$ on that set, and the strict concavity near ψ_0 then gives $\psi^* - \psi_0 = O(\varepsilon^{1/2})$. The critical equation $\nabla\Psi_\beta(\psi^*) = \varepsilon\nabla P_\beta(\psi^*)/\sqrt{P_\beta(\psi^*)}$ has a right side of size $O(\varepsilon)$, and $\nabla^2\Psi_\beta$ is invertible near ψ_0 on the complement of the constants, so $\psi^* - \psi_0 = O(\varepsilon)$. Now we expand to second order around ψ_0. With $g = \nabla(P^{1/2}) = \nabla P/(2\sqrt{\omega_\beta})$, the maximizer is $\psi^* = \psi_0 - 2\varepsilon\mathcal{L}_\beta^+ g + O(\varepsilon^2)$, and the value is $J_G(\beta) - 2\varepsilon\sqrt{\omega_\beta} + 2\varepsilon^2 g^\top\mathcal{L}_\beta^+ g + O(\varepsilon^3)$.

Here $2g^\top \mathcal{L}_\beta^+ g = \nabla P^\top \mathcal{L}_\beta^+ \nabla P/(2\omega_\beta) = \gamma_\beta - q_i$ by (ii). Multiplying by S gives (iii), since $S\varepsilon^3 = x^{3/2}S^{-1/2}$.

(iv) By Proposition 5.2(b) for the face G we may take $r_{\beta,j} = \sqrt{\beta_j/\nu_j}$ and $l_{\beta,j} = \sqrt{\beta_j\nu_j}$. Then $c_j^- l_{\beta,j} = q_{ji}\sqrt{\beta_j\nu_j} = \nu_i q_{ij}\sqrt{\beta_j/\nu_j} = \nu_i c_j^+ r_{\beta,j}$, so $\pi_\beta^{\mathrm{in}} = \pi_\beta^{\mathrm{out}}$ and $\gamma_\beta = q_i$. More is true. For every real $z \geq 0$ the matrix $Q_G(z)$ is reversible with respect to ν_G, since $\nu_j c_j^- c_{j'}^+ = \nu_i q_{ij} q_{ij'}$ is symmetric in j, j'. So by Proposition 5.2(b) its optimal vector is r_β for every z, and $J_G(\beta, z) = \Psi_\beta(r_\beta) - zP_\beta(r_\beta) = J_G(\beta) - z\omega_\beta$. Next $\omega_\beta = \nu_i\big(\sum_j q_{ij}\sqrt{\beta_j/\nu_j}\big)^2 = \big(\sum_j \sqrt{q_{ij}q_{ji}\beta_j}\big)^2$, using $q_{ji} = q_{ij}\nu_i/\nu_j$. Finally (i) gives $\sup_{z>0}[-Sz\omega_\beta - x/z] = -2\sqrt{\omega_\beta S x}$. □

For $1/T \ll x \ll T$ the Bessel kernel is $(4\pi)^{-1/2}(\omega_\beta S)^{1/4}x^{-3/4}\exp(2\sqrt{\omega_\beta S x} - \gamma_\beta x)(1 + O(1/\zeta))$, by $I_1(\zeta) = e^\zeta(2\pi\zeta)^{-1/2}(1 + O(1/\zeta))$ [30, Eq. 10.40.1]. So Proposition 6.4(iii) says that $f_G(s)\mathbf{B}_\beta(x, S)$ and the local law of F, which in occupation coordinates is $T^{-(k-1)/2}C_F(\alpha)e^{-TJ_F(\alpha)}$, have the same exponent up to $O(x^{3/2}S^{-1/2})$. For their powers of x and their constants to agree as well we need $C_F((1-\epsilon)\beta, \epsilon) \sim (4\pi)^{-1/2}\omega_\beta^{1/4}\epsilon^{-3/4}C_G(\beta)$ as $\epsilon \to 0$. The next proposition proves this for reversible Q.

6.5 Prefactor matching

Proposition 6.5 (prefactor matching) *Let Q be reversible on F with respect to ν, and let G satisfy (H), with $|G| \geq 1$. Uniformly for β in a compact subset of Δ_G°, as $\epsilon \to 0$,*

$$C_F((1-\epsilon)\beta, \epsilon) = (4\pi)^{-1/2}\,\omega_\beta^{1/4}\,\epsilon^{-3/4}\,C_G(\beta)\big(1 + O(\sqrt{\epsilon})\big),$$

with $C_G(\beta) = \mu_j$ when $G = \{j\}$.

Proof We use Proposition 5.2(b) at $\alpha = ((1-\epsilon)\beta, \epsilon)$. On G the vectors $r_\alpha = \sqrt{\alpha/\nu}$ and $l_\alpha = \sqrt{\alpha\nu}$ are $\sqrt{1-\epsilon}$ times r_β and l_β, and at i they are $O(\sqrt{\epsilon})$. So $(\mu\cdot r_\alpha)(l_\alpha\cdot\mathbf{1}) = (\mu_G\cdot r_\beta)(l_\beta\cdot\mathbf{1})(1 + O(\sqrt{\epsilon}))$, where $\mu_G \cdot r_\beta > 0$ by (H). Next $\prod_{j\in F}\alpha_j = \epsilon(1-\epsilon)^{k-1}\prod_{j\in G}\beta_j$. The conductances are $c_{jj'}(\alpha) = (1-\epsilon)c_{jj'}(\beta)$ on G, and at i they are $c_{ij}(\alpha) = \sqrt{\epsilon(1-\epsilon)}\sqrt{q_{ij}q_{ji}\beta_j}$. A spanning tree of F with m edges at i has weight $O(\epsilon^{m/2})$. The trees with one edge at i are a spanning tree of G and one edge $\{i, j\}$. So $\mathrm{tree}(c(\alpha)) = \sqrt{\epsilon}\,\sqrt{\omega_\beta}\,\mathrm{tree}(c_G(\beta))(1 + O(\sqrt{\epsilon}))$, by $\sum_j \sqrt{q_{ij}q_{ji}\beta_j} = \sqrt{\omega_\beta}$ from Proposition 6.4(iv). We put these into the formula for C_F and divide by the same formula for C_G, which has one factor $(4\pi)^{1/2}$ fewer. The powers of ϵ are $\epsilon^{1/4}$ from the tree and ϵ^{-1} from the product, which gives $\epsilon^{-3/4}$. For $G = \{j\}$ we use $C_{\{j\}} = \mu_j$ and tree $= 1$ on one vertex. □

The first amplitude correction. Put $c_1(\beta) = \partial_z \log C_G(\beta, z)$ at $z = 0$. For reversible Q,

$$c_1(\beta) = \frac{\mu_i b_\beta}{\mu_G \cdot r_\beta} + \frac{a_\beta}{l_\beta \cdot \mathbf{1}} + \frac{\omega_\beta}{2}\sum_{\{j,j'\}\subseteq G} R_{jj'}\pi_{\beta,j}\pi_{\beta,j'}, \tag{18}$$

with $\pi_\beta = \pi_\beta^{\mathrm{in}} = \pi_\beta^{\mathrm{out}}$ and $R_{jj'}$ the effective resistance for the conductances $l_{\beta,j}q_{jj'}r_{\beta,j'}$. To see this, recall from the proof of Proposition 6.4(iv) that $Q_G(z)$ is reversible with respect to ν_G for real $z \geq 0$, so that its vectors $r_\beta(z) = r_\beta$ and $l_\beta(z) = l_\beta$ do not

move with z. By Proposition 5.2(b) and the formula for $C_G(\beta,z)$ in Section 6.1, for real $z \geq 0$

$$C_G(\beta,z) = \frac{(\mu^{(z)} \cdot r_\beta)(l_\beta \cdot e^{(z)})}{(4\pi)^{(k-2)/2}} \, \frac{\sqrt{\mathrm{tree}(c(z))}}{\prod_{j\in G} \beta_j}, \qquad c_{jj'}(z) = l_{\beta,j} A(z)_{jj'} r_{\beta,j'}.$$

Since $l_{\beta,j} c_j^- = a_\beta \pi_{\beta,j}$ and $c_{j'}^+ r_{\beta,j'} = b_\beta \pi_{\beta,j'}$, we have $c_{jj'}(z) = c_{jj'}(0) + z\omega_\beta \pi_{\beta,j} \pi_{\beta,j'}$. The derivatives at $z = 0$ of the logarithms of $\mu^{(z)} \cdot r_\beta$ and $l_\beta \cdot e^{(z)}$ are the first two terms of (18). The third term is the derivative of $\frac{1}{2} \log \mathrm{tree}(c(z))$, by the rule of Kirchhoff that $\partial \log \mathrm{tree}(c)/\partial c_e = R_e$, which says that $c_e R_e$ is the weighted fraction of the spanning trees that contain the edge e. The function $C_G(\beta,z)$ of Theorem 6.6 below is holomorphic and agrees with this one for real $z \geq 0$, so its derivative at 0 is also given by (18).

6.6 The facet theorem

The layer needs the local law of the weighted chain $Q_G(z)$, uniformly for complex z near 0.

Theorem 6.6 (the weighted facet chain) *Let G satisfy* (H) *with $|G| = k-1 \geq 1$, let $i \notin G$ with $F = G \cup \{i\}$ irreducible, and let $\mathcal{K} \subset \Delta_G^\circ$ be compact. There are a number $z_0 > 0$, not given explicitly, and a constant C, both depending only on Q, μ, i and $\mathcal{K}$, and functions $J_G(\beta,z)$ and $C_G(\beta,z)$ on $\mathcal{K} \times \{|z| < 2z_0\}$, holomorphic in z, jointly continuous, real for real z, equal to the rate and the local-law constant of Section 6.1 for real $0 \leq z < 2z_0$, and with $C_G(\beta,0) = C_G(\beta) > 0$, such that uniformly for $\beta \in \mathcal{K}$, $S \geq 1$ and complex $|z| \leq z_0$*

$$f_G^{(z)}(S\beta) = S^{-(k-2)/2} \, C_G(\beta,z) \, e^{-S\, J_G(\beta,z)} \, (1+\varepsilon), \qquad |\varepsilon| \leq C/S. \tag{19}$$

For $|G| = 1$ this holds with $\varepsilon = 0$.

At $z = 0$ the display is the local law (13), that is Theorem 5.4 for the face G. The proof inherits every constant from $z = 0$ by continuity, which is why z_0 is not explicit, and for larger $|z|$ the complex critical point that it uses can cease to exist. For one or two states the word sum is a closed expression.

Lemma 6.7 (one and two states) *Theorem 6.6 holds when $|G| \leq 2$, with the functions J_G and C_G given in the proof.*

Proof For $G = \{j\}$ the proof of Proposition 6.3 gives $f_G^{(z)}(s) = (\mu_j + z\mu_i q_{ij})(1 + zq_{ji}) e^{-(q_j - z\omega)s}$, which is (19) with $\varepsilon = 0$, $J_G(\beta,z) = q_j - z\omega$ and $C_G(\beta,z) = (\mu_j + z\mu_i q_{ij})(1 + zq_{ji})$. Now let $G = \{1,2\}$, and write $a_{12}(z) = q_{12} + zc_1^- c_2^+$ and $a_{21}(z) = q_{21} + zc_2^- c_1^+$ for the entries of $A(z)$, $\varpi(z) = a_{12}(z) a_{21}(z)$, $p = \mu^{(z)}$ and $e = e^{(z)}$. A word in G with no equal neighbours alternates. The words that start with 1 are $1(21)^n$, with weight $p_1 \varpi^n e_1$ and

$(n+1,n)$ letters, and $(12)^n$, with weight $p_1\varpi^{n-1}a_{12}e_2$ and (n,n) letters, for $n\ge 1$. With $\xi = 2\sqrt{\varpi s_1 s_2}$ the power series of I_1 and I_0 give

$$\sum_{n\ge1} \frac{\varpi^n s_1^n s_2^{n-1}}{n!\,(n-1)!} = \sqrt{\frac{\varpi s_1}{s_2}}\, I_1(\xi), \qquad \sum_{n\ge1} \frac{(\varpi s_1 s_2)^{n-1}}{(n-1)!^2} = I_0(\xi).$$

We do the same for the words that start with 2 and obtain

$$f_G^{(z)}(s) = e^{-\langle d(z),s\rangle}\Big[\Big(p_1e_1\sqrt{\frac{\varpi s_1}{s_2}} + p_2e_2\sqrt{\frac{\varpi s_2}{s_1}}\Big)I_1(\xi) + \big(p_1a_{12}e_2 + p_2a_{21}e_1\big)I_0(\xi)\Big].$$

Both sides are entire in z, so the branch of the square root does not matter. Since G is irreducible, $\varpi(0) = q_{12}q_{21} > 0$. We choose z_0 so that $|\varpi(z) - \varpi(0)| \le \varpi(0)/2$ for $|z| \le 2z_0$ and take the principal branch of $\sqrt{\varpi}$. Then $\xi = 2S\sqrt{\beta_1\beta_2\varpi}$ at $s = S\beta$ satisfies $|\arg\xi| \le \pi/12$ and $|\xi| \ge cS$, uniformly on $\mathcal{K}$. The expansion $I_\nu(\xi) = e^\xi(2\pi\xi)^{-1/2}(1 + O(1/|\xi|))$ holds for $\nu = 0, 1$, uniformly in the sector $|\arg\xi| \le \pi/4$ and $|\xi| \ge 1$ [30, Eq. 10.40.1]. Since $(2\pi\xi)^{-1/2} = S^{-1/2}(4\pi)^{-1/2}(\varpi\beta_1\beta_2)^{-1/4}$, this gives (19) with

$$J_G(\beta,z) = \langle d(z),\beta\rangle - 2\sqrt{\beta_1\beta_2\varpi}, \qquad C_G(\beta,z) = \frac{U+V}{\sqrt{4\pi}\,(\varpi\beta_1\beta_2)^{1/4}},$$

where $U = p_1e_1\sqrt{\varpi\beta_1/\beta_2} + p_2e_2\sqrt{\varpi\beta_2/\beta_1}$ and $V = p_1a_{12}e_2 + p_2a_{21}e_1$, and with $\varepsilon = (U\varepsilon_1 + V\varepsilon_0)/(U+V)$, where $|\varepsilon_\nu| \le C/S$ are the errors of the two expansions. Both functions are holomorphic in z, jointly continuous and real for real z. At $z = 0$ we have $p = \mu_G$ and $e = \mathbf{1}$, so $U + V > 0$ because $\mu(G) > 0$. After shrinking z_0, $|U+V|$ is bounded below and $|U| + |V|$ above on $\mathcal{K}\times\{|z| \le 2z_0\}$, which gives $|\varepsilon| \le C/S$ once $cS \ge 1$. For $1 \le S \le 1/c$ the error ε is continuous on a compact set, so it is bounded, and we enlarge C. For real $z \ge 0$ the rate of the two-state matrix $Q_G(z)$ is $\sup_u[\beta_1(d_1 - a_{12}u_2/u_1) + \beta_2(d_2 - a_{21}u_1/u_2)] = \langle d,\beta\rangle - 2\sqrt{\beta_1\beta_2\varpi}$, by the inequality of the arithmetic and geometric means. The constant agrees with that of Section 6.1, because for real $z \ge 0$ both are the limit of $S^{1/2}e^{SJ_G(\beta,z)}f_G^{(z)}(S\beta)$. For the first this is the paragraph above. For the second it is Lemma A.4 on G for the weights $A(z) \ge 0$, $\mu^{(z)}$ and $e^{(z)}$, since $f_G^{(z)}(s) = e^{-\langle d(z),s\rangle}H_{A(z),\mu^{(z)},e^{(z)}}(s)$ in the notation of Appendix A. □

Facets with two or more states. Let $|G| = k - 1 \ge 2$. We number the states of G as $1,\dots,k-1$ and put $m = k-2$. For $z \in \mathbb{C}$ let $A'(z)$ be $A(z)$ without the row and column of state 1, and let X_z, g_z and b_z be the functions of Lemma A.3 on G for the weights $A(z)$, $\mu^{(z)}$ and $e^{(z)}$. Since c^-, c^+ and μ_i are nonnegative, the weights $A(|z|)$, $\mu^{(|z|)}$ and $e^{(|z|)}$ dominate these in the sense of that lemma. For fixed $\beta \in \Delta_G^\circ$ and $z' \in \mathbb{C}^m$ put

$$\Phi_z(z') = \beta_1 g_z(z') + \sum_{j\ge2}\beta_j z_j.$$

Let $\theta(\beta)$ be the tilt of the face G at β, let $\theta(\beta,z)$ be that of $Q_G(z)$ for real $z \ge 0$, and let $\mathcal{H}(\beta)$ be the Hessian $\mathcal{H}$ of the proof of Theorem 5.4 for the face G at $\alpha = \beta$. On $\mathbb{C}^m$ we use the norm $\|w\| = \max_j|w_j|$ with closed balls $\bar B(w,r)$, and $we^{\mathrm{i}\varphi}$ is the vector with entries $w_je^{\mathrm{i}\varphi_j}$. The proof of Theorem 5.4 for G gives, for every $\beta \in \Delta_G^\circ$,

(P1) $\varrho(A'(0) - \operatorname{diag}\theta'(\beta)) < 0$ and $g_0(\theta'(\beta)) = \theta_1(\beta)$, by (A3),
(P2) $\nabla\Phi_0(\theta'(\beta)) = 0$, and $\nabla^2\Phi_0(\theta'(\beta)) = \mathcal{H}(\beta)$ is real, symmetric and positive definite,
(P3) $\operatorname{Re}\Phi_0(\theta'(\beta)e^{\mathrm{i}\varphi}) \le \Phi_0(\theta'(\beta)) - \sum_{j\ge2}\beta_j\theta_j(\beta)(1-\cos\varphi_j)$ for $\varphi \in \mathbb{T}^m$,
(P4) $b_0(\theta'(\beta)) > 0$.

Also θ and $\mathcal{H}$ are continuous, by Proposition 5.2(a).

Lemma 6.8 (the complex critical point) *Let $|G| \geq 2$ and let $\mathcal{K} \subset \Delta_G^\circ$ be compact. There are $z_1 > 0$, $r_0 \geq r_1 > 0$, $\sigma_0 > 0$ and B_0, which depend only on Q, μ, i and $\mathcal{K}$, such that the following hold for every $\beta \in \mathcal{K}$.*

(a) $r_0 < \frac{1}{4} \min_j \theta_j(\beta)$, and $\varrho(A'(|z|) - \operatorname{diag} s') < 0$ for $|z| \leq z_1$ and real s' with $\|s' - \theta'(\beta)\| \leq 3r_0$.

(b) The functions $g_z(z')$ and $b_z(z')$ are holomorphic in (z, z') and bounded in modulus by B_0 on $\{|z| < z_1\} \times U_\beta$, where U_β is the set of z' with $|z_j| > \theta_j(\beta) - 3r_0$ for all j.

(c) For $|z| < z_1$ there is exactly one $\zeta'(\beta, z) \in \bar{B}(\theta'(\beta), r_1)$ with $\nabla \Phi_z(\zeta'(\beta, z)) = 0$. It is holomorphic in z and jointly continuous in (β, z), $\zeta'(\beta, 0) = \theta'(\beta)$, $\|\zeta'(\beta, z) - \theta'(\beta)\| \leq B_0|z|$, and $\zeta'(\beta, \bar{z}) = \overline{\zeta'(\beta, z)}$.

(d) The matrix $\mathcal{J}(\beta, z) = Z_c \nabla^2 \Phi_z(\zeta'(\beta, z)) Z_c$, where $Z_c = \operatorname{diag} \zeta'(\beta, z)$, is complex symmetric and jointly continuous, and $\mathcal{J}(\beta, 0) = \Theta' \mathcal{H}(\beta) \Theta' \geq \sigma_0 I$, where $\Theta' = \operatorname{diag} \theta'(\beta)$.

(e) For real $0 \leq z < z_1$ we have $\zeta'(\beta, z) = \theta'(\beta, z)$, $\Phi_z(\zeta'(\beta, z)) = \langle \beta, \theta(\beta, z) \rangle$, $\beta_1^2 \det \mathcal{J}(\beta, z) = \prod_{j \geq 2} \theta_j(\beta, z)^2 \det \Sigma_\beta(z)$ and $\beta_1 b_z(\zeta'(\beta, z)) = (\mu^{(z)} \cdot r_\beta(z))(l_\beta(z) \cdot e^{(z)})$.

Proof (a) The Perron root of a Metzler matrix is a continuous function of its entries. By (P1) the function $(\beta, y, s') \mapsto \varrho(A'(y) - \operatorname{diag} s')$ is negative on the compact set of the points $(\beta, 0, \theta'(\beta))$ with $\beta \in \mathcal{K}$. So it is negative on a neighbourhood of this set, and the neighbourhood contains the points with $\beta \in \mathcal{K}$, $0 \leq y \leq z_1$ and $\|s' - \theta'(\beta)\| \leq 3r_0$ when z_1 and r_0 are small. We also take $r_0 < \frac{1}{4} \min_{\mathcal{K}} \min_j \theta_j$, which is positive.

(b) Put $s'_\beta = \theta'(\beta) - 3r_0 \mathbf{1} > 0$. By (a) we may use Lemma A.3 with the dominating weights at $|z|$ and the radii s'_β. So $\mathcal{Z}' - A'(z)$ is invertible on $\{|z| \leq z_1\} \times U_\beta$, and there $|g_z| \leq \bar{g}(s'_\beta)$ and $|b_z| \leq \bar{b}(s'_\beta)$ for the functions $\bar{g}$ and $\bar{b}$ of the weights at $|z|$. The terms of their series do not decrease with $|z|$, so we may take the weights at z_1. Then the bounds are continuous in β, since $(\operatorname{diag} s' - A'(z_1))^{-1}$ is continuous where $\varrho(A'(z_1) - \operatorname{diag} s') < 0$. So they are bounded on $\mathcal{K}$. The weights are affine in z, so the last claim of Lemma A.3 gives the holomorphy.

(c) For $|z| \leq z_1/2$ and $z' \in \bar{B}(\theta'(\beta), r_0)$ the polydisc of radius $\min(z_1/2, r_0)$ about (z, z') lies in $\{|z| < z_1\} \times U_\beta$, since its points have $|z_j| \geq \theta_j - 2r_0$. So by (b) and Cauchy's estimates, the derivatives of $g_z(z')$ and $b_z(z')$ in (z, z') up to order 4 are bounded on $\{|z| \leq z_1/2\} \times \bar{B}(\theta'(\beta), r_0)$ by a constant B_1 that does not depend on β. Let $F(z, z') = \nabla \Phi_z(z') = \beta_1 \nabla g_z(z') + \beta'$. By (P2), $F(0, \theta') = 0$ and $D_{z'} F(0, \theta') = \mathcal{H}(\beta)$, whose inverse has norm at most B_2 on $\mathcal{K}$ by continuity. Put $T_z(z') = z' - \mathcal{H}(\beta)^{-1} F(z, z')$. For $|z| \leq z_1/2$, $r \leq r_0$ and $z' \in \bar{B}(\theta', r)$ we get

$$\|D_{z'} T_z(z')\| = \|\mathcal{H}^{-1}(D_{z'} F(0, \theta') - D_{z'} F(z, z'))\| \leq B_3(|z| + r), \qquad \|T_z(\theta') - \theta'\| \leq B_3|z|,$$

with B_3 depending only on m, B_1 and B_2. Choose $r_1 \leq r_0$ and $z_2 \leq z_1/2$ with $B_3(z_2 + r_1) \leq \frac{1}{2}$ and $B_3 z_2 \leq \frac{1}{2} r_1$. For $|z| \leq z_2$ the map T_z sends $\bar{B}(\theta', r_1)$ into itself and is a contraction with constant $\frac{1}{2}$ there. So $F(z, \cdot)$ has exactly one zero $\zeta'(\beta, z)$ in this ball, the fixed point of T_z, and $\|\zeta' - \theta'\| \leq 2\|T_z(\theta') - \theta'\| \leq 2B_3|z|$. The iterates $T_z^n(\theta')$ are holomorphic in z, continuous in (β, z) and converge uniformly, which gives the regularity of ζ', and $\zeta'(\beta, 0) = \theta'$. The data A_G, $c^\pm$, μ and β are real, so $F(\bar{z}, \bar{w}) = \overline{F(z, w)}$. Hence $\overline{\zeta'(\beta, z)}$ is a zero of $F(\bar{z}, \cdot)$ in the ball, and it equals $\zeta'(\beta, \bar{z})$. We rename z_2 as z_1 and enlarge B_0 to at least $2B_3$.

(d) The Hessian $\nabla^2\Phi_z$ is symmetric and Z_c is diagonal, so $\mathcal{J}$ is complex symmetric, and it is continuous by (b) and (c). At $z=0$, $Z_c=\Theta'$ and (P2) gives $\mathcal{J}(\beta,0)=\Theta'\mathcal{H}(\beta)\Theta'$, which is positive definite and continuous in β. So its least eigenvalue has a positive minimum σ_0 on $\mathcal{K}$.

(e) Let $0\le z<z_1$ be real and write $\zeta'=\zeta'(\beta,z)$. By (c), ζ' is real, and it is positive since $\|\zeta'-\theta'\|\le r_0<\frac14\min_j\theta_j$. The matrix $A(z)=A(|z|)$ is nonnegative, and its graph contains that of A_G, so it is strongly connected. By (a) and the first fact, $X_z(\zeta')\ge0$, and as in the proof of Theorem 5.4 the vector $\tilde r=(1,X_z(\zeta')b_1)\ge0$ lies in the kernel of $A(z)-\operatorname{diag}\tilde\theta$ with $\tilde\theta=(g_z(\zeta'),\zeta')$. By the second fact, $\tilde r>0$ and 0 is the Perron root. Let $\tilde l>0$ be the left Perron vector with $\tilde l\cdot\tilde r=1$, and $\pi=\tilde l\circ\tilde r$. With $\tilde R=\operatorname{diag}\tilde r$, the matrix $\tilde R^{-1}(A(z)-\operatorname{diag}\tilde\theta)\tilde R$ is an irreducible generator with stationary law π, so Lemma A.1 gives $\nabla\lambda(\tilde\theta)=-\pi$ for $\lambda(y)=\varrho(A(z)-\operatorname{diag}y)$. For real z' near ζ' the same argument gives $\lambda(g_z(z'),z')=0$, and differentiating at ζ' gives $\pi_1\partial_jg_z(\zeta')+\pi_j=0$. The critical equation $\beta_1\partial_jg_z(\zeta')+\beta_j=0$ then gives $\pi_j/\pi_1=\beta_j/\beta_1$ for $j\ge2$, and since π and β both sum to 1, $\pi=\beta$. So $\tilde l=\beta/\tilde r$, and $\tilde l^{\top}(A(z)-\operatorname{diag}\tilde\theta)=0$ says that the derivative of $u\mapsto\sum_{j\ne j'}\beta_jA(z)_{jj'}u_{j'}/u_j$ vanishes at $\tilde r$, as before (7). This function is convex in $\log u$, so $\tilde r$ is a minimizer, and Proposition 3.3(b) for $Q_G(z)$ shows that $\tilde r$ is a multiple of $r_\beta(z)$. Hence $\tilde\theta=A(z)\tilde r/\tilde r=\theta(\beta,z)$, and $\Phi_z(\zeta')=\beta_1\theta_1+\sum_{j\ge2}\beta_j\theta_j=\langle\beta,\theta(\beta,z)\rangle$. The last two claims are the identities of Lemma A.4 on G for $A(z)$, $\mu^{(z)}$ and $e^{(z)}$ at $\alpha=\beta$, since $\mathcal{J}(\beta,z)$ is the Hessian $\mathcal{H}$ of that lemma multiplied on both sides by $\operatorname{diag}\theta'(\beta,z)$. □

Proof of Theorem 6.6 For $|G|=1$ this is Lemma 6.7. Let $|G|\ge2$, and let z_1, r_0, r_1, σ_0 and B_0 be as in Lemma 6.8. For $\beta\in\mathcal{K}$ and $|z|<z_1$ write $\zeta'=\zeta'(\beta,z)$ and $\rho_j=|\zeta_j|$, and put

$$\Psi(\beta,z,\varphi)=\Phi_z(\zeta'e^{\mathrm{i}\varphi}),\qquad \chi(\beta,z,\varphi)=b_z(\zeta'e^{\mathrm{i}\varphi})\prod_{j\ge2}\zeta_je^{\mathrm{i}\varphi_j}.$$

Step 1: the constants. Since θ_j is real, $|\rho_j-\theta_j|\le|\zeta_j-\theta_j|\le r_1$. So the points $\zeta'e^{\mathrm{i}\varphi}$ lie in U_β, and by Lemma 6.8(b), (c) the φ-derivatives of Ψ and χ are jointly continuous on the compact set $P_1\times\mathbb{T}^m$, where $P_1=\mathcal{K}\times\{|z|\le z_1/2\}$. Let B_Ψ and B_a bound those of Ψ up to order 4 and those of χ up to order 2. Since $\partial/\partial\varphi_j=\mathrm{i}w_j\,\partial/\partial w_j$ at $w=\zeta'e^{\mathrm{i}\varphi}$ and $\nabla\Phi_z(\zeta')=0$, we have $\nabla_\varphi\Psi(\beta,z,0)=0$ and $-\nabla^2_\varphi\Psi(\beta,z,0)=\mathcal{J}(\beta,z)$, as in the proof of Theorem 5.4. Fix $\varepsilon\in(0,1)$ with $m^3B_\Psi\varepsilon\le\frac34\sigma_0$, and put $2\eta=\beta_{\min}\theta_{\min}(1-\cos\varepsilon)$, where $\beta_{\min}$ and $\theta_{\min}$ are the least entries of β and $\theta(\beta)$ over $\mathcal{K}$. By (P3), $\operatorname{Re}\Psi(\beta,0,\varphi)-\Psi(\beta,0,0)\le-2\eta$ for $\|\varphi\|\ge\varepsilon$. By uniform continuity on $P_1\times\mathbb{T}^m$ we can choose $z_0\in(0,z_1/4]$ such that for $\beta\in\mathcal{K}$ and $|z|\le2z_0$

(i) $\operatorname{Re}\mathcal{J}(\beta,z)\ge\frac12\sigma_0I$, since $\|\mathcal{J}(\beta,z)-\mathcal{J}(\beta,0)\|\le\frac12\sigma_0$ in the operator norm and $\mathcal{J}(\beta,0)$ is real,

(ii) $|\chi(\beta,z,0)|\ge c_\chi$, where $2c_\chi>0$ is the minimum over $\mathcal{K}$ of $\chi(\beta,0,0)=b_0(\theta')\prod_{j\ge2}\theta_j$, which is positive by (P4),

(iii) $\operatorname{Re}\Psi(\beta,z,\varphi)-\operatorname{Re}\Psi(\beta,z,0)\le-\eta$ for $\|\varphi\|\ge\varepsilon$.

The numbers z_0, ε, η and c_χ depend only on Q, μ, i and $\mathcal{K}$.

Step 2: the torus through the critical point. Let $\beta\in\mathcal{K}$, $|z|\le z_0$ and $S\ge1$. Since $\|\rho'-\theta'\|\le r_0$, Lemma 6.8(a) shows that the radii ρ' are admissible in Lemma A.3 with the dominating weights at $|z|$. With $t=S\beta$ it gives

$$e^{S\langle d(z),\beta\rangle}f_G^{(z)}(S\beta)=\frac{1}{(2\pi)^m}\int_{\mathbb{T}^m}\exp\bigl(S\Phi_z(w)\bigr)\,b_z(w)\prod_{j\ge2}w_j\,d\varphi,\qquad w_j=\rho_je^{\mathrm{i}\varphi_j}.$$

The translation of φ by the arguments of the ζ_j preserves the measure and turns w into $\zeta' e^{\mathrm{i}\varphi}$. So the integral is $\int_{\mathbb{T}^m} e^{S\Psi}\chi\, d\varphi$. No contour is moved. The torus representation holds at every admissible radius, and we use it at the radii ρ' of the complex critical point.

Step 3: the Laplace lemma. By Step 1, Lemma B.3 applies on $\mathcal{P} = \mathcal{K} \times \{|z| \le z_0\}$ with $\sigma_{\min} = \frac{1}{2}\sigma_0$, ε and η. Since $\Psi(\beta, z, 0) = \Phi_z(\zeta')$, it gives (19) with

$$J_G(\beta, z) = \langle d(z), \beta\rangle - \Phi_z(\zeta'), \qquad C_G(\beta, z) = (2\pi)^{-m/2} \det \mathcal{J}(\beta, z)^{-1/2}\, b_z(\zeta') \prod_{j\ge 2} \zeta_j,$$

and with $|\varepsilon| \le C/(c_\chi S)$, where C is the constant of Lemma B.3.

Step 4: the functions. We define J_G and C_G on $\mathcal{K} \times \{|z| < 2z_0\}$ by these formulas. Since ζ' is holomorphic in z and $g_z(z')$ and $b_z(z')$ are holomorphic in (z, z'), the numbers $\Phi_z(\zeta')$, $b_z(\zeta')$, $\prod_j \zeta_j$ and the entries of $\mathcal{J}(\beta, z)$ are holomorphic in z and jointly continuous. By (i) the matrix $\mathcal{J}(\beta, z)$ lies in the set $\mathcal{S}_m$ of Appendix B, so $\det \mathcal{J}(\beta, z)^{-1/2}$ is continuous, and as a continuous square root of a holomorphic function without zeros it is holomorphic in z. For real z, ζ' is real by Lemma 6.8(c), so J_G and C_G are real, and $\mathcal{J}(\beta, z)$ is real and positive definite, so the branch is the positive root. For real $0 \le z < 2z_0$, Lemma 6.8(e) gives $J_G(\beta, z) = \langle d(z) - \theta(\beta, z), \beta\rangle$, which is the rate of $Q_G(z)$ by the identity $J = \langle q - \theta, \alpha\rangle$ after Definition 5.1, since the diagonal of $Q_G(z)$ is $-d(z)$. It also gives $C_G(\beta, z) = (\mu^{(z)} \cdot r_\beta(z))(l_\beta(z) \cdot e^{(z)})/((2\pi)^{m/2}\sqrt{\det \Sigma_\beta(z)})$, the constant of Section 6.1. At $z = 0$ it is $C_G(\beta) > 0$, by (H). □

The proof does not use the irreducibility of F, which only makes c^- and c^+ nonzero.

We take the functions of Theorem 6.6 and define for $\beta \in \mathcal{K}$ and $x, S > 0$ the generalized kernel

$$\mathbf{K}_\beta(x, S) = e^{-q_i x}\, \frac{1}{2\pi \mathrm{i}} \oint_{|z|=z_*} e^{x/z}\, \frac{C_G(\beta, z)}{C_G(\beta, 0)}\, e^{S(J_G(\beta) - J_G(\beta, z))}\, \frac{dz}{z^2},$$

for $z_* < 2z_0$. The integral does not depend on the radius below $2z_0$.

Theorem 6.9 (the facet layer) *Let F be irreducible with $k = |F| \ge 2$, let $i \in F$, let $G = F \setminus \{i\}$ satisfy* (H)*, and let $\mathcal{K} \subset \Delta_G^\circ$ be compact. Then there are $\delta > 0$, T_0 and C such that the following hold uniformly for $\beta \in \mathcal{K}$, $T \ge T_0$ and $0 < x \le \delta T$, with $S = T - x$ and $s = S\beta$.*

(a) $f_F(s, x) = f_G(s)\, \mathbf{K}_\beta(x, S)\, (1 + E)$ *with* $|E| \le C/T$.

(b) $f_F(s, x) = f_G(s)\, \mathbf{B}_\beta(x, S)\, e^E$ *with* $|E| \le C\big(1/T + \sqrt{x/T}\,(1 + x)\big)$.

(c) If Q is reversible on F, then $\gamma_\beta = q_i$, (b) holds with $|E| \le C(1/T + \sqrt{x/T})$, and

$$f_F(s, x) = f_G(s)\, \mathbf{B}_\beta(x, S)\Big(1 + c_1(\beta)\, z_* \frac{I_0(\zeta)}{I_1(\zeta)} + E'\Big), \qquad |E'| \le C\, \frac{1 + x}{T}.$$

For $k = 2$, (a) holds with $E = 0$, and Proposition 6.3 gives (b) for all $x > 0$ with an explicit bound.

When Q is not reversible our bound gives the relative form $f_F = f_G \mathbf{B}_\beta(1 + O(1/T + \sqrt{x/T}(1 + x)))$ only for $x \le CT^{1/3}$, and beyond that only the logarithmic form (b) and the kernel of (a). The cause is the cubic term $S\,\partial_z^3 J_G(\beta, 0) z^3/6$ at $z \approx z_*$, of size $x^{3/2}T^{-1/2}$, which the generalized kernel keeps and the Bessel kernel does not. For

the non-reversible cycle $q_{12} = q_{23} = q_{31} = 2$, $q_{21} = q_{32} = q_{13} = 1$ with $G = \{1, 2\}$, $\beta = (\frac{1}{2}, \frac{1}{2})$ and $x = T/20$, the ratio $f_F/(f_G \mathbf{B}_\beta)$ is 19.6, 265 and 48268 at $T = 800$, 1600 and 3200, so the relative form does fail there.

Proof of (a) Theorem 6.6 gives (19), with $\varepsilon = 0$ for $k = 2$. Take $\delta \le \frac{1}{2}$, so $S \ge T/2$. Write $\Lambda(z) = S^{-(k-2)/2} C_G(\beta, z) e^{-S J_G(\beta, z)}$. By (19), $f_G(s) = \Lambda(0)(1 + O(1/S))$, and by (16), for every $\rho \le z_0$,

$$e^{q_i x} f_F(s, x) = \mathcal{A} + \mathcal{D}, \qquad \mathcal{A} = \frac{1}{2\pi \mathrm{i}} \oint_{|z|=\rho} e^{x/z} \Lambda(z) \frac{dz}{z^2}, \qquad |\mathcal{D}| \le \frac{C}{S} \mathcal{A}^{\mathrm{abs}},$$

where $\mathcal{A}^{\mathrm{abs}}$ is the integral of $|e^{x/z}\Lambda(z)||z|^{-2}$ against $|dz|/(2\pi)$. By definition $\mathbf{K}_\beta(x, S) = e^{-q_i x}\mathcal{A}/\Lambda(0)$. So it suffices to show that $\mathcal{A} > 0$ and $\mathcal{A}^{\mathrm{abs}} \le C'\mathcal{A}$. Put

$$\Xi(z) = \frac{C_G(\beta, z)}{C_G(\beta, 0)} e^{S(J_G(\beta) - J_G(\beta, z) - \omega_\beta z)},$$

so that $\Lambda(z)/\Lambda(0) = e^{S\omega_\beta z}\Xi(z)$. By Proposition 6.4(ii), the holomorphy of C_G, and Cauchy estimates on the compact set $\mathcal{K} \times \{|z| \le \frac{3}{2} z_0\}$, the function $R = \log \Xi$ is holomorphic on $|z| \le z_0$ after shrinking z_0, and

$$R(z) = c_1 z - S\omega_\beta(\gamma_\beta - q_i) z^2 + O(|z|^2 + S|z|^3), \qquad |R''(z)| \le C(1 + S).$$

We take δ so small that $z_* \le \sqrt{2\delta/\omega_{\min}} \le z_0/2$, where $\omega_{\min} = \min_{\mathcal{K}} \omega_\beta > 0$.

Few visits, $\zeta \le Z_0$ for a fixed large Z_0. Take $\rho = \max(z_*, 1/S)$, which is at most C/S. On the circle $|x/z| \le Z_0/2$, $|S\omega_\beta z| \le C$ and $|\Xi(z) - 1| \le C/S$. Let $\mathcal{A}_0$ be $\mathcal{A}$ with $\Xi = 1$. By (17) it equals $\Lambda(0)\sqrt{\omega_\beta S/x}\, I_1(\zeta)$, which is at least $\Lambda(0)\omega_\beta S$, because $\sqrt{\omega S/x}\, I_1(\zeta) = \omega S \sum_{n \ge 0} (\zeta/2)^{2n}/(n!(n+1)!)$. Also $|\mathcal{A} - \mathcal{A}_0| \le C\Lambda(0)\rho^{-1}/S \le C\Lambda(0)$ and $\mathcal{A}^{\mathrm{abs}} \le C\Lambda(0)\rho^{-1} \le C\Lambda(0)S$. So $\mathcal{A} \ge \Lambda(0)(\omega_\beta S - C) > 0$ and $\mathcal{A}^{\mathrm{abs}} \le C'\mathcal{A}$ for S large.

Many visits, $\zeta \ge Z_0$. Let $\lambda(z) = x/z + S\omega_\beta z + R(z)$, so that $e^{x/z}\Lambda(z) = \Lambda(0)e^{\lambda(z)}$. It is real on the real axis, and $\lambda'(z) = -x/z^2 + S\omega_\beta + O(1 + S|z|)$. By the implicit function theorem it has a real critical point $\rho_s = z_*(1 + O(\sqrt{x/S} + 1/\zeta))$, and we take $\rho = \rho_s$. For $z = \rho e^{\mathrm{i}\phi}$,

$$\operatorname{Re}\lambda(z) - \lambda(\rho) = (x/\rho + S\omega_\beta \rho)(\cos\phi - 1) + \operatorname{Re}[R(z) - R(\rho)].$$

Since R is real on the real axis, $\operatorname{Re}[R'(\rho)(z - \rho)] = R'(\rho)\rho(\cos\phi - 1)$, and $|R''| \le C(1 + S)$ bounds the rest by $2C(1 + S)\rho^2(1 - \cos\phi)$. Also $|R'(\rho)|\rho \le C(\rho + S\rho^2)$. Now $S\rho^2 \le Cx$, $x/\rho + S\omega_\beta\rho \ge \zeta$, and $x/\zeta \le C\sqrt{\delta}$. So for δ small $\operatorname{Re}\lambda(z) - \lambda(\rho) \le -\frac{\zeta}{2}(1 - \cos\phi)$, and

$$\mathcal{A}^{\mathrm{abs}} \le \Lambda(0)\rho^{-1}e^{\lambda(\rho)} \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{-\frac{\zeta}{2}(1-\cos\phi)} d\phi \le C\Lambda(0)\rho^{-1}e^{\lambda(\rho)}\zeta^{-1/2}.$$

For the lower bound we write $\mathcal{A} = \Lambda(0)\rho^{-1}\frac{1}{2\pi}\int e^{\zeta\ell(\phi)}e^{-\mathrm{i}\phi}d\phi$ with $\ell(\phi) = \lambda(\rho e^{\mathrm{i}\phi})/\zeta$. Here $\ell(0)$ is real, $\ell'(0) = 0$, $-\ell''(0) = \rho^2\lambda''(\rho)/\zeta$ is real and lies in $[\frac{1}{2}, 2]$, the derivatives of ℓ up to order four are bounded uniformly in all parameters, and the gap above holds. The parameters (β, x, S) do not range over a compact set, so we use the uniform form of Lemma B.1 in Remark B.2, with $\sigma_{\min} = \frac{1}{2}$ and the gap $\frac{1}{2}(1 - \cos\varepsilon)$. With $m = 1$, $\Psi = \ell$, $a = e^{-\mathrm{i}\phi}$ and ζ in place of T it gives $\mathcal{A} = \Lambda(0)\rho^{-1}e^{\lambda(\rho)}(2\pi\zeta(-\ell''(0)))^{-1/2}(1 + O(1/\zeta))$, which is positive and at least $c\,\mathcal{A}^{\mathrm{abs}}$. □

Proof of (b) *and* (c) We use two facts on Bessel functions besides (17). For integers $n \ge 1$, $I_n(\zeta) = \frac{1}{\pi}\int_0^\pi e^{\zeta\cos t}\cos(nt)\,dt$ [30, Eq. 10.32.3]. So $I_n \le I_1$ for all $\zeta > 0$, and $|1 - \cos nt -$

$(1-\cos t)| \le n^2t^2/2$ and a Laplace estimate give $|I_1(\zeta) - I_n(\zeta)| \le Cn^2 I_1(\zeta)/\zeta$ for $\zeta \ge 1$. Also $I_0/I_1 \le 1 + 2/\zeta$, as in the proof of Proposition 6.3.

The reversible case. By Proposition 6.4(iv) and holomorphy, $J_G(\beta, z) = J_G(\beta) - z\omega_\beta$ on the disc, so $\Xi(z) = C_G(\beta, z)/C_G(\beta, 0) = 1 + c_1 z + \sum_{n\ge 2} c_n z^n$ with $|c_n| \le C z_0^{-n}$. By (17) the term $c_n z^n$ contributes $c_n z_*^n I_{n-1}(\zeta)/I_1(\zeta)$ relative to the main term $z_*^{-1} I_1(\zeta)$. For $n \ge 2$ this is at most $|c_n| z_*^n$, and the sum over $n \ge 2$ is $O(z_*^2) = O(x/T)$, since $z_* \le z_0/2$. The term $n = 1$ is $c_1 z_* I_0(\zeta)/I_1(\zeta) = O(z_* + 1/(\omega_\beta S)) = O(\sqrt{x/T} + 1/T)$. With (a) this proves both claims of (c), for all $0 < x \le \delta T$.

The general case, $x \le 1$. Write $\Xi(z) = e^{-S\omega_\beta(\gamma_\beta - q_i)z^2}(1 + c_1 z + R_2(z))$ with $|R_2(z)| \le C(|z|^2 + S|z|^3)$. On $|z| = z_*$ the Gaussian factor has modulus at most $e^{(\gamma_\beta - q_i)x} \le C$. We expand it in powers of z^2 and use (17). With $y = (\gamma_\beta - q_i)x$ the term 1 gives the relative factor $\sum_{m\ge 0}(-y)^m I_{2m-1}(\zeta)/(m!\, I_1(\zeta))$. For $\zeta \ge 1$ it differs from e^{-y} by at most $C\sum_{m\ge 1} y^m m^2/(m!\,\zeta) \le Cx/\zeta = O(\sqrt{x/S})$. For $\zeta < 1$ we have $x < 1/(4\omega_\beta S)$, and $|I_{2m-1}/I_1 - 1| \le 1$ bounds the difference by $e^y - 1 = O(1/S)$. In the same way the term $c_1 z$ gives $O(z_*(1+1/\zeta)) = O(\sqrt{x/S} + 1/S)$. The term R_2 is bounded through the modulus of the integrand, and $\frac{1}{2\pi}\int e^{\zeta\cos\phi} d\phi = I_0(\zeta)$ gives $O\big((z_*^2 + Sz_*^3) I_0(\zeta)/I_1(\zeta)\big) = O(1/S + \sqrt{x/S})$. So $\mathbf{K}_\beta = \mathbf{B}_\beta(1 + O(1/T + \sqrt{x/T}))$ for $x \le 1$.

The general case, $1 \le x \le \delta T$. Here $\zeta \ge 2\sqrt{\omega_{\min} S} \ge Z_0$. We use the evaluation of $\mathcal{A}$ in the proof of (a), and the same evaluation with $\Xi = 1$, whose integral is exactly $\Lambda(0) z_*^{-1} I_1(\zeta)$. For $\Xi = 1$ the critical point is z_*, the critical value is ζ, and $-\ell''(0) = 1$. The ratio of the two evaluations is $e^{\lambda(\rho_s) - \zeta}(1 + O(\sqrt{x/S} + 1/\zeta))$, since $\rho_s/z_* - 1$ and the relative change of $-\ell''(0)$ are $O(\sqrt{x/S} + 1/\zeta)$. Now $x/z + S\omega_\beta z$ takes the value ζ at its critical point z_*, and the shift from z_* to ρ_s changes λ only by $O(\lambda''(\rho_s)(\rho_s - z_*)^2) = O(\zeta x/S + 1/\zeta)$. So

$$\lambda(\rho_s) - \zeta = R(z_*) + O(\zeta x/S + 1/\zeta) = -(\gamma_\beta - q_i)x + O\big(\sqrt{x/S}\,(1 + x)\big),$$

because $c_1 z_* = O(\sqrt{x/S})$, $S\omega_\beta z_*^2 = x$, $Sz_*^3 = O(x^{3/2} S^{-1/2})$, $\zeta x/S = O(x^{3/2} S^{-1/2})$ and $1/\zeta \le C\sqrt{x/S}$ for $x \ge 1$. Hence $\mathbf{K}_\beta = \mathbf{B}_\beta \exp(O(\sqrt{x/T}(1 + x)))$, and with (a) this proves (b). □

At $x = \alpha_i T$ with α_i small but fixed, the saddle point of $\mathbf{K}_\beta$ reproduces the exponent $-TJ_F(\alpha)$ exactly, by the Legendre identity of Proposition 6.4(i). So part (a) and Theorem 5.4 overlap, and together they cover the face up to its facet.

6.7 The lattice and the near-vertex case

Take $Q = J - dI$, the uniform start and $\ell = 1$ in Section 8, with $F = \{1, 2\}$, $G = \{1\}$, $i = 2$ and $n = (b, a)$. Then $q_{12} = q_{21} = 1$, $p = 1 - hq_1$ and $u = h/p$, and Lemma 6.2 gives exactly

$$P_N(b, a) = \sum_{K\ge 1} \binom{a-1}{K-1} (1 - hq_2)^{a-K}\, [z^K]\, (\mu_1 + zh\mu_2)(1 + zh)(1 - hq_1 + zh^2)^{b-1}.$$

Its main part is $\mu_1 p^{N-1} \sum_K \binom{a-1}{K-1}\binom{b-1}{K} u^{2K}$, which becomes $B_a(bu^2)$ of (23) when $\binom{b-1}{K}$ is replaced by $b^K/K!$. So for $\ell = 1$ the near-vertex law is the discrete pair-face case, with $\omega = 1$, $S = hb$, $x = ha$ and $\gamma = q_2$. The parameter of Theorem 8.2 is $w = bu^2 = \omega Sh/p^2$, which is ωSh up to the factor $p^{-2} = 1 + O(h)$. There w is of order

one, Lemma 8.1 pins $B_a(w)/(w\mathcal{I}(aw))$ between $1 - w/2$ and 1, and the continuum layer does not apply. When $T^2/N \to 0$ the parameter w tends to 0, and the same lemma shows that the lattice adds no separate regime.

An open problem. A discrete form of Theorem 6.9, with the Laguerre polynomial B_{n_i} in place of the Bessel kernel, would need a discrete form of Theorem 6.6 and the saddle bounds of the proof above on the lattice, and we do not state it.

The scripts of Section 9 check the statements of this section on three chains with three states, including the cycle above, for which $\omega_\beta = 5.1820$ and $\gamma_\beta = 3.4179$ against $q_3 = 3$. The generalized kernel matches f_F/f_G to $O(1/T)$ for x from $0.01/T$ to $T/4$ on all three. They also check Theorem 6.6 on two facets with four states, a facet of a non-reversible face with five states and a path whose jump chain is periodic. There the exact word sum and the prediction differ by a relative error ε with $S|\varepsilon|$ settling at a constant, for instance 0.438 and 0.545 at $z = 0$, with the ratio of the values at $S = 32$ and $S = 16$ equal to 1.002 and 1.007. No statement is deduced from these numbers.

7 The two-state walk and the complete graph

This section applies the second-order theory to the two-state walk and to the complete graph, where every face ties at first order and a few exact facts resolve the tie.

7.1 The two-state walk

We use the polynomials $S_{a,b}$ and S_N, the root u_N, the crossing p_N, $z_N = N(1-p_N)$ and $c_0 = \frac{1}{2}\log(\pi/8)$ of Section 2.1. The first result shows that, at every N, the comparison of a symmetric pair with one of its states is exactly the two-state walk.

Proposition 7.1 (an exact pair identity) *Let $i \neq j$ with $q_{ij} = q_{ji} = a$, $q_i = q_j = q$ and $\mu_i = \mu_j > 0$, and let $F = \{i, j\}$. For every $N \geq 2$, $1 \leq m \leq N - 1$ and $h > 0$ such that $I + hQ$ is stochastic and $qh < 1$,*

$$\frac{P_N(me_i + (N-m)e_j)}{P_N(Ne_i)} = S_{m,N-m}(u), \qquad u = \frac{ah}{1-qh}.$$

So the balanced compositions, with $m = \lfloor N/2 \rfloor$ or $m = \lceil N/2 \rceil$, are the most likely ones with support F, $M_F = M_{\{i\}} S_N(u)$, and $M_F > M_{\{i\}}$ exactly when $u > u_N$.

Proof We group the paths by their number of switches. A path with this composition is a word in i, j with m letters i. Each stay has probability $1 - qh$ and each switch ah, so a word with s switches has probability $\mu_i(1-qh)^{N-1-s}(ah)^s = \mu_i(1-qh)^{N-1}u^s$, while $P_N(Ne_i) = \mu_i(1-qh)^{N-1}$. Sum over these words and divide. This gives $\sum_w u^{s(w)} = S_{m,N-m}(u)$, which is the identity. By [32, Proposition 4.1] the polynomials $S_{m,N-m}$ grow coefficientwise as m moves toward the middle, and $S_{m,N-m} = S_{N-m,m}$. So the balanced m give the largest $S_{m,N-m}(u)$. Finally S_N increases strictly and equals 1 only at u_N (Section 2.1). □

Lean checks the identity for a chain with two states. For $d \geq 3$ the paths with this composition never leave F, so the same computation applies, but this case is not in Lean.

Corollary 7.2 (the two-state walk) *Let $d = 2$, $q_{12} = q_{21} = 1$ and $\mu = (\frac{1}{2}, \frac{1}{2})$, the walk of [32] with $T = N(1-p)$. Then $K_{\{1\}} = K_{\{2\}} = \frac{1}{2}$ and $K_{\{1,2\}} = \sqrt{2/\pi}$, so $b_{FG} = c_0$ for $F = \{1,2\}$ and $G = \{1\}$. For every $N \geq 2$, $M_{\{1,2\}} = M_{\{1\}}$ holds exactly at $T = z_N$, and $z_N + \frac{1}{2}\log z_N = L + c_0 + O(1/L)$.*

Proof Corollary 5.3 with $k = 2$, $\mathrm{tree}(q_F) = 1$ and $\mu(F) = 1$ gives $K_{\{1,2\}}$, and then $b_{FG} = -\log(K_{\{1,2\}}/K_{\{1\}}) = c_0$. Proposition 7.1 with $a = q = 1$ gives $M_{\{1,2\}} = M_{\{1\}} S_N(h/(1-h))$, and $h/(1-h) = u_N$ exactly when $p = p_N$. So for every N the only solution of $M_{\{1,2\}} = M_{\{1\}}$ is $T = z_N$, and by Theorem 5.8(i) it lies in $[L/2, 2L] \subset [1, N^{1/3}]$ for large N. Theorem 5.8(ii) with $\Delta = \delta = 1$ gives the equation. □

The persistent-walk paper [32] proves the next term $1/(8z_N)$ of this equation and bounds its remainder at every N.

7.2 The complete graph

From now on in this section $Q = J - dI$. The chain switches to each other state with probability h (the r of [33]) and keeps its state with probability $p = 1-(d-1)h$. The following symbols of [33] are local to this section and to Appendix C. Put

$$\sigma = p - h, \qquad u = \frac{h}{p}, \qquad v = \frac{h}{\sigma} = \frac{u}{1-u}, \qquad V_N = \frac{p^{N-1}}{d}.$$

So u is the odds of a switch to one given state against no switch, and V_N is the probability of a vertex Ne_i for the uniform start, that is, $M_{\{i\}}$. For a composition n we write $S_n(u)$ for the sum of u^j over the words with the counts n, where j is the number of switches of the word, and $W_n(j)$ for the number of these words with j switches. For a composition with some zero coordinates, S_n is the polynomial of its positive coordinates, which does not depend on d. By (2), the uniform start has

$$P_N(n) = V_N S_n(u) \qquad (0 < p \leq 1), \tag{20}$$

and so does every start that is constant on $\operatorname{supp} n$, with V_N replaced by $\mu_a p^{N-1}$ for $a \in \operatorname{supp} n$. A composition is *balanced* if its positive coordinates differ by at most 1. For the uniform start all faces of k states have the same face maximum M_F, and all balanced compositions with k positive coordinates have the same probability. For $p \geq 1/d$ this probability is M_F, by Theorem 7.6 below. (The preprint [33] writes $C_{N,k}$ for it.) Put

$$a_k = \frac{1}{2}\log(4\pi) - \frac{k+\frac{1}{2}}{k-1}\log k, \qquad D_k = e^{-(k-1)a_k} = \frac{k^{k+1/2}}{(4\pi)^{(k-1)/2}} \qquad (k \geq 2),$$

and $D_1 = 1$. Numerically $a_2 = -0.4674$, $a_3 = -0.6571$ and $a_4 = -0.8139$.

Corollary 7.3 (the complete graph) *A face F of k states has $\lambda_F = d - k$ and α_F^* uniform, and if $\mu(F) > 0$ then $K_F = \frac{\mu(F)}{k} D_k$. So $\Phi_1(F) = d - 1$ for every face, and all faces tie at $\tau = 1$. With the uniform start, the crossing of a face of $k \geq 2$ states with a vertex has $\Delta = \delta = k - 1$ and $b_{FG} = a_k$.*

Proof Each Q_F is symmetric with row sums $-(d-k)$, and the complete graph on F has k^{k-2} spanning trees by Lemma 5.12. So Corollary 5.3 gives λ_F, α_F^* and

$$K_F = \mu(F) k^{(k+1)/2} k^{(k-2)/2} (4\pi)^{-(k-1)/2} = \frac{\mu(F)}{k} D_k.$$

A vertex has $K = \frac{1}{d}$ and $\lambda = d - 1$ for the uniform start, so $b_{FG} = -\frac{1}{k-1} \log(dK_F) = -\frac{1}{k-1} \log D_k = a_k$. The arithmetic that gives b_{FG} from K_F here and in Corollary 7.2 is checked in Lean. □

Remark 7.4 (first-order winners on other graphs) Theorem 4.4 reads the first-order winners of any chain off the exit rates of its admissible faces. Two cases have short proofs. On the complete graph a face of k states has $\Phi_\tau(F) = \tau(d-k) + k - 1 = (d-1)\tau + (k-1)(1-\tau)$ by Corollary 7.3. So the vertices win for $\tau < 1$, the whole set $[d]$ wins for $\tau > 1$, and all faces tie at $\tau = 1$. Now let Q be symmetric and fix a state x. Rayleigh–Ritz with the unit vector f that equals $(d-1)^{-1/2}$ off x and 0 at x gives $\lambda_{[d]\setminus\{x\}} \leq q_x/(d-1)$, with equality exactly when f is an eigenvector of $-Q$ restricted to $[d] \setminus \{x\}$. That matrix maps f to the vector with entries $q_{yx}(d-1)^{-1/2}$, so equality holds exactly when $q_{yx} = q_x/(d-1)$ for all $y \neq x$. Take unit rates on a connected graph and a start of full support, and let x have the least degree δ. The vertices of degree δ and the face $[d]$ tie at $\tau_0 = (d-1)/\delta$. The restricted matrix is block diagonal, so a component C of $[d] \setminus \{x\}$ of least exit rate has $\lambda_C = \lambda_{[d]\setminus\{x\}}$ and $\Phi_{\tau_0}(C) \leq \tau_0 \lambda_{[d]\setminus\{x\}} + d - 2 \leq d - 1$. Equality needs $q_{yx} = \delta/(d-1)$ for every $y \neq x$, which for unit rates means that the graph is complete. So the winners go from the vertices straight to $[d]$ exactly when the graph is complete. On every other connected graph a face that avoids x beats both at τ_0, and the mode can move to a community that does not contain its first vertex.

By Example 5.13 at the center, $D_k = dK_F = k/\sqrt{\det(2\pi\Sigma_k)}$ with $\Sigma_k = \Sigma_{\mathbf{1}/k} = (2/k^2)(I - J/k)$ in the coordinates α'. So D_k is the mass k of a face, in units of a vertex, over the Gaussian volume of the fluctuations of its fractions. The preprint [33, Proposition 3.7] also obtains it by summing the exact law over the whole face. Example 5.13, Theorem 5.4 and Lemma 5.5 give every composition whose fractions are bounded away from the boundary of its face. Let $\alpha = n/N$ have k positive coordinates, all at least ε, and put $A = \sum_i \sqrt{\alpha_i}$. Uniformly for such compositions and for $1 \leq T \leq N^{1/3}$,

$$\begin{aligned} \frac{P_N(n)}{V_N} &= D(\alpha) \, \frac{T^{(k-1)/2} e^{(A^2-1)T}}{N^{k-1}} \Big(1 + O_\varepsilon\Big(\frac{1}{T} + \frac{T^2}{N}\Big)\Big), \\ D(\alpha) &= \frac{A^{k/2+1}}{(4\pi)^{(k-1)/2} (\prod_i \alpha_i)^{3/4}}, \end{aligned} \tag{21}$$

for the uniform start. The relation between $D(\alpha)$ and $C_F(\alpha)$ is given in Example 5.13, and $V_N = \frac{1}{d} e^{-(d-1)T}(1 + O(T^2/N))$. This is [33, Theorem 3.3], and at the center

$D(\mathbf{1}/k) = D_k$. By the Cauchy–Schwarz inequality $A^2 \le k$, with equality only at the center.

Theorem 7.5 (balanced crossings) *Let μ be uniform and let F be a face of $2 \le k \le d$ states.*

(i) For every $N \ge k$ there is exactly one $p = p_{N,k} \in (0,1)$ with $M_F = V_N$. The face maximum M_F is larger than V_N below $p_{N,k}$ and smaller above it, and so is the probability of a balanced composition of F. The crossing $p_{N,k}$ is strictly increasing in N, $p_{k,k} = 1/(1+(d-1)(k!)^{-1/(k-1)})$, and every crossing has $u \le (k!)^{-1/(k-1)} < 1$, so $p_{N,k} > 1/d$.

(ii) For large N, uniformly for $1 \le T \le N^{1/3}$,

$$\log \frac{M_F}{V_N} = (k-1)\Big[T - L + \frac{1}{2}\log T - a_k\Big] + O\Big(\frac{1}{T} + \frac{T^2}{N}\Big).$$

With $T = L - \frac{1}{2}\log L + t$, the ratio M_F/V_N tends to $e^{(k-1)(t-a_k)}$, uniformly for t in compact sets.

(iii) The value $T_{N,k} = N(1-p_{N,k})/(d-1)$ of T at the crossing satisfies

$$T_{N,k} + \frac{1}{2}\log T_{N,k} = L + a_k + O(1/L),$$

and

$$T_{N,k} = L - \frac{1}{2}\log L + a_k + \frac{\log L}{4L} + O\Big(\frac{1}{L}\Big). \tag{22}$$

(iv) $a_2 > a_3 > a_4 > \cdots$.

For $d = 2$, $T_{N,2}$ is the crossing z_N of Corollary 7.2, and $a_2 = c_0$. The proof is in Appendix C. The exact part uses that S_n has nonnegative coefficients and the first nonzero coefficient $W_n(k-1) = k!$, from the words that use each state in a single run. The next coefficient $W_n(k) = k!(k-1)(N-k)/2$ grows with N, which moves the root. The asymptotic part is Theorems 5.7 and 5.8. The equation in (iii) has the next term $1/(8T_{N,k})$ on its right side. The persistent-walk paper [32] proves it for $k = 2$, and the preprint [33, Theorem 3.2] proves it for every $k \ge 2$ through a Bessel majorant. We do not reprove that term here.

A count of run lengths alone does not explain the next theorem. For fixed numbers of runs the choices of run lengths are most numerous at the balanced split when all states have the same number of runs, but not in general. If one state has a single run and another has several, the count grows when visits move to the second state. What works is to sum over the numbers of runs first. The law then becomes an integral of a product of one polynomial per state, and this polynomial is log-concave in the count of its state.

Theorem 7.6 (exact balancing) *Let $1/d \le p < 1$, and let μ be constant and positive on a face F. Moving one unit from a coordinate b of a composition with support F to another coordinate a, where $n_b \ge n_a + 2$ and $n_a \ge 1$, strictly increases the probability of the composition. So the most likely compositions with support F are exactly the balanced ones, and they all have the same probability.*

For an integer $n \geq 1$ put

$$B_n(w) = \sum_{m=1}^{n} \binom{n-1}{m-1} \frac{w^m}{m!} = \frac{w}{n} L_{n-1}^{(1)}(-w), \tag{23}$$

where $L_{n-1}^{(1)}$ is the Laguerre polynomial [30, Eq. 18.5.12]. The proof in Appendix C rests on the log-concavity of $B_n(w)$ in n for $w > 0$. The condition $p \geq 1/d$ holds for $T \leq N/d$, and so at every crossing by Theorem 7.5(i).

Corollary 7.7 (exactly one mode change) *Let μ be uniform, and fix $d \geq 2$ and $C > 1$. There is N_0 such that for $N \geq N_0$ and every p with $0 < T \leq CL$ the following hold.*

(i) If $p > p_{N,d}$, the global modes are exactly the d vertices.

(ii) If $p < p_{N,d}$, they are exactly the balanced compositions with d positive coordinates.

(iii) If $p = p_{N,d}$, they are these two classes and nothing else.

Moreover $p_{N,2} < p_{N,3} < \cdots < p_{N,d}$ and $p_{N,d} - p_{N,k} = (d-1)(a_k - a_d + o(1))/N$. So all the crossings of balanced compositions with the vertices lie in a p-interval of length $(d-1)(a_2 - a_d + o(1))/N$, and as p decreases the balanced compositions with d positive coordinates cross first.

On the triangle the middle composition $(\frac{N}{3}, \frac{N}{3}, \frac{N}{3})$ overtakes the vertices while the middles of the edges are still below them by the factor $e^{-(a_2-a_3)} = 0.827$. The proof is in Appendix C. In the tie window at $\tau^* = 1$ this is Theorem 5.9, whose envelope E has a single kink, at $t = a_d$.

Remark 7.8 (an intermediate face can be a mode at finite N) Corollary 7.7 does not hold for every N. Take $d = 3$, $N = 4$ and $p = 10/17$, so $u = 7/20$. For an edge center and a full center, counting the words gives

$$\frac{P_4(2,2,0)}{V_4} = 2u + 2u^2 + 2u^3 = \frac{4123}{4000} > 1, \qquad \frac{P_4(2,1,1)}{V_4} = 6u^2 + 6u^3 = \frac{3969}{4000} < 1.$$

Here $p > 1/3$, so by Theorem 7.6 every global mode is a vertex or a balanced composition, and the three edge centers are the only global modes. Also $0 < T = 14/17 < \log 4$, so this p lies in the range of Corollary 7.7 for every $C > 1$. Hence $N_0 > 4$ when $d = 3$. These values are checked in Lean.

7.3 A start

By Corollary 7.3 the start enters K_F only through $\mu(F)$. It changes the constants by an amount of order 1, the order at which the faces are separated, so a start can decide which face wins.

Corollary 7.9 (crossings with a start) *Let F be a face of $k \geq 2$ states with $\mu(F) > 0$, and let i be a state with $\mu_i > 0$. Put $a_k(\mu) = a_k + \frac{1}{k-1} \log \frac{k\mu_i}{\mu(F)}$. For large N the equation $M_F = M_{\{i\}}$*

has a root in $[1, N^{1/3}]$*, and every such root satisfies* $T + \frac{1}{2}\log T = L + a_k(\mu) + O(1/L)$*. With* $T = L - \frac{1}{2}\log L + t$*,* $\log(M_F/M_{\{i\}}) = (k-1)(t - a_k(\mu)) + o(1)$*, uniformly for* t *in compact sets.*

Proof By Corollary 7.3, $\log(K_F/K_{\{i\}}) = \log D_k + \log \frac{\mu(F)}{k\mu_i} = -(k-1)a_k(\mu)$, and $\Delta = \delta = k - 1$. So Theorem 5.8 gives the root and its equation. For the last claim, Theorem 5.9(i) at $\tau^* = 1$ gives $\log(M_F/M_{\{i\}}) = \ell_F(t) - \ell_{\{i\}}(t) + O(\log L/L)$, and $\ell_F(t) - \ell_{\{i\}}(t) = \log(K_F/\mu_i) + (k-1)t$. □

So a start changes only the constant of each crossing, as in [33, Theorem 4.4].

Example 7.10 (the start decides which face wins) Let $d = 3$ and $\mu = (\frac{1}{2}, \frac{1}{2}, 0)$, and put $t_{\text{ef}} = \log\frac{3}{2} + 2a_3 - a_2 = \frac{1}{2}\log\frac{32\pi}{243}$. Fix t, put $T = L - \frac{1}{2}\log L + t$ and let N be large. If $t < a_2$, the global modes are $(N, 0, 0)$ and $(0, N, 0)$. If $a_2 < t < t_{\text{ef}}$, they are exactly the balanced compositions of the edge $\{1, 2\}$. If $t > t_{\text{ef}}$, every global mode has three positive coordinates. The window $(a_2, t_{\text{ef}}) = (-0.4673558, -0.4412978)$ has width $\log\frac{16}{9\sqrt{3}} = 0.0260580$, while with the uniform start no proper face wins (Corollary 7.7).

Proof By Lemma 2.2(b) and (c) the admissible faces are the faces that meet $\{1, 2\}$, and they satisfy (H). By Corollary 7.3 they all tie at $\tau^* = 1$, so Theorem 5.9 applies with $m^* = 2$. For a face F of k states its line is $\ell_F(t) = \log K_F - (3-k)t$, and $\ell_F(t) + 2t$ equals $\log\mu_i$ for $F = \{i\}$ and $\log(\mu(F)/k) + (k-1)(t - a_k)$ for $k \geq 2$. So $\ell_F + 2t$ is $g_1 = \log\frac{1}{2}$ for the vertices 1 and 2, $g_2 = \log\frac{1}{2} + t - a_2$ for the edge $\{1, 2\}$, $g_2 - \log 2$ for the other two edges and $g_3 = \log\frac{1}{3} + 2(t - a_3)$ for the full face. Hence $E + 2t$ is the upper envelope of g_1, g_2 and g_3. Here $g_2 > g_1$ exactly when $t > a_2$, $g_2 > g_3$ exactly when $t < t_{\text{ef}}$, and $g_3 > g_1$ exactly when $t > t^* = a_3 + \frac{1}{2}\log\frac{3}{2}$. By the definition of a_k, $t_{\text{ef}} - a_2 = \log\frac{3}{2} + 2(a_3 - a_2) = 4\log 2 - \frac{5}{2}\log 3 = \log\frac{16}{9\sqrt{3}}$, which is positive since $16^2 = 256 > 243 = (9\sqrt{3})^2$. Also $t^* - a_2 = \frac{1}{2}(t_{\text{ef}} - a_2)$, so $a_2 < t^* < t_{\text{ef}}$. Hence the kinks of E are a_2 and t_{ef}, and E is the line of the two vertices for $t < a_2$, of the edge $\{1, 2\}$ for $a_2 < t < t_{\text{ef}}$, and of the full face for $t > t_{\text{ef}}$. For fixed t that is not a kink, the consequence in Theorem 5.9 puts every global mode on a face with that line. The vertices Ne_1 and Ne_2 have the same probability $\frac{1}{2}p^{N-1}$. On the edge, μ is constant and $p > 1/3$ for large N, so by Theorem 7.6 the most likely compositions with support $\{1, 2\}$ are exactly the balanced ones, and they all have the same probability. □

The comparison of the lines and the value of the width are checked in Lean.

7.4 A weak field

Proposition 5.15 applies to the complete graph with the uniform start. There α_F^* is uniform, so the field shifts the line of a face S by the average $\bar{\xi}_S = \frac{1}{|S|}\sum_{i\in S}\xi_i$ of the field over S. In the Potts chain the field ξ_i/N acts at each site in state i. The scores $\ell_S(t, \xi)$ below are the lines ℓ_S^ξ of Proposition 5.15(c) at $\tau^* = 1$, measured from the line $\ell_{\{1\}}(t) = -\log d - (d-1)t$ of a vertex without field, that is $\ell_S(t, \xi) = \ell_S^\xi(t) - \ell_{\{1\}}(t)$. The orientation and the slope convention are those of Theorem 5.9, with $\lambda_S = d - |S|$ and $\log K_S = \log D_{|S|} - \log d$.

Corollary 7.11 (weak-field modal phase diagram) *Let μ be uniform, put $T = L - \frac{1}{2}\log L + t$ and define the face scores*

$$\ell_S(t,\xi) = \begin{cases} \xi_i, & S = \{i\}, \\ (|S|-1)(t - a_{|S|}) + \bar{\xi}_S, & |S| \geq 2. \end{cases}$$

Uniformly for (t,ξ) in compact sets,

$$\log \max_n \frac{e^{\langle \xi, n\rangle/N} P_N(n)}{V_N} \longrightarrow \max_{\varnothing \neq S \subseteq [d]} \ell_S(t,\xi). \tag{24}$$

For fixed (t,ξ) and all large N, every global mode of $\widetilde{P}_{N,\xi}$ has a support that maximizes the limiting score. If the maximizing support is unique, all global modes have exactly that support, and if it has $|S| \geq 2$ states they satisfy $n/N = \mathbf{1}_S/|S| + O(1/L)$. Equal limiting scores do not imply that both supports hold modes at finite N.

If the fields are sorted as $\xi_1 \geq \cdots \geq \xi_d$, the maximum of the scores is the upper envelope of the d lines $\ell_k(t) = \ell_{\{1,\dots,k\}}(t,\xi)$, namely

$$\ell_1(t) = \xi_1, \qquad \ell_k(t) = (k-1)(t - a_k) + \frac{1}{k}\sum_{i=1}^{k} \xi_i, \quad 2 \leq k \leq d. \tag{25}$$

Away from ties, the winning support size is nondecreasing in t. If the fields are strictly ordered, the winning supports are the nested sets $\{1,\dots,k\}$. Every support size $k \in \{2,\dots,d-1\}$ is the unique winning size on an open interval of t for suitable fixed fields. So proper faces can hold the mode under a field of order $1/N$ per step.

Proof Step 1. The best composition on a support. Let S have $k \geq 2$ states. By Theorem 7.5(ii), $\log(M_S/V_N) = (k-1)(t-a_k) + o(1)$, uniformly for t in a compact set. Proposition 5.15(b), with α_S^* uniform, gives $\log(M_S^\xi/V_N) = \ell_S(t,\xi) + o(1)$, uniformly for (t,ξ) in compact sets. A vertex has $M_{\{i\}}^\xi = e^{\xi_i} V_N$ exactly. There are finitely many supports, which proves (24).

Step 2. The modes. A support whose score is below the maximum by a fixed amount has M_S^ξ smaller than the largest face maximum for large N, so it holds no global mode. A global mode with support S maximizes w_ξ on S, and Proposition 5.15(b) gives its location.

Step 3. The lines. Among the supports of size k the largest $\bar{\xi}_S$ is the average of the k largest fields, which gives (25). The line ℓ_k has slope $k-1$. If size k_1 wins at t_1 and size k_2 at $t_2 > t_1$, adding $\ell_{k_1}(t_1) \geq \ell_{k_2}(t_1)$ and $\ell_{k_2}(t_2) \geq \ell_{k_1}(t_2)$ gives $(k_2 - k_1)(t_2 - t_1) \geq 0$, so $k_2 \geq k_1$, as in Theorem 4.4(b). If the fields are strictly ordered, the unique best support of size k is $\{1,\dots,k\}$.

Step 4. Every intermediate size. To realize a given $2 \leq k < d$, set $\xi_1 = \cdots = \xi_k = 0$ and $\xi_{k+1} = \cdots = \xi_d = -H$, and take $t > a_k$ so close to a_k that $t < a_{k-1}$ if $k > 2$. Then the score of $\{1,\dots,k\}$ is positive, the largest vertex score is 0, and every face with $2 \leq j < k$ states scores at most $(j-1)(t-a_j) < 0$, by Theorem 7.5(iv). A face with $j > k$ states scores at most $(j-1)(t-a_j) - H(j-k)/j$, and another face with k states at most $(k-1)(t-a_k) - H/k$, so both lose once H is large. The comparisons are strict, so they hold on an open interval of t, and small changes make the fields strictly ordered without removing the phase. □

One field can make every support size win in turn, from one state to all d. This is the sequence of phases in Figure 1.

Corollary 7.12 (one field realizes the full face hierarchy) *Let μ be uniform, fix $d \geq 2$ and $H > 3/4$, and let $\xi_i = -H(i-1)^2$ for $1 \leq i \leq d$. In particular $\xi_i = -(i-1)^2$ works in every fixed dimension. Put*

$$t_k = \log D_k - \log D_{k+1} + \frac{H(4k-1)}{6} \quad (1 \leq k < d),$$

and $t_0 = -\infty$, $t_d = +\infty$. Then $t_1 < \cdots < t_{d-1}$, and for every $k \in \{1, \ldots, d\}$ the support $\{1, \ldots, k\}$ is the unique winning support on the nonempty interval $t_{k-1} < t < t_k$. So for each fixed t in this interval and all large N, every global mode of $\widetilde{P}_{N,\xi}$ has exactly those k positive coordinates.

Proof Since $\frac{1}{k}\sum_{i=1}^{k}(i-1)^2 = (k-1)(2k-1)/6$ and $-(k-1)a_k = \log D_k$, the lines (25) are $\ell_k(t) = (k-1)t + \log D_k - H(k-1)(2k-1)/6$, also for $k = 1$, and $\ell_{k+1}(t) - \ell_k(t) = t - t_k$. Extend $\log D_k$ to the smooth function $c(x) = (x + \frac{1}{2})\log x - \frac{x-1}{2}\log(4\pi)$ for $x \geq 1$. Then $0 < c''(x) = (x - \frac{1}{2})/x^2 \leq \frac{1}{2}$, and integrating twice over unit intervals gives $\log D_{k+2} - 2\log D_{k+1} + \log D_k \leq \frac{1}{2}$. Hence

$$t_{k+1} - t_k = \frac{2H}{3} - (\log D_{k+2} - 2\log D_{k+1} + \log D_k) > 0.$$

For $t_{k-1} < t < t_k$, summing the differences $\ell_{j+1}(t) - \ell_j(t) = t - t_j$ shows that line k strictly exceeds every other line. The fields are strictly decreasing, so the unique best support of size k is $\{1, \ldots, k\}$, and Corollary 7.11 gives the conclusion for finite N. Most of this arithmetic is checked in Lean. □

8 Near a vertex

This section is the worked case of the boundary layer of Section 6 (Theorem 6.9 and Proposition 6.3) on the complete graph near a vertex, and Section 6.7 shows that with one rare count it is the discrete pair-face layer. The vertex law covers the regime in which T is of order $\sqrt{N}$, where the lattice parameter is of order one and the Laguerre kernel cannot be replaced by the Bessel kernel of the continuum layer.

We keep the notation of Section 7.2, namely $Q = J - dI$, the uniform start, p, u, v, and $S_n(u) = P_N(n)/V_N$ by (20). Let one count b be large and let the other positive counts $a_1, \ldots, a_\ell$ be small compared with b, with $\ell \geq 1$ fixed. We put $|a| = a_1 + \cdots + a_\ell$ and write $S_{(b,a)}$ for S_n with $n = (b, a_1, \ldots, a_\ell)$. These rare counts are not the constants a_k of Section 7.2. Since S_n does not depend on the start, $P_N(n) \leq (\max_i \mu_i)p^{N-1}S_n(u)$, so upper bounds on S_n hold for every start.

A rare run away from the ends needs two switches, which gives a factor u^2, and it can go in about b places. So the rare counts enter through bu^2, and for fixed rare counts the crossing moves to odds u of order $b^{-1/2}$, that is, to T of order $\sqrt{N}$. Use B_a from (23) and put

$$\mathcal{I}(x) = \sum_{j \geq 0} \frac{x^j}{j!\,(j+1)!} = \frac{I_1(2\sqrt{x})}{\sqrt{x}},$$

where I_1 is the modified Bessel function [30, Eq. 10.25.2], and $\mathcal{I}(0) = 1$. By the Laguerre form in (23), the next lemma is a quantitative form of the limit of Laguerre polynomials to Bessel functions.

Lemma 8.1 (a uniform Laguerre–Bessel inequality) *For every integer $a \geq 1$ and real $w > 0$,*

$$0 \leq 1 - \frac{B_a(w)}{w\,\mathcal{I}(aw)} \leq \frac{w}{2}. \tag{26}$$

The proof is in Appendix D. Put

$$\mathcal{B}(y) = \prod_{i=1}^{\ell} B_{a_i}(y), \qquad \kappa(y) = \frac{y\,\mathcal{B}'(y)}{\mathcal{B}(y)}.$$

The polynomial $\mathcal{B}$ has positive coefficients, $\mathcal{B}(0) = 0$ and $\mathcal{B}(1) \geq 1$, so $\mathcal{B}(y_a) = 1$ has exactly one positive root y_a, and $y_a \leq 1$. The polynomial $S_{(b,a)}$ has nonnegative coefficients, zero constant term and a positive coefficient at u^ℓ, so $S_{(b,a)}(u) = 1$ has exactly one positive root $u_{b,a}$, the crossing odds of the composition $(b, a_1, \ldots, a_\ell)$ with a vertex.

Theorem 8.2 (the vertex law) *Fix $\ell \geq 1$ and put*

$$c_\ell = \ell + \sqrt{\ell} + 1, \qquad \varepsilon = \sqrt{\frac{|a|+1}{b}}, \qquad \delta = 16 c_\ell \varepsilon.$$

For all positive counts $b, a_1, \ldots, a_\ell$ with $\delta \leq 1/2$,

$$(1-\delta)\sqrt{y_a/b} \;\leq u_{b,a} \leq (1+\delta)\sqrt{y_a/b}, \qquad \mathcal{B}(y_a) = 1. \tag{27}$$

In particular

$$\left| \frac{u_{b,a}}{\sqrt{y_a/b}} - 1 \right| \leq 16 c_\ell \sqrt{\frac{|a|+1}{b}} \qquad \text{whenever } \sqrt{\frac{|a|+1}{b}} \leq \frac{1}{32 c_\ell}, \tag{28}$$

so $u_{b,a} = \sqrt{y_a/b}\,(1 + o(1))$ as $(|a|+1)/b \to 0$. The constants are not optimized.

For fixed rare counts Lemma D.2 gives $S_{(b,a)}(\sqrt{y/b}) \to \mathcal{B}(y)$ as $b \to \infty$, and the theorem shows that the crossing stays near $\sqrt{y_a/b}$ uniformly as long as $|a| = o(b)$. For $d = 2$ and $\ell = 1$ its leading term is that of the fixed-edge expansion of [32, Theorem 4.10]. The proof is in Appendix D. It rests on two facts. First, $\log \mathcal{B}$ is concave, and for $0 < y \leq 1$ its elasticity $\kappa(y)$ lies between $\max\{\ell, \frac{1}{2}\sqrt{|a|y}\}$ and $\ell + \sqrt{\ell |a| y}$ (Lemma D.1). Second, $S_{(b,a)}(u) = (1-u)^{|a|}\mathbb{E}[\mathcal{B}(v\mathcal{X})]$ exactly, where $\mathcal{X}$ is a gamma variable plus a compound binomial sum with mean $2 + (b-1)u$ (Lemma D.2). Jensen's inequality then gives a lower bound that exceeds 1 at $u = (1+\delta)\sqrt{y_a/b}$, and the tangent line of $\log \mathcal{B}$ at bu^2 with the moment generating function of $\mathcal{X}$ gives an upper bound that is below 1 at $u = (1-\delta)\sqrt{y_a/b}$.

Remark 8.3 (the boundary of the simplex) Away from a vertex, the preprint [33] gives a Bessel majorant for every positive count vector [33, Theorem 6.2] and shows that, when at least two coordinates are of order N, a small coordinate starts to matter at size of order $N/(\log N)^2$ [33, Remark 6.3 and Theorem C.1]. We do not use these results.

9 Verification and open questions

Every theorem, proposition, lemma and corollary of this paper is proved in the text or in the appendices. We quote results of [32] and [33] where we say so. No proof rests on a computation. The formal proofs and the computations below check parts of the paper independently.

Formal proofs

Many finite and algebraic steps are proved in Lean 4 (v4.33.1, with Mathlib) in the folders lean/PaperC and lean/PaperB of the repository named under Code availability, and the file lean/PaperC/MAP.md matches each statement of the paper to its Lean theorems. No proof there uses `sorry` or `admit`. The files cover the run expansion (Lemma 2.3), the lower bound and the reversible form of the rate (Proposition 3.3), the restriction identity (Proposition 3.1), the finite form of Theorem 4.4, the tree formula of Proposition 5.2(b) for two and three states, the exact parts of Example 5.14, and the exact statements on the complete graph and near a vertex in Sections 7 and 8. Mathlib has no Perron and Frobenius theorem for irreducible Metzler matrices, so a positive Perron vector is a hypothesis of the Lean statements where one is needed, and the asymptotic results are not in Lean.

Computations

The scripts are in the folder named under Code availability. Each check is an assertion, and the checks in floating point use a tolerance. None of them is interval-certified. The script `verify_all.py` runs all eight with `py -I` and writes each output, headed by the sha256 of its script, to the folder data. The scripts `verify_first_order.py`, `verify_second_order.py`, `verify_forest.py`, `verify_rate_flows.py`, `verify_weak_field.py` and `verify_examples.py` compute the law in two independent ways and check Sections 2 to 5 and 7. They were written for an earlier version of the paper, and a few of their checks concern statements that this version no longer makes. The script `verify_layer.py` checks the constants and kernels of Section 6, including the inequality $I_0/I_1 \le 1 + 2/\zeta$ and the ratios for the non-reversible cycle after Theorem 6.9. The script `verify_layer_general.py` checks Theorem 6.6 on two facets with four states, a facet of a non-reversible face with five states and a path with a periodic jump chain. It computes the word sum $f_G^{(z)}(S\beta)$ exactly by a recursion over the words, compares it with $S^{-3/2}C_G(\beta, z)e^{-SJ_G(\beta,z)}$ computed from the complex critical point of Lemma 6.8, and prints S times the relative error for S doubling from 4 to 32 at real and complex z. The products settle to constants, as (19) predicts. The figures illustrate the theorems, and no statement is deduced from them.

Open questions

A local law up to the lower-dimensional corners of a face is open. For a reducible face we expect the power of T in M_F to be $(k^* - 1)/2 + k_F - k^*$, between the bounds of Theorem 4.2(ii). The crossings of Theorem 5.8 are known to be unique only in special

cases. A discrete form of Theorem 6.9, with the Laguerre polynomials of Section 8 in place of the Bessel kernel, is not stated.

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Statements and Declarations

- **Funding.** No funding was received for this work.
- **Competing interests.** The author has no competing interests to declare.
- **Ethics approval and consent to participate.** Not applicable.
- **Consent for publication.** Not applicable.
- **Data availability.** The outputs of the computations described in Section 9 are in the folder kagey-problems/problem131/paperC/data of the repository named under Code availability, each headed by the sha256 of the script that wrote it.
- **Materials availability.** Not applicable.
- **Code availability.** All code, data and Lean files supporting this paper are in the folder kagey-problems/problem131 of https://github.com/apemm/Kagey-Problems. The scripts of this paper are in the subfolder paperC. To rerun the checks, run `py -I verify_all.py` in that subfolder, which takes about nine minutes and writes the outputs to paperC/data. The option `--full` adds the largest cases of four of the scripts and takes much longer. Each script also runs on its own, and its docstring says what it checks and how long it takes. The scripts need Python 3 with `numpy`, `scipy`, `sympy` and `mpmath`. The Lean files are in the subfolder lean and build with Lean 4 v4.33.1 and Mathlib, and the file lean/PaperC/MAP.md matches the statements of this paper to their Lean theorems.
- **Author contribution.** I am the sole author and am responsible for the whole paper.

Appendix A Proofs of the local law and of the first-order bounds

This appendix proves Proposition 5.2(a), Lemma 4.1, Theorem 5.4, Lemma 5.5 and Theorem 4.2(ii). Its last section proves Lemma A.5, which is used for Proposition 5.11. Except in Section A.6, F is a face with $k \geq 2$ states, numbered $1, \dots, k$. For $x \in \mathbb{C}^F$ we write $x' = (x_2, \dots, x_k)$. We write i for the imaginary unit, since i denotes a state, and $\mathbb{T}^m = (\mathbb{R}/2\pi\mathbb{Z})^m$. We use two standard facts [2]. First, let $B \geq 0$ be square and S diagonal with positive diagonal. Then $\varrho(B - S) < 0$ if and only if $\varrho(S^{-1}B) < 1$, and also if and only if $(S - B)v > 0$ for some $v > 0$. In that case $(S - B)^{-1} = \sum_{j \geq 0} (S^{-1}B)^j S^{-1} \geq 0$. Second, if M is an irreducible Metzler matrix, then every nonnegative eigenvector of M is positive and belongs to $\varrho(M)$, and $\varrho(M)$ is larger than the Perron root of each proper principal submatrix of M (Lemma 3.2).

A.1 The Perron root under a diagonal perturbation

An irreducible generator on F is a Metzler matrix G with $G\mathbf{1} = 0$ whose graph is strongly connected. It has a unique stationary law $\pi > 0$. The argument after Definition 5.1, with (G, π) in place of $(\widehat{Q}_\alpha, \alpha)$, shows that $Z = (\mathbf{1}\pi^\top - G)^{-1} - \mathbf{1}\pi^\top$ exists and satisfies (8). Put $D = \operatorname{diag}\pi$ and $\Gamma = DZ + Z^\top D$.

Lemma A.1 (the Perron root) *Let G, π, Z and Γ be as above. Then $\kappa(\eta) = \varrho(G + \operatorname{diag}\eta)$ is real-analytic near 0, with $\kappa(0) = 0$, $\nabla\kappa(0) = \pi$ and $\nabla^2\kappa(0) = \Gamma$. For $\eta \in \mathbb{R}^F$ and $u = Z\eta$,*

$$\eta^\top \Gamma \eta = \sum_{i \neq j} \pi_i G_{ij} (u_i - u_j)^2. \tag{A1}$$

So Γ is positive semidefinite with kernel $\mathbb{R}\mathbf{1}$, and Σ, which is Γ without the row and column of state 1, is positive definite.

Proof We differentiate the eigenvalue equation twice at $\eta = 0$. Since $G + \operatorname{diag}\eta$ is irreducible, its Perron root is a simple eigenvalue. So by the implicit function theorem, κ and the right eigenvector $v(\eta)$ with $\pi^\top v = 1$ are real-analytic near 0, with $\kappa(0) = 0$ and $v(0) = \mathbf{1}$. We write subscripts for derivatives at $\eta = 0$. Differentiating $(G + \operatorname{diag}\eta)v = \kappa v$ in η_j gives $Gv_j + e_j = \kappa_j \mathbf{1}$, and multiplying by $\pi^\top$ gives $\kappa_j = \pi_j$, since $\pi^\top G = 0$. Then $v_j = Ze_j$, because by (8) both vectors solve $Gx = \pi_j\mathbf{1} - e_j$ with $\pi^\top x = 0$, and the kernel of G is $\mathbb{R}\mathbf{1}$. Differentiating again, in η_i, gives $Gv_{ij} + \operatorname{diag}(e_i)v_j + \operatorname{diag}(e_j)v_i = \kappa_{ij}\mathbf{1} + \kappa_i v_j + \kappa_j v_i$. We multiply by $\pi^\top$ again. The terms with G, κ_i and κ_j vanish, and we get $\kappa_{ij} = \pi_i Z_{ij} + \pi_j Z_{ji} = \Gamma_{ij}$.

For (A1) let $\bar\eta = \eta - (\pi \cdot \eta)\mathbf{1}$. By (8), $-Gu = \bar\eta$ and $\pi \cdot u = 0$, so $\eta^\top\Gamma\eta = 2\sum_i \pi_i \eta_i u_i = 2\sum_i \pi_i \bar\eta_i u_i = -2\sum_i \pi_i u_i (Gu)_i$. Now expand the squares in (A1). By $G\mathbf{1} = 0$ and $\pi^\top G = 0$ the terms with u_i^2 and with u_j^2 each sum to $-\sum_i \pi_i G_{ii} u_i^2$, and the cross terms sum to $-2\sum_i \pi_i u_i (Gu)_i + 2\sum_i \pi_i G_{ii} u_i^2$. So both sides of (A1) agree. Also $\Gamma\mathbf{1} = 0$ by (8). If $\eta^\top\Gamma\eta = 0$, then u is constant along every edge of the strongly connected graph of G, so u is constant. Then $u = 0$ because $\pi \cdot u = 0$, so $\bar\eta = 0$ and $\eta \in \mathbb{R}\mathbf{1}$. A nonzero vector that vanishes at state 1 is not in $\mathbb{R}\mathbf{1}$, which gives the claim about Σ. □

Proof of Proposition 5.2(a) Step 1: the derivatives of the Perron root. For $\vartheta \in \mathbb{R}^F$ let $\Lambda(\vartheta) = \varrho(Q_F + \operatorname{diag}\vartheta)$, so that $\Lambda(\vartheta + c\mathbf{1}) = \Lambda(\vartheta) + c$ (the shift property). It is a simple eigenvalue with positive right and left eigenvectors $\hat r, \hat l$, normalized by $\sum_i \hat r_i = 1$ and $\hat l \cdot \hat r = 1$, and by the implicit function theorem Λ, $\hat r$ and $\hat l$ are real-analytic in ϑ. Put $\pi = \hat l \circ \hat r$ and $G = \hat R^{-1}(Q_F + \operatorname{diag}\vartheta - \Lambda(\vartheta)I)\hat R$ with $\hat R = \operatorname{diag}\hat r$. Then $G\mathbf{1} = 0$ and $\pi^\top G = 0$ by the two eigenvector equations, so G is an irreducible generator with stationary law π. Since $\hat R$ commutes with diagonal matrices, $\Lambda(\vartheta + \eta) = \Lambda(\vartheta) + \varrho(G + \operatorname{diag}\eta)$. So by Lemma A.1, $\nabla\Lambda(\vartheta) = \pi$ and $\nabla^2\Lambda(\vartheta) = \Gamma$, with Γ built from (G, π).

Step 2: the gradient can be inverted. Let $\Lambda_0(\beta) = \Lambda(0, \beta)$ for $\beta \in \mathbb{R}^{k-1}$. Its Hessian is Γ without the row and column of state 1, which is positive definite by Lemma A.1. So Λ_0 is strictly convex, $\nabla\Lambda_0$ is injective, and by the inverse function theorem it has a real-analytic inverse on its open image.

Step 3: the tilt that belongs to α. Fix $\alpha \in \Delta_F^\circ$ and put $\vartheta(\alpha) = q - \theta(\alpha)$. Then $Q_F + \operatorname{diag}\vartheta(\alpha) = A_F - \operatorname{diag}\theta(\alpha)$ has the positive right and left eigenvectors r_α and l_α for the eigenvalue 0, by Definition 5.1 and (7). So its Perron root is 0, and at $\vartheta(\alpha)$ we have $\pi = \alpha$,

$G = \widehat{Q}_\alpha$ and $\Gamma = \Gamma_\alpha$. Let $\beta(\alpha)$ be the last $k-1$ coordinates of $\vartheta(\alpha) - \vartheta_1(\alpha)\mathbf{1}$. A shift along $\mathbf{1}$ does not change the derivatives of Λ, so $\nabla\Lambda_0(\beta(\alpha)) = \alpha'$ and $\nabla^2\Lambda_0(\beta(\alpha)) = \Sigma_\alpha$. Hence Σ_α is positive definite, and $\beta(\alpha) = (\nabla\Lambda_0)^{-1}(\alpha')$ is real-analytic. The shift property and $\Lambda(\vartheta(\alpha)) = 0$ give $\vartheta_1(\alpha) = -\Lambda_0(\beta(\alpha))$. So $\theta(\alpha)$ is real-analytic, and then so are its Perron vectors and Σ_α.

Step 4: the Hessian of J_F. Since $\vartheta(\alpha) = \vartheta_1(\alpha)\mathbf{1} + (0, \beta(\alpha))$ and $\sum_i \alpha_i = 1$, we have $J_F(\alpha) = \langle \vartheta(\alpha), \alpha\rangle = \langle \beta(\alpha), \alpha'\rangle - \Lambda_0(\beta(\alpha))$. We differentiate in α'. The terms with the derivative of β cancel because $\nabla\Lambda_0(\beta(\alpha)) = \alpha'$, and $\nabla J_F = \beta$ remains. So $\nabla^2 J_F$ is the Jacobian of $(\nabla\Lambda_0)^{-1}$ at α', which is Σ_α^{-1}. The formula for C_F follows from Definition 5.1. □

A.2 The torus representation

The generating function of the words in H is Good's resolvent formula [20, (2)], and the torus integral below is his constant-term formula, here for the killed matrix A_F and with the circle of one state integrated out. We split

$$A_F = \begin{pmatrix} 0 & a_1^\top \\ b_1 & A' \end{pmatrix}, \qquad \mu_F = (\mu_1, \mu'),$$

so $a_1 = (q_{1j})_{j\ge 2}$, $b_1 = (q_{i1})_{i\ge 2}$ and $A' = (q_{ij})_{i,j\ge 2}$. For $z' \in \mathbb{C}^{k-1}$ let $\mathcal{Z}' = \operatorname{diag}(z')$. Where $\mathcal{Z}' - A'$ is invertible we put $X(z') = (\mathcal{Z}' - A')^{-1}$, $g(z') = a_1^\top X(z') b_1$ and

$$b(z') = \big(\mu_1 + \mu'^\top X(z') b_1\big)\big(1 + a_1^\top X(z')\mathbf{1}\big).$$

Lemma A.2 (torus representation) *Let $s' \in (0,\infty)^{k-1}$ satisfy $\varrho(A' - \operatorname{diag} s') < 0$. On the set of $z' \in \mathbb{C}^{k-1}$ with $|z_i| \ge s_i$ for every $i \ge 2$, the matrix $\mathcal{Z}' - A'$ is invertible, and X, g and b are given by absolutely convergent power series in $1/z_2, \dots, 1/z_k$ with nonnegative coefficients. Hence $|g(z')| \le g(s')$ and $|b(z')| \le b(s')$ there. For every $t \in (0,\infty)^F$,*

$$H(t) = \frac{1}{(2\pi)^{k-1}} \int_{\mathbb{T}^{k-1}} \exp\Big(t_1 g(z') + \sum_{i\ge 2} t_i z_i\Big)\, b(z') \prod_{i\ge 2} z_i \, d\varphi, \qquad z_i = s_i e^{\mathrm{i}\varphi_i}.$$

In words, $H(t)$ is a Cauchy integral over $k-1$ circles, one for each state of F except state 1. The circle of state 1 has already been integrated out, and it leaves the factor $e^{t_1 g(z')}$.

Proof Step 1: the series. By the first fact, $\varrho(\operatorname{diag}(s')^{-1} A') < 1$. So the series $X(z') = \sum_{j\ge 0} (\mathcal{Z}'^{-1} A')^j \mathcal{Z}'^{-1}$ converges absolutely for $|z_i| \ge s_i$, since its terms are bounded entry-wise in modulus by those at $z' = s'$. As a_1, b_1 and μ are nonnegative, this gives the claims about X, g and b.

Step 2: a radius for state 1. Choose $s_1 > g(s')$ and put $S = \operatorname{diag}(s_1, s')$. We show that $\varrho(S^{-1}A_F) < 1$. For $\varepsilon > 0$ let $v = (1, v')$ with $v' = X(s')(b_1 + \varepsilon\mathbf{1}) > 0$. Then $((S - A_F)v)_i = \varepsilon$ for $i \ge 2$, and $((S - A_F)v)_1 = s_1 - g(s') - \varepsilon a_1^\top X(s')\mathbf{1}$. For small ε both are positive, and the first fact gives the claim.

Step 3: k circles. Let $\mathcal{Z} = \operatorname{diag}(z)$ with $|z_i| = s_i$ for every $i \in F$. Since $\varrho(A_F S^{-1}) = \varrho(S^{-1}A_F) < 1$ (the two matrices are similar), expanding $(\mathcal{Z} - A_F)^{-1} = \mathcal{Z}^{-1}(I - A_F\mathcal{Z}^{-1})^{-1}$

gives

$$\mu_F^\top (\mathcal{Z} - A_F)^{-1}\mathbf{1} = \sum_c \mu_{c_1} \prod_{j<M} q_{c_j c_{j+1}} \prod_{i\in F} z_i^{-m_i(c)},$$

absolutely and uniformly on the torus. Here $c = c_1 \cdots c_M$ runs over the words in F with $M \geq 1$ and $c_j \neq c_{j+1}$, and $m_i(c)$ is the number of letters i in c. For an integer m and $t_i > 0$, Cauchy's formula gives $\frac{1}{2\pi}\int_0^{2\pi} e^{t_i z_i} z_i^{1-m}\, d\varphi_i = t_i^{m-1}/(m-1)!$ if $m \geq 1$ and 0 if $m \leq 0$. We multiply the series by $e^{\langle t,z\rangle}\prod_i z_i$ and integrate term by term. The words that miss a state of F drop out, and grouping the others by $m(c)$ gives

$$H(t) = \frac{1}{(2\pi)^k}\int_{\mathbb{T}^k} e^{\langle t,z\rangle}\, \mu_F^\top (\mathcal{Z} - A_F)^{-1}\mathbf{1} \prod_{i\in F} z_i\, d\varphi. \tag{A2}$$

Step 4: the first circle. The Schur complement $z_1 - g(z')$ of $\mathcal{Z}' - A'$ in $\mathcal{Z} - A_F$ is not 0, since $|g(z')| \leq g(s') < s_1$. Block inversion gives

$$\mu_F^\top (\mathcal{Z} - A_F)^{-1}\mathbf{1} = \frac{b(z')}{z_1 - g(z')} + \mu'^\top X(z')\mathbf{1}.$$

We integrate over φ_1 in (A2) and divide by 2π. The second term does not depend on z_1, so it contributes $\frac{1}{2\pi}\int_0^{2\pi} e^{t_1 z_1} z_1\, d\varphi_1 = 0$. In the first term $g(z')$ lies inside the circle $|z_1| = s_1$, so Cauchy's formula gives $e^{t_1 g(z')} b(z')$. The factors without z_1 are unchanged, and this proves the formula. □

Section 6 needs the same formula for complex weights and an end vector other than $\mathbf{1}$. Let A be a complex matrix on F with zero diagonal and let $\nu, e \in \mathbb{C}^F$. Let $W_{A,\nu,e}(m)$ be the sum over the words c of $W(m)$ of $\nu_{c_1}\prod_{j<M} A_{c_j c_{j+1}} e_{c_M}$, and $H_{A,\nu,e}(t)$ the sum $H(t)$ with $W_{A,\nu,e}$ in place of W. So $H = H_{A_F,\mu,\mathbf{1}}$. We split A and ν as above and define X and g by the same formulas, and

$$b(z') = \big(\nu_1 + \nu'^\top X(z')b_1\big)\big(e_1 + a_1^\top X(z')e'\big).$$

Lemma A.3 (complex weights) *Let $\bar{A} \geq 0$ be a matrix on F with zero diagonal and $\bar{\nu}, \bar{e} \in [0,\infty)^F$ with $|A_{ij}| \leq \bar{A}_{ij}$, $|\nu_i| \leq \bar{\nu}_i$ and $|e_i| \leq \bar{e}_i$ for all i, j, and let $\bar{X}$, $\bar{g}$ and $\bar{b}$ be the functions above for $(\bar{A}, \bar{\nu}, \bar{e})$. Let $s' \in (0,\infty)^{k-1}$ satisfy $\varrho(\bar{A}' - \operatorname{diag} s') < 0$. Then Lemma A.2 holds for $H_{A,\nu,e}$ with these X, g and b, except that their series are dominated term by term in modulus by those of $\bar{X}$, $\bar{g}$ and $\bar{b}$ at s'. In particular $|g(z')| \leq \bar{g}(s')$ and $|b(z')| \leq \bar{b}(s')$ when $|z_i| \geq s_i$ for every $i \geq 2$. Where $\mathcal{Z}' - A'$ is invertible, X, g and b are rational functions of z' and of the entries of A, ν and e.*

Proof We follow the four steps of the proof of Lemma A.2. In Step 1 the first fact gives $\varrho(\operatorname{diag}(s')^{-1}\bar{A}') < 1$, and for $|z_i| \geq s_i$ the entries of $\mathcal{Z}'^{-1}A'$ are bounded in modulus by those of $\operatorname{diag}(s')^{-1}\bar{A}'$. So the series of X converges absolutely, and its terms are dominated by those of $\bar{X}(s') \geq 0$. Since a_1, b_1, ν and e are dominated by their barred versions, the same holds for g and for the two factors of b. Step 2 applies to $\bar{A}$ as it stands and gives $s_1 > \bar{g}(s')$ with $\varrho(S^{-1}\bar{A}) < 1$. In Step 3 the entries of $A\mathcal{Z}^{-1}$ are bounded in modulus by those of $\bar{A}S^{-1}$, so the word expansion of $\nu^\top(\mathcal{Z} - A)^{-1}e$ converges absolutely and uniformly on the torus, and the same integration gives (A2) with A, ν and e in place of A_F, μ_F and $\mathbf{1}$. In Step 4 the Schur complement $z_1 - g(z')$ is not 0, since $|g(z')| \leq \bar{g}(s') < s_1$, and block inversion gives $\nu^\top(\mathcal{Z} - A)^{-1}e = b(z')/(z_1 - g(z')) + \nu'^\top X(z')e'$. The rest of Step 4 is unchanged. The last claim is Cramer's rule. □

A.3 Proof of Lemma 4.1

We bound the sum of Lemma 2.3 term by term by a continuum sum H_σ with slightly larger rates, and H_σ by the largest modulus of the integrand of Lemma A.2.

Step 1: the radii. Let F be irreducible and $\theta \in V_F$. Then $A_F - \operatorname{diag}\theta$ is an irreducible Metzler matrix with Perron root 0. So it has a right eigenvector $u > 0$ for 0, and $\theta_i = (A_F u)_i / u_i$. Each row of A_F has a positive entry, since G_F is strongly connected and $k \geq 2$, so $\theta > 0$. By the second fact, $\varrho(A' - \operatorname{diag}\theta') < 0$. The rows $i \geq 2$ of $(A_F - \operatorname{diag}\theta)u = 0$ give $u' = u_1 X(\theta') b_1$, and then row 1, which reads $a_1^\top u' = \theta_1 u_1$, gives $\theta_1 = g(\theta')$. Hence

$$\theta > 0, \qquad \varrho(A' - \operatorname{diag}\theta') < 0, \qquad g(\theta') = \theta_1 \qquad \text{for every } \theta \in V_F. \tag{A3}$$

Step 2: a continuum sum. Fix n with support F and h with $hq_{\max} \leq \frac{1}{2}$, and put $|n| = \sum_i n_i$, $t = hn$ and $\sigma = e^{hq_{\max}}$. Lemma 2.3 with $|n|$ in place of N gives the probability $P_{|n|}(n)$ in the lemma. Only terms with $m_i \leq n_i$ for every i are nonzero, and in them $h^{m_i - 1}\binom{n_i - 1}{m_i - 1} \leq t_i^{m_i - 1}/(m_i - 1)!$ and $(1 - hq_i)^{n_i - m_i} \leq e^{-hq_i(n_i - m_i)} \leq e^{-q_i t_i}\sigma^{m_i}$. Note that $h^{M-1} = h^{k-1}\prod_i h^{m_i - 1}$ and $\sigma^M W(m) = \sigma W_\sigma(m)$, where W_σ is W for σA_F in place of A_F (a word with M letters has $M - 1$ switches). So

$$P_{|n|}(n) \leq \sigma\, h^{k-1} e^{-\langle q, t\rangle} H_\sigma(t), \tag{A4}$$

where H_σ is H for σA_F.

Step 3: the bound for H_σ. We use Lemma A.2 for σA_F with the radii $\sigma\theta'$, which is allowed since $\varrho(\sigma A' - \operatorname{diag}\sigma\theta') = \sigma\varrho(A' - \operatorname{diag}\theta') < 0$. The functions of Lemma A.2 for σA_F satisfy $X_\sigma(\sigma z') = \sigma^{-1} X(z')$, so $g_\sigma(\sigma z') = \sigma g(z')$ and $b_\sigma(\sigma z') = b(z')$. So on the torus with radii $\sigma\theta'$ we have $|g_\sigma| \leq \sigma g(\theta') = \sigma\theta_1$ by (A3), $|b_\sigma| \leq b(\theta')$ and $|z_i| = \sigma\theta_i$. Hence the integrand has modulus at most $e^{\sigma\langle t, \theta\rangle} b(\theta')\sigma^{k-1}\prod_{i\geq 2}\theta_i$, and for every $\sigma > 0$

$$H_\sigma(t) \leq \sigma^{k-1} b(\theta') \prod_{i \geq 2} \theta_i\, e^{\sigma\langle t, \theta\rangle}. \tag{A5}$$

With (A4) this gives $P_{|n|}(n) \leq \sigma^k b(\theta')\prod_{i\geq 2}\theta_i\, h^{k-1}\exp(-h\langle n, q - \theta\rangle + (\sigma - 1)h\langle n, \theta\rangle)$. Finally $\sigma \leq e^{1/2}$ and $\sigma - 1 \leq 2hq_{\max}$, so the lemma holds with

$$C_\theta = \max\Big(e^{k/2} b(\theta') \prod_{i \geq 2} \theta_i,\ 2q_{\max} \max_i \theta_i\Big).$$

By Lemma A.2, b is continuous on the open set of s' with $\varrho(A' - \operatorname{diag} s') < 0$, which contains θ' for every $\theta \in V_F$ by (A3). So C_θ is bounded on compact subsets of V_F. □

A.4 Proof of Theorem 5.4

The field cancels from the two sides of Theorem 5.4, so we take $v = 0$. For $k = 1$ the formula is part of Lemma 2.4. Let $k \geq 2$ and $\alpha \in \mathcal{K}$, and write $\theta = \theta(\alpha)$, $r = r_\alpha$, $l = l_\alpha$

and $\Theta' = \operatorname{diag}(\theta')$. By (A3) we may use Lemma A.2 with the radii θ' and $t = T\alpha$. With Lemma 2.4 this gives

$$f_T(\alpha) = \frac{T^{k-1} e^{-T\langle q,\alpha\rangle}}{(2\pi)^{k-1}} \int_{\mathbb{T}^{k-1}} e^{T\Psi(\alpha,\varphi)} \chi(\alpha,\varphi)\, d\varphi, \tag{A6}$$

where $z_i = \theta_i e^{\mathrm{i}\varphi_i}$, $\Psi(\alpha,\varphi) = \alpha_1 g(z') + \sum_{i\ge 2} \alpha_i z_i$ and $\chi(\alpha,\varphi) = b(z') \prod_{i\ge 2} z_i$. We check the hypotheses of Lemma B.1 with $m = k-1$.

Hypothesis (iii). By (A3), $\Psi(\alpha,0) = \langle \alpha, \theta\rangle$ is real. By Lemma A.2, $|g(z')| \le g(\theta') = \theta_1$ on the torus, so

$$\operatorname{Re}\Psi(\alpha,\varphi) \le \Psi(\alpha,0) - \sum_{i\ge 2} \alpha_i\theta_i(1-\cos\varphi_i).$$

All α_i and θ_i are positive, so the sum is positive unless $\varphi = 0$.

Hypotheses (i) *and* (ii). We need the first two derivatives of g at θ'. We get them by differentiating a relation $\lambda(g(z'), z') = 0$ for a Perron root λ, whose derivatives are known from Lemma A.1. For $z \in \mathbb{R}^F$ near θ let $\lambda(z) = \varrho(A_F - \operatorname{diag} z)$. With $R = \operatorname{diag} r$ we have $A_F - \operatorname{diag}(\theta - \eta) = R(\widehat{Q}_\alpha + \operatorname{diag}\eta)R^{-1}$, and $\widehat{Q}_\alpha$ is an irreducible generator with stationary law α. So $\lambda(\theta-\eta) = \varrho(\widehat{Q}_\alpha + \operatorname{diag}\eta)$, and Lemma A.1 gives $\nabla\lambda(\theta) = -\alpha$ and $\nabla^2\lambda(\theta) = \Gamma_\alpha$. For real z' near θ' we have $\varrho(A' - \operatorname{diag} z') < 0$, so $(1, X(z')b_1) \ge 0$ by the first fact. This vector lies in the kernel of $A_F - \operatorname{diag}(g(z'), z')$ (rows $i \ge 2$ by the definition of X, and row 1 by the definition of g). By the second fact a nonnegative eigenvector belongs to the Perron root, so $\lambda(g(z'), z') = 0$.

Write g_j, g_{ij} for the derivatives of g at θ' and λ_j, λ_{ij} for those of λ at θ. Differentiating $\lambda(g(z'), z') = 0$ in z_j gives $\lambda_1 g_j + \lambda_j = 0$, and differentiating again in z_i gives $\lambda_1 g_{ij} + \lambda_{11} g_i g_j + \lambda_{1j} g_i + \lambda_{i1} g_j + \lambda_{ij} = 0$. Put $w_j = e_j - (\alpha_j/\alpha_1)e_1 \in \mathbb{R}^F$ for $j \ge 2$. Then the first equation is $g_j = -\alpha_j/\alpha_1$, so $w_j = e_j + g_j e_1$, and the last four terms of the second equation are $w_i^\top \Gamma_\alpha w_j$. So

$$g_j = -\frac{\alpha_j}{\alpha_1}, \qquad \alpha_1 g_{ij} = w_i^\top \Gamma_\alpha w_j.$$

Hence $\partial\Psi/\partial z_j = \alpha_1 g_j + \alpha_j = 0$ at θ', and the Hessian $\mathcal{H}$ of Ψ in z' at θ' is $V^\top \Gamma_\alpha V$, where V has columns $w_2, \dots, w_k$. Since $\partial/\partial\varphi_j = \mathrm{i} z_j\, \partial/\partial z_j$ and the first derivatives vanish, $\nabla_\varphi \Psi(\alpha, 0) = 0$. We saw above that $\Psi(\alpha,0)$ is real, so this is hypothesis (i). Also $-\nabla_\varphi^2 \Psi(\alpha,0) = \Theta'\mathcal{H}\Theta'$, which is real and symmetric, as (ii) requires.

For the rest of (ii), namely that this matrix is positive definite, let $B = I + \mathbf{1}\alpha'^\top/\alpha_1$. For $y \in \mathbb{R}^{k-1}$ the vector Vy has the entry $-\alpha'^\top y/\alpha_1$ at state 1 and the entries y elsewhere, so $Vy + (\alpha'^\top y/\alpha_1)\mathbf{1} = (0, By)$. Since $\Gamma_\alpha \mathbf{1} = 0$, we may replace Vy by $(0, By)$ in $y^\top V^\top \Gamma_\alpha V y$, which gives $\mathcal{H} = B^\top \Sigma_\alpha B$. The matrix determinant lemma gives $\det B = 1 + \alpha'^\top \mathbf{1}/\alpha_1 = 1/\alpha_1$. By Lemma A.1, Σ_α is positive definite, and hence so are $\mathcal{H}$ and $\Theta'\mathcal{H}\Theta'$. Also $\det \mathcal{H} = \det \Sigma_\alpha / \alpha_1^2$.

Regularity. By Proposition 5.2(a), $\theta(\alpha)$ is real-analytic. Fix $\alpha_0 \in \mathcal{K}$. By (A3) and continuity there is $\delta \in (0, \min_i \theta_i(\alpha_0))$ with $\varrho(A' - \operatorname{diag}(\theta'(\alpha_0) - \delta\mathbf{1})) < 0$. By Lemma A.2, X, g and b are holomorphic where $|z_i| > \theta_i(\alpha_0) - \delta$ for every $i \ge 2$, and

this holds on the torus of radii $\theta'(\alpha)$ for α near α_0. So Ψ and χ are real-analytic near $\{\alpha_0\}\times\mathbb{T}^{k-1}$. Covering $\mathcal{K}$ by finitely many such sets shows that their φ-derivatives are jointly continuous on $\mathcal{K}\times\mathbb{T}^{k-1}$.

The constant. The matrix $\mathcal{J} = \Theta'\mathcal{H}\Theta'$ of Lemma B.1 has $\det\mathcal{J} = \prod_{i\ge2}\theta_i^2\,\det\Sigma_\alpha/\alpha_1^2$. So the lemma gives, uniformly for $\alpha\in\mathcal{K}$ and $T\ge1$,

$$\int_{\mathbb{T}^{k-1}} e^{T\Psi}\chi\,d\varphi = e^{T\langle\alpha,\theta\rangle}\Big(\frac{2\pi}{T}\Big)^{\frac{k-1}{2}}\frac{\alpha_1}{\prod_{i\ge2}\theta_i\sqrt{\det\Sigma_\alpha}}\Big(\chi(\alpha,0)+O\Big(\frac{1}{T}\Big)\Big).$$

Here $\chi(\alpha,0)=b(\theta')\prod_{i\ge2}\theta_i$. As in the derivation of (A3), $r'=r_1X(\theta')b_1$. In the same way the columns $j\ge2$ of $l^\top(A_F-\operatorname{diag}\theta)=0$, which is (7), give $l'=l_1X(\theta')^\top a_1$. Hence $\mu\cdot r=r_1(\mu_1+\mu'^\top X(\theta')b_1)$ and $l\cdot\mathbf{1}=l_1(1+a_1^\top X(\theta')\mathbf{1})$. Since $l_1r_1=\alpha_1$,

$$b(\theta')=\frac{(\mu\cdot r_\alpha)(l_\alpha\cdot\mathbf{1})}{\alpha_1}.$$

As $r_\alpha,l_\alpha>0$ and $\mu(F)>0$ by (H), $\chi(\alpha,0)$ is positive and continuous, hence bounded below on $\mathcal{K}$. So the last bracket is $\chi(\alpha,0)(1+O(1/T))$. Putting this into (A6) and using $J_F(\alpha)=\langle q-\theta,\alpha\rangle$ gives $f_T(\alpha)=T^{(k-1)/2}C_F(\alpha)e^{-TJ_F(\alpha)}(1+O(1/T))$, uniformly on $\mathcal{K}$. □

The proof uses A_F, μ and the end vector $\mathbf{1}$ only through their signs and the graph of A_F. Section 6 needs it for other weights.

Lemma A.4 (general weights) *Let $A\ge0$ be a matrix on F with zero diagonal and a strongly connected graph, let $\nu,e\in[0,\infty)^F$ be nonzero, and let $\theta(\alpha)$, r_α, l_α and Σ_α be the objects of Definition 5.1 for A in place of A_F. Let g and b be the functions of Lemma A.3 for (A,ν,e). Then the identities (A3), $g_j=-\alpha_j/\alpha_1$, $\mathcal{H}=B^\top\Sigma_\alpha B$, $\det\mathcal{H}=\det\Sigma_\alpha/\alpha_1^2$ and $\alpha_1 b(\theta')=(\nu\cdot r_\alpha)(l_\alpha\cdot e)$ of the proof above hold, and uniformly for α in a compact $\mathcal{K}\subset\Delta_F^\circ$ and $T\ge1$*

$$H_{A,\nu,e}(T\alpha)=(2\pi T)^{-(k-1)/2}e^{T\langle\alpha,\theta(\alpha)\rangle}\,\frac{(\nu\cdot r_\alpha)(l_\alpha\cdot e)}{\sqrt{\det\Sigma_\alpha}}\,\big(1+O(1/T)\big).$$

Proof Definition 5.1, Proposition 5.2(a) and Lemma A.1 involve A_F only as a nonnegative matrix with zero diagonal and a strongly connected graph, so they hold for A. We repeat the proof above with Lemma A.3, for $(\bar A,\bar\nu,\bar e)=(A,\nu,e)$, in place of Lemma A.2. The paragraphs on the hypotheses and on the regularity do not change. In the paragraph on the constant the identities $r'=r_1X(\theta')b_1$ and $l'=l_1X(\theta')^\top a_1$ give $\nu\cdot r_\alpha=r_1(\nu_1+\nu'^\top X(\theta')b_1)$ and $l_\alpha\cdot e=l_1(e_1+a_1^\top X(\theta')e')$, and $l_1r_1=\alpha_1$ gives the formula for $b(\theta')$. It is positive since $r_\alpha,l_\alpha>0$ and $\nu,e\ge0$ are nonzero. Then (A6) without the factor $T^{k-1}e^{-T\langle q,\alpha\rangle}$ and the Laplace lemma give the display. □

A.5 Proof of Lemma 5.5

Idea of the proof. The discrete law of Lemma 2.3 and the continuum density of Lemma 2.4 are sums over the same vectors m of run counts, and we compare them

term by term. The two terms for m differ by a factor κ_m that is close to 1 when the number of runs M is small compared with $\sqrt{N}$, and under the continuum weights a path has of order T runs.

Step 1: the ratio of the terms. Let $t = hn$ and $\alpha = n/N \in \mathcal{K}$. Lemma 2.4 gives the identity $N^{-(k-1)} f_T(n/N) = h^{k-1} e^{-\langle q,t\rangle} H(t) = \sum_m v_m$ with

$$v_m = W(m)\, h^{k-1} \prod_{i\in F} e^{-q_i t_i} \frac{t_i^{m_i-1}}{(m_i-1)!}, \qquad m \in \mathbb{Z}^F_{\geq 1}.$$

Since $h^{m_i-1}\binom{n_i-1}{m_i-1} = \frac{t_i^{m_i-1}}{(m_i-1)!}\prod_{1\leq j<m_i}(1-j/n_i)$, the m-th term of Lemma 2.3 is $\kappa_m v_m$ with

$$\kappa_m = \prod_{i\in F}\Big[\prod_{1\leq j<m_i}\Big(1-\frac{j}{n_i}\Big)\Big](1-hq_i)^{n_i-m_i}e^{hq_in_i}.$$

Let $w_m = v_m/\sum_{m'} v_{m'}$, a probability on the vectors m since $H(t) > 0$ under (H). We write $\mathbb{E}_w$ and $\mathbb{P}_w$ for it and regard $M = \sum_i m_i$ as a random variable. By Lemma 2.3 the ratio in the lemma is $\sum_m \kappa_m v_m / \sum_m v_m = \mathbb{E}_w \kappa_m$.

Step 2: bounds for κ_m. Let $n_{\min} = \min_i n_i$. We claim that for every m

$$\Big(1-\frac{M^2}{2n_{\min}}\Big)e^{-q_{\max}^2 T^2/N} \leq \kappa_m \leq e^{hq_{\max}M}. \tag{A7}$$

First let $m_i \leq n_i$ for every i. The factors $1-j/n_i$ are at most 1, and $(1-hq_i)^{n_i-m_i} \leq e^{-hq_i(n_i-m_i)}$, so $\kappa_m \leq e^{hq_{\max}M}$. For the lower bound, $\log(1-x) \geq -x-x^2$ for $0 \leq x \leq \frac12$ gives $(1-hq_i)^{n_i-m_i}e^{hq_in_i} \geq e^{-n_ih^2q_i^2}$, and the product of these over i is at least $e^{-q_{\max}^2T^2/N}$. The factors $1-j/n_i$ lie in $[0,1]$, so their product is at least 1 minus the sum of the j/n_i, and hence at least $1-\sum_i m_i^2/(2n_i) \geq 1-M^2/(2n_{\min})$. Now let $m_i > n_i$ for some i. Then $\kappa_m = 0$, while $M \geq n_{\min}+1$ gives $M^2/(2n_{\min}) \geq 2$, so the left side of (A7) is negative.

By (A7) it remains to bound $\mathbb{E}_w e^{hq_{\max}M}$ and $\mathbb{E}_w M^2$ from above.

Step 3: the number of runs under w. Since W for σA_F is $\sigma^{M-1}W(m)$, we have $\mathbb{E}_w\sigma^{M-1} = H_\sigma(t)/H(t)$. So the function $\Upsilon(c) = \log \mathbb{E}_w e^{c(M-1)} = \log H_{e^c}(t) - \log H(t)$ is finite. It is convex with $\Upsilon(0) = 0$, as the logarithm of a moment generating function, and $\Upsilon(1) \geq 0$ since $M \geq 1$. So $\Upsilon(c) \leq c\,\Upsilon(1)$ for $0 \leq c \leq 1$. We show that $\Upsilon(1) \leq C_{\mathcal{K}}T$, with an upper bound for H_e and a lower bound for H at $t = T\alpha$. By (A5) with $\sigma = e$ and $\theta = \theta(\alpha)$, $H_e(T\alpha) \leq e^{k-1}b(\theta')\prod_{i\geq 2}\theta_i\, e^{eT\langle\alpha,\theta\rangle}$. For the lower bound note that $H(T\alpha)e^{-T\langle\alpha,\theta\rangle} = T^{-(k-1)}e^{TJ_F(\alpha)}f_T(\alpha)$, by the definition of f_T and $J_F(\alpha) = \langle q-\theta,\alpha\rangle$. By Theorem 5.4 this is at least $\frac12 T^{-(k-1)/2}C_F(\alpha)$ for $T \geq T_1$, and for $1 \leq T \leq T_1$ it is continuous and positive on $[1,T_1]\times\mathcal{K}$. So in both cases $H(T\alpha) \geq c_{\mathcal{K}}T^{-(k-1)/2}e^{T\langle\alpha,\theta\rangle}$. Since $\theta(\alpha)$ and $b(\theta'(\alpha))$ are bounded on $\mathcal{K}$, dividing the two bounds gives $\Upsilon(1) \leq C_{\mathcal{K}}T$.

Step 4: the two averages. Let $x = hq_{\max} \leq \frac12$. Then $\mathbb{E}_w e^{xM} = e^{x+\Upsilon(x)} \leq e^{x(1+C_{\mathcal{K}}T)} = 1 + O(T^2/N)$, because $x \leq q_{\max}T/N$ and $T^2/N \leq 1$. For the second moment we use the Chernoff bound $\mathbb{P}_w(M-1 \geq y) \leq e^{\Upsilon(1)-y}$ for $y \geq \Upsilon(1)$ and the

bound 1 for $y < \Upsilon(1)$. This gives

$$\mathbb{E}_w(M-1)^2 = \int_0^\infty 2y\,\mathbb{P}_w(M-1 \ge y)\,dy$$
$$\le \Upsilon(1)^2 + \int_{\Upsilon(1)}^\infty 2ye^{\Upsilon(1)-y}\,dy = \Upsilon(1)^2 + 2\Upsilon(1) + 2,$$

so $\mathbb{E}_w M^2 = O(T^2)$. Finally $n_{\min} \ge \varepsilon_{\mathcal{K}} N$ with $\varepsilon_{\mathcal{K}} = \min_{\alpha\in\mathcal{K}} \min_i \alpha_i > 0$. So by (A7), $\mathbb{E}_w \kappa_m$ lies between $e^{-q_{\max}^2 T^2/N}(1 - \mathbb{E}_w M^2/(2\varepsilon_{\mathcal{K}} N))$ and $\mathbb{E}_w e^{hq_{\max} M}$, and both are $1 + O(T^2/N)$, uniformly for $\alpha \in \mathcal{K}$ and $1 \le T \le N^{1/2}$. □

A.6 Proof of Theorem 4.2(ii)

Let F be admissible, with components $C_1, \dots, C_r$. If F is irreducible, then $r = 1$, and the argument below uses only Lemma 4.1 and an explicit family of paths, not the local law.

Idea of the proof. A path with support F visits the components one after the other, in r blocks, and each passage to the next block is one switch. For the upper bound we apply Lemma 4.1 to every block. For the lower bound we choose the block lengths ourselves. A component of least exit rate gets almost all the time, and every other component C gets only about N/T steps, the time scale of a single switch. Such a short block has probability bounded below, and it has only about $(N/T)^{|C|-1}$ compositions instead of $N^{|C|-1}$.

If $Q = 0$, only singletons with positive start mass are admissible, and the bounds follow from $M_{\{i\}} = \mu_i$. Otherwise let a be the least positive rate q_{ij}. Since $T \le N^{1/3}$, we have $h \le N^{-2/3}$ and $T^2/N \le 1$, and we take N_0 so large that h is as small as we need. As in the proof of Lemma 2.2, a path of positive probability with support F visits each component in one block of consecutive times, and it passes from one block to the next by one switch. We call the order of the blocks, the first state and these switches the pattern of the path. There are finitely many patterns.

Upper bound. Fix a pattern whose j-th block lies in the component D_j and starts at y_j, and whose switch out of block j is $x_j \to y_{j+1}$ for $j < r$. So block j ends at x_j for $j < r$, and the pattern does not fix the end of the last block. Let $\operatorname{supp} n = F$. Block j has length $N_j = \sum_{i\in D_j} n_i$. By the Markov property the paths with this pattern and composition n have total probability $\mu_{y_1} \prod_{j<r} hq_{x_j y_{j+1}} \prod_{j\le r} \varpi_j$. Here ϖ_j is the probability that the chain started at y_j stays in D_j for N_j visits, with the composition that n prescribes on D_j, and for $j < r$ also ends at x_j. We drop the condition on the end. Then ϖ_j is at most the probability that $X_1, \dots, X_{N_j}$ have that composition for the chain started at y_j. If $|D_j| \ge 2$, Lemma 4.1 with start y_j and $\theta = q - \lambda_{D_j}\mathbf{1}$ gives $\varpi_j \le \kappa_1 h^{|D_j|-1} e^{-h\lambda_{D_j} N_j + \kappa_1 h^2 N_j}$, where κ_1 is the largest of these finitely many constants. If $D_j = \{i\}$, then $\lambda_{D_j} = q_i$ and $\varpi_j = (1 - hq_i)^{N_j - 1} \le e^{1/2 - h\lambda_{D_j} N_j}$. Now we multiply over j. The powers of h add up to $\sum_j (|D_j| - 1) + r - 1 = k_F - 1$. In the exponent, $\lambda_{D_j} \ge \lambda_F$ and $\sum_j N_j = N$ give $h \sum_j \lambda_{D_j} N_j \ge T\lambda_F$ and $\kappa_1 h^2 \sum_j N_j = \kappa_1 T^2/N \le \kappa_1$. Summing over the patterns gives $P_N(n) \le \kappa_2 (T/N)^{k_F - 1} e^{-T\lambda_F}$.

Lower bound. We write down a family of paths with support F, bound its probability from below and count its compositions. By Lemma 2.2(a) we can number the components so that there are $y_1 \in C_1$ with $\mu_{y_1} > 0$ and $x_j \in C_j$, $y_{j+1} \in C_{j+1}$ with $q_{x_j y_{j+1}} > 0$ for $j < r$. Take any $x_r \in C_r$, choose j^* with $\lambda_{C_{j^*}} = \lambda_F$ and $|C_{j^*}| = k^*$, and put $m = \lceil N/T \rceil$, so that $1 \le hm \le 2$.

Claim A. There is $c_A > 0$ such that for every component C and all $y, x \in C$, the chain started at y stays in C for its first m visits, visits every state of C and has $X_m = x$ with probability at least c_A. Take a walk $y = v_0, \dots, v_J = x$ through every state of C with $q_{v_s v_{s+1}} > 0$ and $J \le |C|^2$. Consider the paths $X_1, \dots, X_m$ that make exactly these J switches, in this order, at any J of the $m-1$ transition times. There are $\binom{m-1}{J}$ of them, and each has probability at least $(ha)^J (1 - hq_{\max})^m$. For large N we have $m > |C|^2 \ge J$. Then $\binom{m-1}{J} h^J \ge (hm - Jh)^J / J! \ge (1 - Jh)^J / J!$ and $(1 - hq_{\max})^m \ge e^{-2hq_{\max} m} \ge e^{-4q_{\max}}$, since $1 \le hm \le 2$. So their total probability is at least $(1 - Jh)^J a^J e^{-4q_{\max}} / J!$, and $(1 - Jh)^J \ge 2^{-J}$ for small h.

Claim B. There is $c_B > 0$ such that for all $y, x \in C_{j^*}$ and $\ell \ge 2m$, the event of Claim A, with C_{j^*} in place of C and ℓ visits in place of m, has probability at least $c_B e^{-h\lambda_F \ell - h^2 \lambda_F^2 \ell}$. Let $w > 0$ satisfy $Q_{C_{j^*}} w = -\lambda_F w$ and let $B = I + hQ_{C_{j^*}} \ge 0$. Then $(B^s)_{zz'}$ is the probability of going from z to z' in s steps inside C_{j^*}. Since $B^s w = (1 - h\lambda_F)^s w$, the probability of staying in C_{j^*} for s steps from z is $\sum_{z'} (B^s)_{zz'} \ge (B^s w)_z / \max w \ge (1 - h\lambda_F)^s \min w / \max w$. We use Claim A for the first m visits, from y back to y, then $s = \ell - 2m + 1$ steps inside C_{j^*} to any state z', then Claim A for the last m visits, from z' to x. By the Markov property this gives at least $c_A^2 (\min w / \max w)(1 - h\lambda_F)^\ell$. Also $\lambda_F \le q_{\max}$, because the Perron root of $Q_{C_{j^*}}$ is at least each diagonal entry. So $h\lambda_F \le \frac{1}{2}$ and $\log(1 - h\lambda_F) \ge -h\lambda_F - h^2 \lambda_F^2$.

The family of paths. Give block $j \ne j^*$ the length m and block j^* the length $\ell = N - (r-1)m$, which is at least $2m$ for $T \ge T_0 = r+2$ and $N \ge N_0$. Consider the paths that go from y_j to x_j in block j as in the claims and pass from block j to block $j+1$ by the switch $x_j \to y_{j+1}$. They all have support F. By the Markov property, and since $h\ell \le T$ and $h^2 \ell \le T^2 / N \le 1$, their total probability is at least $\mu_{y_1} (ha)^{r-1} c_A^{r-1} c_B e^{-T\lambda_F - q_{\max}^2}$. The composition of such a path is fixed by its restrictions to the blocks. On the block in C_j the restriction is a vector of $|C_j|$ positive integers whose sum is m for $j \ne j^*$ and ℓ for $j = j^*$. A vector of positive integers with a given sum is fixed by all but one of its entries, and each entry is at most the sum. Also $\ell \le N$. So these paths have at most $(N+1)^{k^*-1} \prod_{j \ne j^*} (m+1)^{|C_j|-1}$ compositions, and $m + 1 \le 3N/T$. Since $\sum_{j \ne j^*} (|C_j| - 1) = k_F - k^* - (r-1)$ and $h = T/N$, dividing the total probability by the number of compositions gives $M_F \ge c\, N^{-(k_F - 1)} T^{k_F - k^*} e^{-T\lambda_F}$. □

A.7 A forest formula for the covariance

For weights $f_{ij} \ge 0$ on the ordered pairs $i \ne j$ of F and a state $i \in F$ let

$$\mathrm{arb}_i(f) = \sum_{\mathcal{T}} \prod_{a \ne i} f_{a\, p_{\mathcal{T}}(a)},$$

the sum over the spanning trees $\mathcal{T}$ of the complete graph on F, where $p_{\mathcal{T}}(a)$ is the neighbor of a on the path from a to i in $\mathcal{T}$. So $\mathrm{arb}_i(f)$ is the weighted count of the spanning trees oriented toward the root i. If f is symmetric, then $\mathrm{arb}_i(f) = \mathrm{tree}(f)$ for every i.

Lemma A.5 (a forest formula for the covariance) *Let G be an irreducible generator on F with stationary law π, where $k = |F| \geq 2$, and let D, Z and Γ be as at the start of this appendix, and Σ as in Lemma A.1 (namely Γ without the row and column of state 1). Put $f_{ij} = \pi_i G_{ij}$ for $i \neq j$, the stationary flow from i to j.*

(i) The number $\mathrm{arb}_i(f)$ does not depend on i. We write $\mathrm{arb}(f)$ for it. Also $\det(-G)_{(i)} = \pi_i \,\mathrm{arb}(f)/\prod_j \pi_j$ for every i, where (i) means that row and column i are deleted, and

$$\det(\mathbf{1}\pi^\top - G) = \sum_{i\in F} \det(-G)_{(i)} = \frac{\mathrm{arb}(f)}{\prod_j \pi_j}.$$

(ii) Let $S = -(DG + G^\top D)$. Then S is the Laplacian of the conductances $f_{ij} + f_{ji}$, and $\Gamma = Z^\top S Z$.

(iii) With $\mathrm{tree}(f + f^\top)$ the weighted count of spanning trees for the conductances $f_{ij} + f_{ji}$,

$$\det \Sigma = \frac{\mathrm{tree}(f + f^\top)}{\det(\mathbf{1}\pi^\top - G)^2} = \frac{(\prod_j \pi_j)^2 \,\mathrm{tree}(f + f^\top)}{\mathrm{arb}(f)^2}.$$

For two states with rates a and b this gives $\det \Sigma = 2ab/(a+b)^3$.

Proof We use three facts about the adjugate $\mathrm{adj}\, M$ of a $k \times k$ matrix M. First, $\mathrm{adj}(AB) = \mathrm{adj}(B)\,\mathrm{adj}(A)$ for all square matrices A and B, by the Cauchy–Binet formula. Second, $\det(M + uv^\top) = \det M + v^\top \mathrm{adj}(M)\, u$ for all vectors u and v. Third, suppose that M has rank $k-1$ and that $Mu = 0$ and $v^\top M = 0$ with u and v nonzero. Then $\mathrm{adj}\, M = m\, uv^\top$ for some number m. Indeed $M\,\mathrm{adj}\, M = \mathrm{adj}(M)M = (\det M)I = 0$, so every column of $\mathrm{adj}\, M$ lies in the kernel $\mathbb{R}u$ of M, and every row lies in the left kernel $\mathbb{R}v^\top$. A matrix with these two properties is a multiple of $uv^\top$.

(i) Let $\mathcal{L} = -DG$. Its off-diagonal entries are $-f_{ij}$, its row sums vanish because $G\mathbf{1} = 0$, and its column sums vanish because $\pi^\top G = 0$. The kernel of G is $\mathbb{R}\mathbf{1}$ because G is irreducible, so $\mathcal{L}$ has rank $k-1$. By the third fact with $u = v = \mathbf{1}$, all entries of $\mathrm{adj}\,\mathcal{L}$ are equal. By the matrix-tree theorem for directed graphs [10, 37], the diagonal entry $\det \mathcal{L}_{(i)}$ of $\mathrm{adj}\,\mathcal{L}$ is $\mathrm{arb}_i(f)$. So $\mathrm{arb}_i(f)$ does not depend on i. Deleting row and column i from $-G = D^{-1}\mathcal{L}$ gives $\det(-G)_{(i)} = \mathrm{arb}(f)/\prod_{j\neq i}\pi_j = \pi_i\,\mathrm{arb}(f)/\prod_j \pi_j$, and the sum over i is $\mathrm{arb}(f)/\prod_j \pi_j$ because $\sum_i \pi_i = 1$. Next, $-G$ has rank $k-1$, right kernel $\mathbb{R}\mathbf{1}$ and left kernel $\mathbb{R}\pi^\top$. So $\mathrm{adj}(-G) = g\,\mathbf{1}\pi^\top$ for some number g by the third fact. The trace of $\mathbf{1}\pi^\top$ is 1, so g is the trace of $\mathrm{adj}(-G)$, which is $\sum_i \det(-G)_{(i)}$. Since $\det(-G) = 0$, the second fact gives

$$\det(\mathbf{1}\pi^\top - G) = \pi^\top \mathrm{adj}(-G)\,\mathbf{1} = g\,(\pi^\top \mathbf{1})(\pi^\top \mathbf{1}) = g.$$

(ii) The matrix S is symmetric. For $i \neq j$ its entry is $-(\pi_i G_{ij} + \pi_j G_{ji}) = -(f_{ij} + f_{ji})$, and $S\mathbf{1} = -DG\mathbf{1} - G^\top \pi = 0$. So S is the Laplacian of the conductances $f_{ij} + f_{ji}$. By (8), $-GZ = I - \mathbf{1}\pi^\top$ and $\pi^\top Z = 0$. Hence

$$Z^\top(-DG)Z = Z^\top D(I - \mathbf{1}\pi^\top) = Z^\top D - (\pi^\top Z)^\top \pi^\top = Z^\top D,$$

where we used $D\mathbf{1} = \pi$. The transpose of this identity is $Z^\top(-G^\top D)Z = DZ$. Adding the two gives $Z^\top S Z = Z^\top D + DZ = \Gamma$.

(iii) Put $w = \det(\mathbf{1}\pi^\top - G)$, which is not 0 because Z exists. By the matrix-tree theorem [6] every diagonal entry of $\operatorname{adj} S$ is $\operatorname{tree}(f + f^\top)$. This number is positive because the graph of G is strongly connected. So S has rank $k-1$, and the third fact with $u = v = \mathbf{1}$ gives $\operatorname{adj} S = \operatorname{tree}(f + f^\top)\,\mathbf{1}\mathbf{1}^\top$. Now we turn to Z. By (8), $Z\mathbf{1} = 0$ and $\pi^\top Z = 0$. The matrix $Z + \mathbf{1}\pi^\top = (\mathbf{1}\pi^\top - G)^{-1}$ is invertible, so Z has rank $k-1$, and $\operatorname{adj} Z = z\,\mathbf{1}\pi^\top$ for some number z by the third fact. Since $\det Z = 0$, the second fact gives

$$\frac{1}{w} = \det(Z + \mathbf{1}\pi^\top) = \pi^\top \operatorname{adj}(Z)\,\mathbf{1} = z.$$

Also $\operatorname{adj}(Z^\top) = (\operatorname{adj} Z)^\top = z\,\pi\mathbf{1}^\top$. By (ii) and the first fact,

$$\operatorname{adj}\Gamma = \operatorname{adj}(Z)\operatorname{adj}(S)\operatorname{adj}(Z^\top) = z^2\,\operatorname{tree}(f + f^\top)\,\mathbf{1}(\pi^\top\mathbf{1})(\mathbf{1}^\top\pi)\mathbf{1}^\top = \frac{\operatorname{tree}(f + f^\top)}{w^2}\,\mathbf{1}\mathbf{1}^\top.$$

The entry of $\operatorname{adj}\Gamma$ at state 1 is $\det\Sigma$. This is the first formula in (iii), and the second follows from (i). □

Appendix B A Laplace lemma with a parameter

This appendix proves the Laplace lemma that Appendix A uses for Theorem 5.4, and its form with a complex Hessian that Section 6 uses for Theorem 6.6. There the parameter is $\alpha \in \mathcal{K}$. A field needs no parameter, since it enters the density as the exact factor $e^{Tv\cdot\alpha}$, and the compact space $\mathcal{K}$ can include the rates and the start when constants are needed uniformly over a class of chains. On $\mathbb{T}^m = (\mathbb{R}/2\pi\mathbb{Z})^m$ we use $\|\varphi\| = \max_i \operatorname{dist}(\varphi_i, 2\pi\mathbb{Z})$. For $x \in \mathbb{R}^m$ we write $\|x\| = \max_i |x_i|$ and $|x|$ for the Euclidean norm. Repeated indices are summed from 1 to m.

Lemma B.1 (Laplace integrals with a parameter) *Let $\mathcal{K}$ be a compact metric space and $m \geq 1$. Let $\Psi, a : \mathcal{K}\times\mathbb{T}^m \to \mathbb{C}$ be such that the φ-derivatives of Ψ up to order 4 and of a up to order 2 exist and are jointly continuous on $\mathcal{K}\times\mathbb{T}^m$. Assume that for every $\alpha \in \mathcal{K}$*

(i) the value $\Psi(\alpha,0)$ is real and $\nabla_\varphi\Psi(\alpha,0) = 0$,

(ii) the matrix $\mathcal{J}(\alpha) = -\nabla^2_\varphi\Psi(\alpha,0)$ is real, symmetric and positive definite, and

(iii) the inequality $\operatorname{Re}\Psi(\alpha,\varphi) < \Psi(\alpha,0)$ holds for every $\varphi \neq 0$ in $\mathbb{T}^m$.

Then, uniformly for $\alpha \in \mathcal{K}$ and $T \geq 1$,

$$\int_{\mathbb{T}^m} e^{T\Psi(\alpha,\varphi)}a(\alpha,\varphi)\,d\varphi = e^{T\Psi(\alpha,0)}\Big(\frac{2\pi}{T}\Big)^{m/2}\det\mathcal{J}(\alpha)^{-1/2}\Big(a(\alpha,0) + O\Big(\frac{1}{T}\Big)\Big).$$

Idea of the proof. By (iii) and compactness the part of the torus away from $\varphi = 0$ is exponentially small. Near 0 we substitute $\varphi = x/\sqrt{T}$ and expand. The integrand becomes a Gaussian weight times $a(\alpha,0)$, plus a term of order $T^{-1/2}$ that is odd in x and integrates to 0, plus a remainder of order $1/T$. All constants depend on α only through bounds that hold on the whole of $\mathcal{K}$.

Proof By compactness $B_\Psi = \max_{|\beta|\leq 4}\sup|\partial^\beta_\varphi\Psi|$ and $B_a = \max_{|\beta|\leq 2}\sup|\partial^\beta_\varphi a|$ are finite, the least eigenvalue of $\mathcal{J}(\alpha)$ has a positive minimum $\sigma_{\min}$ over $\mathcal{K}$, and $\det\mathcal{J}(\alpha) \leq (mB_\Psi)^m$.

Below, C depends only on m, $\sigma_{\min}$, B_Ψ, B_a and the numbers ε and η chosen below, and it may change from line to line. So every bound below is uniform on $\mathcal{K}$.

Localization. Choose $\varepsilon \in (0,1)$ with $m^3 B_\Psi \varepsilon \le \frac{3}{2}\sigma_{\min}$. The set of (α,φ) with $\|\varphi\| \ge \varepsilon$ is compact, and by (iii) the continuous function $\operatorname{Re}\Psi(\alpha,\varphi) - \Psi(\alpha,0)$ is negative on it. So it is at most $-\eta$ there for some $\eta > 0$, and the integral over this set has modulus at most $(2\pi)^m B_a e^{T(\Psi(\alpha,0)-\eta)} \le CT^{-1-m/2} e^{T\Psi(\alpha,0)}$.

Scaling. Reduction modulo 2π identifies the cube $\|x\| \le \varepsilon$ in $\mathbb{R}^m$ with the set $\|\varphi\| \le \varepsilon$, and it preserves measure. Let $R(\alpha,\varphi) = \Psi(\alpha,\varphi) - \Psi(\alpha,0) + \frac{1}{2}\varphi^\top \mathcal{J}(\alpha)\varphi$. By (i) and (ii), R and its derivatives of order 1 and 2 vanish at $\varphi = 0$. Taylor's theorem with integral remainder, which holds for complex values, gives on the cube $R = \frac{1}{6}\Psi_{ijk}\varphi^i\varphi^j\varphi^k + \omega_0$ with $|\omega_0| \le C\|\varphi\|^4$, where Ψ_{ijk} are the third derivatives at $(\alpha,0)$. It also gives $|R| \le \frac{m^3}{6} B_\Psi \|\varphi\|^3 \le \frac{1}{4}\sigma_{\min}|\varphi|^2$, since $\sum_{|\beta|=3} 1/\beta! = m^3/6$ and $\|\varphi\|^3 \le \varepsilon|\varphi|^2$. We substitute $\varphi = x/\sqrt{T}$ and put $w(x) = TR(\alpha, x/\sqrt{T})$. The integral over the cube is

$$T^{-m/2} e^{T\Psi(\alpha,0)} \int_{\|x\| \le \varepsilon\sqrt{T}} e^{-\frac{1}{2}x^\top \mathcal{J} x}\, e^{w(x)} a(\alpha, x/\sqrt{T})\, dx,$$

and $|w(x)| \le \frac{1}{4}\sigma_{\min}|x|^2$.

Taylor expansion. Let $P(x) = \frac{1}{6}\Psi_{ijk}x^i x^j x^k$, $a_0 = a(\alpha,0)$ and $a_i = \partial_{\varphi_i} a(\alpha,0)$. Then $w = T^{-1/2}P + \omega_1$ with $|\omega_1| \le CT^{-1}|x|^4$, and $|w| \le CT^{-1/2}|x|^3$. Since $|e^w - 1 - w| \le \frac{1}{2}|w|^2 e^{|w|}$, this gives $|e^w - 1 - T^{-1/2}P| \le CT^{-1}(|x|^4 + |x|^6) e^{\sigma_{\min}|x|^2/4}$. Taylor's theorem for a to orders 1 and 2 gives $|a(\alpha, x/\sqrt{T}) - a_0| \le CT^{-1/2}|x|$ and $|a(\alpha, x/\sqrt{T}) - a_0 - T^{-1/2}a_i x^i| \le CT^{-1}|x|^2$. Multiplying out,

$$e^w a(\alpha, x/\sqrt{T}) = a_0 + T^{-1/2}\big(a_0 P(x) + a_i x^i\big) + \omega(x), \qquad |\omega(x)| \le CT^{-1}(1+|x|)^6 e^{\sigma_{\min}|x|^2/4}.$$

Integration. Since $x^\top \mathcal{J} x \ge \sigma_{\min}|x|^2$, the term ω contributes at most CT^{-1} to the integral, because $(1+|x|)^6 e^{-\sigma_{\min}|x|^2/4}$ is integrable. Outside the cube $|x| > \varepsilon\sqrt{T}$, so extending the integrals of the other two terms to $\mathbb{R}^m$ changes them by at most $Ce^{-\sigma_{\min}\varepsilon^2 T/4} \le CT^{-1}$. On $\mathbb{R}^m$ we have $\int e^{-\frac{1}{2}x^\top \mathcal{J} x} dx = (2\pi)^{m/2} \det \mathcal{J}^{-1/2}$, and the odd polynomial $a_0 P(x) + a_i x^i$ integrates to 0 against this weight. This is why the error is $O(1/T)$ and not $O(T^{-1/2})$. So the integral over the cube is $e^{T\Psi(\alpha,0)} T^{-m/2} \big[(2\pi)^{m/2} \det \mathcal{J}^{-1/2} a_0 + O(T^{-1})\big]$. Since $\det \mathcal{J} \le (mB_\Psi)^m$, this error and the one from the localization are $O(1/T)$ relative to $e^{T\Psi(\alpha,0)}(2\pi/T)^{m/2} \det \mathcal{J}^{-1/2}$. □

Remark B.2 (the uniform form) The proof uses compactness and joint continuity only to obtain the bounds B_Ψ and B_a on the derivatives, the lower bound $\sigma_{\min}$ on the least eigenvalue of $\mathcal{J}$, and a gap $\operatorname{Re}\Psi(\alpha,\varphi) - \Psi(\alpha,0) \le -\eta$ for $\|\varphi\| \ge \varepsilon$, where $\varepsilon \in (0,1)$ satisfies $m^3 B_\Psi \varepsilon \le \frac{3}{2}\sigma_{\min}$. So the following holds with the same proof. Let $\mathcal{P}$ be any set, and let $\Psi, a : \mathcal{P} \times \mathbb{T}^m \to \mathbb{C}$ satisfy (i) and (ii) for every $\alpha \in \mathcal{P}$, with φ-derivatives of Ψ up to order 4 and of a up to order 2 bounded by B_Ψ and B_a on $\mathcal{P} \times \mathbb{T}^m$, with $\mathcal{J}(\alpha) \ge \sigma_{\min} I$ for all α, and with the gap above for all α. Then the expansion of Lemma B.1 holds uniformly for $\alpha \in \mathcal{P}$ and $T \ge 1$, with a constant that depends only on m, B_Ψ, B_a, $\sigma_{\min}$, ε and η.

Section 6 needs the same expansion when $\Psi(\alpha,0)$ and the Hessian are complex. Let $\mathcal{S}_m$ be the set of complex symmetric $m \times m$ matrices M whose real part is positive definite. For $M \in \mathcal{S}_m$ and $x \in \mathbb{C}^m$ with $x \ne 0$ we have $\operatorname{Re}(\bar{x}^\top M x) = \bar{x}^\top (\operatorname{Re} M) x > 0$, because $\bar{x}^\top P x$ is real for a real symmetric P. So $\det M \ne 0$. The set $\mathcal{S}_m$ is convex, so

$1/\det M$ has exactly two continuous square roots on it. We write $\det M^{-1/2}$ for the one that is positive on the real matrices of $\mathcal{S}_m$. Then

$$\int_{\mathbb{R}^m} e^{-\frac{1}{2}x^\top M x}\,dx = (2\pi)^{m/2}\det M^{-1/2} \qquad (M \in \mathcal{S}_m).$$

Indeed both sides are holomorphic functions of the entries M_{jk}, $j \le k$, on the open set $\mathcal{S}_m$. For the left side we may differentiate under the integral, since $|e^{-x^\top M x/2}| = e^{-x^\top (\operatorname{Re} M)x/2}$ is bounded by a fixed Gaussian for M near a point of $\mathcal{S}_m$. The right side is a continuous square root of a holomorphic function without zeros. The two sides agree on the real positive definite matrices, which form a nonempty open subset of the real points of $\mathcal{S}_m$. A holomorphic function on a connected open set that vanishes on such a subset vanishes everywhere, since its Taylor coefficients at a real point are real derivatives. So the two sides agree on $\mathcal{S}_m$.

Lemma B.3 (a complex Hessian) *Let $\mathcal{P}$ be a set and $m \ge 1$. Let $\Psi, a : \mathcal{P} \times \mathbb{T}^m \to \mathbb{C}$ have φ-derivatives up to order 4 and 2 that are continuous in φ and bounded in modulus by B_Ψ and B_a on $\mathcal{P} \times \mathbb{T}^m$. Let $\sigma_{\min} > 0$, $\eta > 0$ and $\varepsilon \in (0,1)$ with $m^3 B_\Psi \varepsilon \le \frac{3}{2}\sigma_{\min}$. Assume that for every $p \in \mathcal{P}$*

(i) $\nabla_\varphi \Psi(p,0) = 0$,

(ii) the complex symmetric matrix $\mathcal{J}(p) = -\nabla_\varphi^2 \Psi(p,0)$ *satisfies* $\operatorname{Re}\mathcal{J}(p) \ge \sigma_{\min} I$, *and*

(iii) $\operatorname{Re}\Psi(p,\varphi) - \operatorname{Re}\Psi(p,0) \le -\eta$ *for* $\|\varphi\| \ge \varepsilon$.

Then the expansion of Lemma B.1 holds uniformly for $p \in \mathcal{P}$ and $T \ge 1$, with the branch of $\det \mathcal{J}^{-1/2}$ defined above and a constant that depends only on m, B_Ψ, B_a, $\sigma_{\min}$, ε and η.

Proof The proof of Lemma B.1 in the form of Remark B.2 goes through with three changes. First, the factor $e^{T\Psi(p,0)}$ is now complex, and the localization uses only its modulus $e^{T\operatorname{Re}\Psi(p,0)}$. By (iii) the integral over $\|\varphi\| \ge \varepsilon$ has modulus at most $(2\pi)^m B_a e^{-\eta T}|e^{T\Psi(p,0)}|$. Second, for real x the number $x^\top(\operatorname{Im}\mathcal{J})x$ is real, so $|e^{-x^\top \mathcal{J}x/2}| = e^{-x^\top(\operatorname{Re}\mathcal{J})x/2} \le e^{-\sigma_{\min}|x|^2/2}$. This is the only bound on the Gaussian weight that the integration step uses, and the weight is still even and integrable, so the odd polynomial still integrates to 0. Third, the Gaussian integral over $\mathbb{R}^m$ is $(2\pi)^{m/2}\det\mathcal{J}^{-1/2}$ by the identity above, and Hadamard's inequality gives $|\det\mathcal{J}| \le (mB_\Psi)^m$, so the errors are $O(1/T)$ relative to the main term. The scaling and the Taylor expansion use only the bounds on the derivatives, Taylor's theorem for complex values and $|w(x)| \le \frac{1}{4}\sigma_{\min}|x|^2$, and they do not change. □

Appendix C Proofs for the complete graph

This appendix proves Theorems 7.5 and 7.6 and Corollary 7.7, with the notation of Section 7.2. The first lemma writes the law with blocks between refreshes in place of runs.

Lemma C.1 (the refresh form) *Let $N \ge 1$ and let n be a composition with support F. Sums over m run over the integer vectors with $1 \le m_i \le n_i$ for $i \in F$, and $M = \sum_i m_i$.*

(a) With $0^0 = 1$,

$$P_N(n) = \sum_m \Big(\sum_{i \in F} \mu_i m_i \Big) \frac{(M-1)!}{\prod_i m_i!} \, h^{M-1} \sigma^{N-M} \prod_{i \in F} \binom{n_i - 1}{m_i - 1}.$$

(b) Let $1/d < p < 1$, *so that* $0 < u < 1$ *and* $v > 0$. *Then*

$$S_n(u) = (1-u)^{N-1} \sum_m M! \, v^{M-1} \prod_{i \in F} \frac{1}{m_i!} \binom{n_i - 1}{m_i - 1}, \tag{C1}$$

which is a sum of positive terms, and

$$S_n(u) = \frac{(1-u)^{N-1}}{v} \int_0^\infty e^{-t} \prod_{i \in F} B_{n_i}(vt) \, dt, \tag{C2}$$

where B_n *is defined in* (23).

Proof (a) The transition matrix is $I + hQ = \sigma I + hJ$, so a word $x_1 \cdots x_N$ has probability $\mu_{x_1} \prod_{s<N} (\sigma \mathbf{1}\{x_s = x_{s+1}\} + h)$. We expand each factor into its two terms and call the steps where the term h is chosen the refreshes. If there are $M-1$ refreshes, they cut the word into M blocks, and a term of the expansion is not zero only if the word is constant on each block. So $P_N(n)$ is the sum, over the sequences of M block labels and over the block lengths, of $\mu_{\ell_1} h^{M-1} \sigma^{N-M}$, where ℓ_1 is the label of the first block. If m_i blocks carry the label i, there are $m_i(M-1)!/\prod_j m_j!$ sequences of labels that start with i. Consecutive labels may be equal. Given the labels, the lengths of the m_i blocks of the label i form an ordered sum of n_i with m_i positive parts, in $\binom{n_i-1}{m_i-1}$ ways. Multiplying and summing gives (a).

(b) For the uniform start $\sum_i \mu_i m_i = M/d$. By (20) we divide (a) by $V_N = p^{N-1}/d$, and $h^{M-1}\sigma^{N-M}/p^{N-1} = u^{M-1}(1-u)^{N-M} = (1-u)^{N-1} v^{M-1}$, since $\sigma/p = 1-u$. This gives (C1). We write $M! = \int_0^\infty e^{-t} t^M \, dt$ and sum over each m_i separately. All terms are nonnegative, so the sum and the integral can be exchanged, and this gives (C2). □

When $\sigma \geq 0$, part (a) says that the chain keeps its state with probability σ and otherwise draws a fresh state uniformly from all d states, possibly the same one. The proof does not need $\sigma \geq 0$.

C.1 Proof of Theorem 7.5

Let F be a face of k states and n a balanced composition of F, so that $P_N(n)/V_N = S_n(u)$ by (20), with $u = (1-p)/((d-1)p)$.

Part (i), one crossing. A word with j switches is a sequence of $j+1$ run colors in which successive colors differ, together with run lengths. If color i appears m_i times, its run lengths can be chosen in $\binom{n_i-1}{m_i-1}$ ways. So every coefficient of S_n is nonnegative, and the first nonzero one is $W_n(k-1) = k!$, from the $k!$ orders of k single runs. Hence S_n increases strictly from 0 to ∞ on $u > 0$ and equals 1 exactly once. As p increases over $(0,1)$, u decreases over $(0,\infty)$. This gives $p_{N,k}$ and the two inequalities for the balanced composition.

Part (i), monotonicity in N. Choose the balanced compositions for increasing N by adding 1 to one of the smallest coordinates. Each coefficient $W_n(j)$ is then nondecreasing. Indeed, inserting an extra copy of the incremented color just before its

first occurrence keeps the number of switches, and deleting the first letter of the first run of that color inverts it. The coefficient of u^k is

$$W_n(k) = \frac{k!(k-1)}{2} \sum_i (n_i - 1) = \frac{k!(k-1)}{2}(N-k),$$

since exactly one run color appears twice, and for each choice of it there are $(k+1)!/2 - k! = k!(k-1)/2$ admissible sequences of run colors (all orders, minus those with the two copies adjacent) and $n_i - 1$ ways to split its n_i visits into two runs. This coefficient increases strictly with N. So at every $u > 0$ the value $S_n(u)$ increases strictly with N, the root of $S_n(u) = 1$ decreases strictly, and $p = 1/(1 + (d-1)u)$ increases.

Part (i), the first crossing. If $N = k$, every coordinate is 1 and $S_n(u) = k!u^{k-1}$, which gives $p_{k,k}$. Since the root decreases with N, every root has $u \le (k!)^{-1/(k-1)} < 1$, and $u < 1$ means $h < p$, that is, $p > 1/d$.

Part (i), the face maximum. For $p \ge 1/d$, Theorem 7.6 gives $M_F = P_N(n)$. For $p < 1/d$ we have $u > 1$, which exceeds the root of $S_n(u) = 1$, so $M_F \ge P_N(n) > V_N$. Hence $M_F = V_N$ exactly at $p_{N,k}$, with the same two inequalities.

Part (ii). By (9) with $K_F = D_k/d$ and $\lambda_F = d - k$ (Corollary 7.3), uniformly for $1 \le T \le N^{1/3}$,

$$\log M_F = -(k-1)L + \tfrac{k-1}{2} \log T + \log(D_k/d) - (d-k)T + O(1/T + T^2/N).$$

By (1), $\log V_N = -\log d - (d-1)T + O(T^2/N)$. Subtracting, and using $\log D_k = -(k-1)a_k$, gives the display. With $T = L - \frac{1}{2}\log L + t$ and t in a compact set, $T - L + \frac{1}{2}\log T = t + \frac{1}{2}\log(T/L) = t + O(\log L/L)$, and the error tends to 0.

Part (iii). Let G be a vertex, so that $M_G = V_N$, $\Delta = \delta = k-1$ and $\log(K_F/K_G) = \log D_k = -(k-1)a_k$. By Theorem 5.8(i) the function $\log(M_F/M_G)$ has a zero in $[L/2, 2L]$, and by (i) this zero is $T_{N,k}$. Dividing (11) by $k-1$ gives the implicit equation. For the explicit form put $z_* = L - \frac{1}{2}\log L + a_k + e_L/L$ with $e_L = \frac{1}{4}\log L - \frac{1}{2}a_k$. Since $\log z_* = \log L - \frac{\log L}{2L} + \frac{a_k}{L} + O(\frac{(\log L)^2}{L^2})$, we get $z_* + \frac{1}{2}\log z_* = L + a_k + O((\log L)^2/L^2)$. The derivative of $z + \frac{1}{2}\log z$ is at least 1, so $T_{N,k} - z_* = O(1/L)$, which is (22).

Part (iv). We have $a_k = \frac{1}{2}\log(4\pi) - f(k)$ with $f(x) = (x + \frac{1}{2})\log x/(x-1)$. For $x \ge 2$ the sign of $f'(x)$ is the sign of $\Delta(x) = 2x^2 - x - 1 - 3x\log x$. Here $\Delta(2) = 5 - 6\log 2 > 0$, and $\Delta'(x) = 4x - 4 - 3\log x > 0$ for $x \ge 2$, since $\Delta'(2) > 0$ and $\Delta''(x) = 4 - 3/x > 0$. So f increases strictly and a_k decreases strictly. This order is checked in Lean. □

C.2 Proof of Theorem 7.6

Let $\mu_a = \mu_0$ for $a \in F$. By (2), $P_N(n) = \mu_0 p^{N-1} S_n(u)$ for every composition with support F, so it is enough to show that the move strictly increases $S_n(u)$. The rest follows, since S_n is symmetric in the coordinates of n and repeated moves reach a balanced composition. At $p = 1/d$ we have $u = 1$, $S_n(1) = N!/\prod_i n_i!$, and the move multiplies it by $n_b/(n_a + 1) > 1$. Let $1/d < p < 1$, so that $v > 0$.

Log-concavity. We show that for each $w > 0$ the sequence $B_n(w)$, $n \geq 1$, is strictly log-concave. Write

$$\frac{B_n(w)}{w} = b_{n-1}, \qquad b_j = \sum_{i=0}^{j} \binom{j}{i} \rho_i, \quad \rho_i = \frac{w^i}{(i+1)!}, \qquad c_j = \sum_{i=0}^{j} \binom{j}{i} \rho_{i+1}.$$

Pascal's identity gives $b_{j+1} = b_j + c_j$. The ratio c_j/b_j is the mean of the strictly decreasing sequence $f_i = \rho_{i+1}/\rho_i = w/(i+2)$ under the weights $\omega_i = \binom{j}{i}\rho_i$, $0 \leq i \leq j$, normalized. Let $\omega_i' = \binom{j+1}{i}\rho_i$, $0 \leq i \leq j+1$, be the weights for $j+1$, and put $\omega_{j+1} = 0$. On $0 \leq i \leq j$ the ratio $\omega_i'/\omega_i = (j+1)/(j+1-i)$ increases in i. So for $i < l$ we have $\omega_i'\omega_l - \omega_l'\omega_i \leq 0$, with strict inequality when $l = j+1$. Hence

$$\sum_i \omega_i' f_i \sum_l \omega_l - \sum_i \omega_i' \sum_l \omega_l f_l = \sum_{i<l} (\omega_i'\omega_l - \omega_l'\omega_i)(f_i - f_l) < 0,$$

since $f_i > f_l$ for $i < l$. So $c_{j+1}/b_{j+1} < c_j/b_j$, and $b_{j+1}/b_j = 1 + c_j/b_j$ decreases strictly with j. By the Laguerre form in (23), this is a Turán-type inequality for the Laguerre polynomials.

The transfer. Let $n_b \geq n_a + 2$. By log-concavity, $B_{n_a+1}(w)B_{n_b-1}(w) > B_{n_a}(w)B_{n_b}(w)$. We apply this for $w = vt > 0$ in (C2) and integrate. □

C.3 Proof of Corollary 7.7

For large N, $T \leq CL$ forces $h \to 0$ and so $p > 1/d$. By Theorem 7.6 every global mode is a vertex or a balanced composition. So we compare the ratios M_F/V_N, which equal $S_n(u)$ at the balanced compositions n of F and depend on F only through $k = |F|$, with 1 and with each other. Fix $0 < c < 1$.

Case $T \leq cL$. The ratio $S_n(u)$ increases with u, and so with T. So it is at most its value at $T = cL$, which is $\exp(-(k-1)(1-c)L + O(\log L)) \to 0$ by Theorem 7.5(ii). So the vertices are the only modes, and $p > p_{N,d}$.

Case $cL \leq T \leq CL$. Theorem 7.5(ii) gives $\log(M_F/V_N) = (k-1)(y - a_k) + o(1)$ uniformly, with $y = T - L + \frac{1}{2}\log T$ the same for every k. Fix $0 < \delta < (a_{d-1} - a_d)/2$ when $d \geq 3$, and any $\delta > 0$ when $d = 2$. If $y \leq a_d - \delta$, every face ratio is at most $e^{-(k-1)\delta + o(1)} < 1$, so only the vertices are modes. If $y \geq a_d + \delta$, the full center beats the vertices, and it beats a balanced composition with $2 \leq k < d$ positive coordinates by

$$(d-k)(y - a_d) + (k-1)(a_k - a_d) + o(1) > 0.$$

If $|y - a_d| < \delta$, a proper face with $k \geq 2$ states has $(k-1)(y-a_k) + o(1) \leq -(k-1)\delta + o(1) < 0$, since $a_k - a_d > 2\delta$. So it loses to the vertices, and the comparison of the vertices with the full centers is decided by the unique crossing $p_{N,d}$ of Theorem 7.5(i). In each subcase the conclusion agrees with (i) to (iii).

The last claim follows from (22) and $p_{N,k} = 1 - (d-1)T_{N,k}/N$, since $a_2 > \cdots > a_d$. □

Appendix D Proofs near a vertex

We prove Lemma 8.1 and Theorem 8.2, with the notation of Section 8.

Proof of Lemma 8.1 By definition, $B_a(w)/w = \sum_{j=0}^{a-1} \binom{a-1}{j} w^j/(j+1)!$. So the coefficient of $(aw)^j/[j!(j+1)!]$ in $B_a(w)/w$ is $\binom{a-1}{j} j!/a^j = \prod_{l=1}^{j}(1-l/a)_+$, also for $j \ge a$, where it is 0. By the clipped products of [32, Lemma 3.2(b)], its deficit from 1 lies between 0 and $j(j+1)/(2a)$. Also $\sum_{j\ge1} j(j+1)x^j/[j!(j+1)!] = \sum_{j\ge1} x^j/[(j-1)!\,j!] = x\,\mathcal{I}(x)$. So

$$0 \le \mathcal{I}(aw) - \frac{B_a(w)}{w} \le \frac{1}{2a} \sum_{j\ge0} \frac{j(j+1)(aw)^j}{j!(j+1)!} = \frac{w}{2}\,\mathcal{I}(aw).$$

□

Lemma D.1 (elasticity and concavity of the rare-count polynomial) *The polynomial $\mathcal{B}$ is convex, $\log \mathcal{B}$ is concave on $(0,\infty)$, and κ is nondecreasing. Also*

$$\frac{1}{|a|+1} \le y_a \le 1, \qquad \kappa(y) \le \ell + \sqrt{\ell|a|y} \quad (y>0). \tag{D1}$$

For $0 < y \le 1$ we also have $\kappa(y) \ge \max\{\ell, \frac{1}{2}\sqrt{|a|y}\}$.

Proof Step 1. One factor as a law. The polynomial $\mathcal{B}$ has nonnegative coefficients, so it is convex. For one factor B_a and $y > 0$, let J take values in $\{0,\dots,a-1\}$ with $\mathbb{P}(J=j)$ proportional to $\binom{a-1}{j} y^j/(j+1)!$. Since $B_a(y) = \sum_j y\binom{a-1}{j} y^j/(j+1)!$, the elasticity of B_a is $\kappa_a = yB_a'/B_a = \mathbb{E}[J+1]$. The ratio

$$\mathcal{R}(j) = \frac{(j+1)\mathbb{P}(J=j+1)}{\mathbb{P}(J=j)} = \frac{y(a-1-j)}{j+2}, \qquad \mathcal{R}(a-1) = 0,$$

is nonincreasing in j. Summing the ratio identity gives $\mathbb{E}J = \mathbb{E}\mathcal{R}(J)$ and $\mathbb{E}J^2 = \mathbb{E}[(J+1)\mathcal{R}(J)]$. So $\operatorname{Var} J = \mathbb{E}J + \operatorname{Cov}(J, \mathcal{R}(J)) \le \mathbb{E}J$, since for an independent copy J' twice the covariance is $\mathbb{E}[(J-J')(\mathcal{R}(J)-\mathcal{R}(J'))] \le 0$.

Step 2. Concavity and monotonicity. Since $y^2 B_a''/B_a = \mathbb{E}[J(J+1)]$, we get

$$(\log B_a)'' = \frac{\operatorname{Var} J - \mathbb{E}[J+1]}{y^2} < 0, \qquad y\kappa_a' = \operatorname{Var} J \ge 0.$$

Summing over the factors proves the concavity of $\log \mathcal{B}$ and the monotonicity of $\kappa = \sum_i \kappa_{a_i}$.

Step 3. The elasticity bounds. The differential equation $yB_a'' + yB_a' = aB_a$, which we check on the coefficients, gives

$$ay = \kappa_a^2 + \operatorname{Var} J + (y-1)\kappa_a.$$

Using $0 \le \operatorname{Var} J \le \kappa_a - 1$ we get

$$\kappa_a \le 1 + \sqrt{ay}, \qquad \kappa_a^2 + y\kappa_a \ge ay + 1.$$

For $0 < y \le 1$ the second inequality gives $(\kappa_a + \frac{1}{2})^2 > ay$, and $\kappa_a \ge 1$. So $\kappa_a \ge \max\{1, \sqrt{ay} - 1/2\} \ge \frac{1}{2}\sqrt{ay}$. Summing over the factors proves the elasticity bounds, using the Cauchy–Schwarz inequality $\sum_i \sqrt{a_i} \le \sqrt{\ell|a|}$ for the upper bound and $\sqrt{\sum_i x_i} \le \sum_i \sqrt{x_i}$ for the lower one.

Step 4. The root. We have $B_a(1) \ge 1$, from its term $m=1$, and $B_a(y) \le y\,\mathcal{I}(ay) \le ye^{ay/2}$ by Lemma 8.1 and $(j+1)! \ge 2^j$. So $\mathcal{B}(1) \ge 1$, while, since $|a|+1 \ge 2$, $\mathcal{B}(1/(|a|+1)) \le (|a|+1)^{-\ell} e^{|a|/(2(|a|+1))} \le \frac{1}{2}e^{1/2} < 1$. □

Lemma D.2 (an exact compound-binomial representation) *For $0 < u < 1$, put $v = u/(1-u)$. Let Γ_2 have the gamma distribution with shape 2 and rate 1, let $E_1, \dots, E_{b-1}$ be independent rate-one exponential variables, and let $Z_1, \dots, Z_{b-1}$ be independent Bernoulli variables with parameter u. Take all these variables independent and set*

$$\mathcal{X} = \Gamma_2 + \sum_{j=1}^{b-1} Z_j E_j.$$

Then

$$S_{(b,a)}(u) = (1-u)^{|a|}\mathbb{E}[\mathcal{B}(v\mathcal{X})], \qquad \mathbb{E}e^{s\mathcal{X}} = (1-s)^{-2}\left(1 + \frac{us}{1-s}\right)^{b-1} \quad (s<1). \tag{D2}$$

In particular, set $y = bu^2$ and $\kappa = \kappa(y)$. For every $0 < u < 1$,

$$(1-u)^{|a|}\mathcal{B}(y) \le S_{(b,a)}(u). \tag{D3}$$

If also $s = \kappa/[bu(1-u)] < 1$, then

$$S_{(b,a)}(u) \le (1-u)^{|a|}\mathcal{B}(y)e^{-\kappa}(1-s)^{-2}\left(1 + \frac{us}{1-s}\right)^{b-1}. \tag{D4}$$

Proof For $0 < u < 1$ the stay probability $p = 1/(1 + (d-1)u)$ lies in $(1/d, 1)$, so by (C2) with $N = b + |a|$,

$$S_{(b,a)}(u) = \frac{(1-u)^{b+|a|-1}}{v}\int_0^\infty e^{-t}B_b(vt)\,\mathcal{B}(vt)\,dt.$$

The integral of $e^{-t}B_b(vt)$ is $\sum_m \binom{b-1}{m-1}v^m = v(1+v)^{b-1}$, and $(1-u)^{b-1}(1+v)^{b-1} = 1$. So $e^{-t}B_b(vt)$, normalized, is the density $\sum_{j=0}^{b-1}\binom{b-1}{j}u^j(1-u)^{b-1-j}\,t^{j+1}e^{-t}/(j+1)!$, a mixture of gamma densities with shape $J+2$, where J has the binomial law with parameters $b-1$ and u. This is exactly the law of $\mathcal{X}$, which proves the identity and its moment generating function. Since $\mathbb{E}\mathcal{X} = 2+(b-1)u$, we have $v\mathbb{E}\mathcal{X} - bu^2 = u(2-u+bu^2)/(1-u) > 0$, so $v\mathbb{E}\mathcal{X} \ge y$. Jensen's inequality for the convex function $\mathcal{B}$, and the monotonicity of $\mathcal{B}$, prove (D3). Concavity of $\log\mathcal{B}$ gives, for $w > 0$,

$$\mathcal{B}(w) \le \mathcal{B}(y)\exp\left(\frac{\kappa}{y}(w-y)\right).$$

Insert $w = v\mathcal{X}$, note that $\kappa v/y = s$, and use the moment generating function to prove (D4). □

Proof of Theorem 8.2 Write $S = S_{(b,a)}$, $u_0 = \sqrt{y_a/b}$ and $\kappa_0 = \kappa(y_a)$. Since S increases on $u > 0$, it suffices to show $\log S(u_+) > 0$ and $\log S(u_-) < 0$ at $u_\pm = (1 \pm \delta)u_0$. Here $\delta \le 1/2$ forces $\varepsilon \le 1/(32c_\ell) \le 1/96$, and $u_0 \le \sqrt{1/b} \le \varepsilon$.

Step 1. The upper point. We have $u_+ \le \frac{3}{2}\varepsilon < 1/4$. Since $\log\mathcal{B}(y_a) = 0$ and $(\log\mathcal{B})' = \kappa/y$, the monotonicity of κ gives $\log\mathcal{B}(bu_+^2) \ge \kappa_0\log(bu_+^2/y_a) = 2\kappa_0\log(1+\delta)$. The lower bound in Lemma D.1, applied at $y_a \le 1$, gives $|a|u_0/\kappa_0 \le 2\sqrt{|a|/b} \le 2\varepsilon$. Hence, by (D3) and $-\log(1-x) \le x/(1-x)$,

$$\log S(u_+) \ge 2\kappa_0\log(1+\delta) - \frac{|a|u_+}{1-u_+} \ge \kappa_0\left(\frac{4}{3}\delta - 4\varepsilon\right) > 0.$$

Here we used $2\log(1+\delta) \ge \frac{4}{3}\delta$ for $\delta \le \frac{1}{2}$, and $|a|u_+/(1-u_+) \le \frac{4}{3}\cdot\frac{3}{2}|a|u_0 \le 4\kappa_0\varepsilon$. The last inequality holds since $\frac{4}{3}\delta = \frac{64}{3}c_\ell\varepsilon > 4\varepsilon$.

Step 2. The lower point. Write $y = bu_-^2$ and $\kappa = \kappa(y)$. The inequalities $y_a \geq 1/(|a|+1)$, $\delta \leq 1/2$ and the upper bound in (D1) imply

$$u_- \leq \varepsilon, \qquad bu_- \geq \frac{1}{2\varepsilon}, \qquad s = \frac{\kappa}{bu_-(1-u_-)} \leq 2(2\ell + \sqrt{\ell})\varepsilon \leq 4c_\ell \varepsilon \leq \frac{1}{8}.$$

For the bound on s, note that $\kappa/(bu_-) \leq \ell/(bu_-) + \sqrt{\ell|a|y}/(bu_-) \leq 2\ell\varepsilon + \sqrt{\ell|a|/b} \leq (2\ell + \sqrt{\ell})\varepsilon$, and $1 - u_- \geq \frac{1}{2}$. We use $\log(1+x) \leq x$, $-\log(1-s) \leq s/(1-s)$ and $u_-, s \leq 1/4$ in (D4), and we drop the nonpositive term $|a|\log(1-u_-)$. This gives

$$\begin{aligned} \log \frac{S(u_-)}{\mathcal{B}(y)} &\leq -\kappa + \frac{2s}{1-s} + \frac{bu_- s}{1-s} \\ &= \kappa \left(\frac{1}{(1-u_-)(1-s)} - 1 \right) + \frac{2s}{1-s} \\ &\leq 5\kappa(u_- + s) \leq 25 c_\ell \kappa \varepsilon. \end{aligned}$$

Here the equality uses $bu_- s = \kappa/(1-u_-)$. The next inequality uses $(1-u_-)(1-s) \geq \frac{21}{32}$, so the first term is at most $\frac{32}{21}\kappa(u_- + s)$, and $\kappa \geq \ell \geq 1$, so the second is at most $\frac{16}{7}\kappa s$. The last one uses $u_- + s \leq 5c_\ell\varepsilon$. Again by the monotonicity of κ,

$$\log \mathcal{B}(y) = -\int_y^{y_a} \frac{\kappa(x)}{x}\, dx \leq 2\kappa \log(1-\delta) \leq -2\kappa\delta = -32 c_\ell \kappa \varepsilon.$$

Hence $\log S(u_-) \leq \kappa(-32c_\ell + 25c_\ell)\varepsilon < 0$. □

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